

Asterion Supply Chain Ontology

Submission bundle for AWS Context Ontology Accelerator

Generated from 01_base_tables_and_data.sql — 37 classes covering all 37 tables in Data/*.csv

README.md

Overview: what this bundle is and how it maps to the AWS Context Ontology Accelerator

Supply Chain Ontology – AWS Context Ontology Accelerator submission bundle

Generated from the authoritative DDL in
[``01_base_tables_and_data.sql``](../01_base_tables_and_data.sql)
by [``generate_ontology.py``](generate_ontology.py). The 37 classes here cover every table in that
DDL 1:1 with the 37 CSVs in [``Data/``](../Data) – this bundle is the ontology description of that
dataset, structured the way ``aws/context-ontology-accelerator`` (COA) actually ingests ontologies.

Note: [aws/context-ontology-accelerator](https://github.com/aws/context-ontology-accelerator) is
published as a **read-only mirror** – it does not accept pull requests. "Submitting" this bundle
means calling a deployed COA instance's Control Plane API (Scan → **Model** → Serve), not committing
a file into that repo.

Files

```
| File | Purpose | COA target |
|---|---|---|
| `supply_chain.ttl` | OWL/RDFS/SKOS T-Box: 37 classes, 57 object properties (FKs), 157 datatype
  properties, `owl:hasKey`/cardinality restrictions, `skos:altLabel` synonyms | `POST
  /namespaces/{ns}/ontologies/{ontologyId}/upload` (`UploadOntology`) |
| `supply_chain_r2rml.ttl` | R2RML mapping from each class back to its Snowflake table/columns,
  incl. FK join conditions | `POST /namespaces/{ns}/proposals/update` (`UpdateProposal.r2rmlTurtle`)
  – this is what makes a class `coa:isMapped` and queryable via the VKG (Tier-2 NL→SPARQL→SQL);
  without it the T-Box only serves Tier-3 retrieval |
| `supply_chain_catalog.json` | Data-catalog / schema-induction input: tables, columns, types,
  constraints (PK/FK), synonyms, sampled distinct values | Mock data-catalog source / induction
  input for the `ontology-engine`, namespace `ns-asterion-supplychain` |
```

Conventions (per ``packages/ontology-engine/docs/ontology-data-model.md``)

- Property IRIs are ``camelCase(Class)_camelCase(Column)`` so identically-named columns on different
classes never collide into one multi-domain property.
- FKs emit an ``owl:ObjectProperty`` in the T-Box (what the relationship **means**) and a Referencing
Object Map in the R2RML (the actual join keys) – join keys never appear in the T-Box.
- No ``owl:imports`` (Ontop network-resolves imports at VKG load and fails every query in the
namespace on a non-dereferenceable IRI) and no ``coa:isMapped`` (server-derived from R2RML
``rr:class``
membership; caller-supplied triples are stripped at ingest).

Regenerating

```
```bash
python ontology/generate_ontology.py
```
```

Re-parses the DDL and rewrites all three files. Do not hand-edit them – add table/column business
names, synonyms, enums, or relationship labels to the tables in ``generate_ontology.py`` instead.

Scope

Namespace: ``ns-asterion-supplychain``. Datasource: ``ds-scs-snowflake`` (``SCS.INVENTORY`` schema,
served via Athena federation – no direct Snowflake driver on the query path). Covers supplier
master and approvals, purchase orders and receipts, bills of material, work orders, inventory
snapshots, shipments with lane/carrier detail, IoT cold-chain telemetry, and quality inspections
and nonconformances. All data is synthetic.

supply_chain.ttl

OWL / RDFS / SKOS T-Box — target: UploadOntology

```
# Supply Chain Ontology – Asterion Devices (synthetic)
# Generated by generate_ontology.py from 01_base_tables_and_data.sql. Do not hand-edit.
#
# Target: AWS Context Ontology Accelerator
#   POST /namespaces/{ns}/ontologies/{ontologyId}/upload
#
# Conventions (packages/ontology-engine/docs/ontology-data-model.md):
#   - owl:imports is never emitted: Ontop network-resolves imports at VKG load
#   - and fails every query in the namespace on a non-dereferenceable IRI.
#   - coa:isMapped is never emitted: it is server-derived from R2RML rr:class
#   - membership and caller-supplied triples are stripped at ingest.
#   - FK join keys live only in the R2RML; the T-Box carries what the
#     relationship means (domain, range, coa:fkProvenance).

@prefix owl: <http://www.w3.org/2002/07/owl#> .
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
@prefix skos: <http://www.w3.org/2004/02/skos/core#> .
@prefix coa: <http://coa.amazon.com/vocab/coa#> .
@prefix sc: <https://asterion.example.com/ontology/supplychain#> .

<https://asterion.example.com/ontology/supplychain#> a owl:Ontology ;
  rdfs:label "Asterion Supply Chain Ontology" ;
  rdfs:comment "Electronics manufacturing supply chain: sourcing, procurement, inbound quality,
  manufacturing, inventory, logistics and demand. Induced from the SCS.INVENTORY Snowflake
  schema – 37 tables covering supplier master and approvals, purchase orders and receipts, bills
  of material, work orders, inventory snapshots, shipments with lane and carrier detail, IoT
  cold-chain telemetry, and quality inspections and nonconformances. All data is synthetic." ;
  owl:versionInfo "1.0.0" .

#-----
# CalendarDate <- SCS.INVENTORY.DIM_DATE
#-----

sc:CalendarDate a owl:Class ;
  rdfs:label "Calendar Date" ;
  rdfs:comment "Calendar dimension from 2022-01-01 to 2026-12-31" ;
  skos:altLabel "date" ;
  skos:altLabel "day" ;
  skos:altLabel "calendar" ;
  skos:altLabel "DIM_DATE" ;
  skos:altLabel "time dimension" ;
  owl:hasKey ( sc:calendardate_dateKey ) ;
  rdfs:subClassOf
    [ a owl:Restriction ; owl:onProperty sc:calendardate_dateKey ; owl:cardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:calendardate_year ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:calendardate_quarter ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:calendardate_month ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:calendardate_monthName ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:calendardate_weekOfYear ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:calendardate_dayOfWeek ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:calendardate_dayName ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:calendardate_isWeekend ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:calendardate_fiscalYear ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:calendardate_fiscalQuarter ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] .

sc:calendardate_dateKey a owl:DatatypeProperty ;
  rdfs:label "DATE_KEY" ;
  rdfs:comment "DIM_DATE.DATE_KEY" ;
  rdfs:domain sc:CalendarDate ;
  rdfs:range xsd:date .

sc:calendardate_year a owl:DatatypeProperty ;
```

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    rdfs:label "YEAR" ;
    rdfs:comment "DIM_DATE.YEAR" ;
    rdfs:domain sc:CalendarDate ;
    rdfs:range xsd:integer .

sc:calendardate_quarter a owl:DatatypeProperty ;
    rdfs:label "QUARTER" ;
    rdfs:comment "DIM_DATE.QUARTER" ;
    rdfs:domain sc:CalendarDate ;
    rdfs:range xsd:integer .

sc:calendardate_month a owl:DatatypeProperty ;
    rdfs:label "MONTH" ;
    rdfs:comment "DIM_DATE.MONTH" ;
    rdfs:domain sc:CalendarDate ;
    rdfs:range xsd:integer .

sc:calendardate_monthName a owl:DatatypeProperty ;
    rdfs:label "MONTH_NAME" ;
    rdfs:comment "DIM_DATE.MONTH_NAME" ;
    rdfs:domain sc:CalendarDate ;
    coa:distinctValues "Apr" ;
    coa:distinctValues "Aug" ;
    coa:distinctValues "Dec" ;
    coa:distinctValues "Feb" ;
    coa:distinctValues "Jan" ;
    coa:distinctValues "Jul" ;
    coa:distinctValues "Jun" ;
    coa:distinctValues "Mar" ;
    coa:distinctValues "May" ;
    coa:distinctValues "Nov" ;
    coa:distinctValues "Oct" ;
    coa:distinctValues "Sep" ;
    rdfs:range xsd:string .

sc:calendardate_weekOfYear a owl:DatatypeProperty ;
    rdfs:label "WEEK_OF_YEAR" ;
    rdfs:comment "DIM_DATE.WEEK_OF_YEAR" ;
    rdfs:domain sc:CalendarDate ;
    rdfs:range xsd:integer .

sc:calendardate_dayOfWeek a owl:DatatypeProperty ;
    rdfs:label "DAY_OF_WEEK" ;
    rdfs:comment "DIM_DATE.DAY_OF_WEEK" ;
    rdfs:domain sc:CalendarDate ;
    rdfs:range xsd:integer .

sc:calendardate_dayName a owl:DatatypeProperty ;
    rdfs:label "DAY_NAME" ;
    rdfs:comment "DIM_DATE.DAY_NAME" ;
    rdfs:domain sc:CalendarDate ;
    coa:distinctValues "Fri" ;
    coa:distinctValues "Mon" ;
    coa:distinctValues "Sat" ;
    coa:distinctValues "Sun" ;
    coa:distinctValues "Thu" ;
    coa:distinctValues "Tue" ;
    coa:distinctValues "Wed" ;
    rdfs:range xsd:string .

sc:calendardate_isWeekend a owl:DatatypeProperty ;
    rdfs:label "IS_WEEKEND" ;
    rdfs:comment "DIM_DATE.IS_WEEKEND" ;
    rdfs:domain sc:CalendarDate ;
    rdfs:range xsd:boolean .

sc:calendardate_fiscalYear a owl:DatatypeProperty ;
    rdfs:label "FISCAL_YEAR" ;
    rdfs:comment "DIM_DATE.FISCAL_YEAR" ;
    rdfs:domain sc:CalendarDate ;
    rdfs:range xsd:integer .

sc:calendardate_fiscalQuarter a owl:DatatypeProperty ;
    rdfs:label "FISCAL_QUARTER" ;
    rdfs:comment "DIM_DATE.FISCAL_QUARTER" ;
    rdfs:domain sc:CalendarDate ;
    rdfs:range xsd:integer .

```

```

#-----
# Region <- SCS.INVENTORY.DIM_GEO_REGION

```

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#-----
sc:Region a owl:Class ;
  rdfs:label "Region" ;
  rdfs:comment "Geographic regions" ;
  skos:altLabel "geography" ;
  skos:altLabel "geo region" ;
  skos:altLabel "DIM_GEO_REGION" ;
  skos:altLabel "world region" ;
  owl:hasKey ( sc:region_regionId ) ;
  rdfs:subClassOf
    [ a owl:Restriction ; owl:onProperty sc:region_regionId ; owl:cardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:region_regionName ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] .

sc:region_regionId a owl:DatatypeProperty ;
  rdfs:label "REGION_ID" ;
  rdfs:comment "DIM_GEO_REGION.REGION_ID" ;
  rdfs:domain sc:Region ;
  rdfs:range xsd:integer .

sc:region_regionName a owl:DatatypeProperty ;
  rdfs:label "REGION_NAME" ;
  rdfs:comment "DIM_GEO_REGION.REGION_NAME" ;
  rdfs:domain sc:Region ;
  rdfs:range xsd:string .

#-----
# Country <- SCS.INVENTORY.DIM_GEO_COUNTRY
#-----

sc:Country a owl:Class ;
  rdfs:label "Country" ;
  rdfs:comment "Countries mapped to regions" ;
  skos:altLabel "nation" ;
  skos:altLabel "DIM_GEO_COUNTRY" ;
  skos:altLabel "country of origin" ;
  owl:hasKey ( sc:country_countryId ) ;
  rdfs:subClassOf
    [ a owl:Restriction ; owl:onProperty sc:country_countryId ; owl:cardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:country_countryName ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:country_countryCode ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:country_regionId ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] .

sc:country_countryId a owl:DatatypeProperty ;
  rdfs:label "COUNTRY_ID" ;
  rdfs:comment "DIM_GEO_COUNTRY.COUNTRY_ID" ;
  rdfs:domain sc:Country ;
  rdfs:range xsd:integer .

sc:country_countryName a owl:DatatypeProperty ;
  rdfs:label "COUNTRY_NAME" ;
  rdfs:comment "DIM_GEO_COUNTRY.COUNTRY_NAME" ;
  rdfs:domain sc:Country ;
  rdfs:range xsd:string .

sc:country_countryCode a owl:DatatypeProperty ;
  rdfs:label "COUNTRY_CODE" ;
  rdfs:comment "DIM_GEO_COUNTRY.COUNTRY_CODE" ;
  rdfs:domain sc:Country ;
  rdfs:range xsd:string .

sc:country_regionId a owl:ObjectProperty ;
  rdfs:label "in region" ;
  rdfs:comment "Geographic region the country belongs to" ;
  rdfs:domain sc:Country ;
  rdfs:range sc:Region ;
  coa:fkProvenance "MANY_TO_ONE" .

#-----
# Commodity <- SCS.INVENTORY.DIM_COMMODITY
#-----

sc:Commodity a owl:Class ;
  rdfs:label "Commodity" ;

```

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    rdfs:comment "Top-level commodity classification" ;
    skos:altLabel "category" ;
    skos:altLabel "commodity group" ;
    skos:altLabel "DIM_COMMODITY" ;
    skos:altLabel "spend category" ;
    owl:hasKey ( sc:commodity_commodityId ) ;
    rdfs:subClassOf
      [ a owl:Restriction ; owl:onProperty sc:commodity_commodityId ; owl:cardinality
        "1"^^xsd:nonNegativeInteger ] ,
      [ a owl:Restriction ; owl:onProperty sc:commodity_commodityName ; owl:minCardinality
        "1"^^xsd:nonNegativeInteger ] .

sc:commodity_commodityId a owl:DatatypeProperty ;
  rdfs:label "COMMODITY_ID" ;
  rdfs:comment "DIM_COMMODITY.COMMODITY_ID" ;
  rdfs:domain sc:Commodity ;
  rdfs:range xsd:integer .

sc:commodity_commodityName a owl:DatatypeProperty ;
  rdfs:label "COMMODITY_NAME" ;
  rdfs:comment "DIM_COMMODITY.COMMODITY_NAME" ;
  rdfs:domain sc:Commodity ;
  rdfs:range xsd:string .

#-----
# SubCommodity <- SCS.INVENTORY.DIM_SUB_COMMODITY
#-----

sc:SubCommodity a owl:Class ;
  rdfs:label "Sub-Commodity" ;
  rdfs:comment "Sub-commodity classification" ;
  skos:altLabel "subcategory" ;
  skos:altLabel "sub commodity" ;
  skos:altLabel "DIM_SUB_COMMODITY" ;
  owl:hasKey ( sc:subcommodity_subCommodityId ) ;
  rdfs:subClassOf
    [ a owl:Restriction ; owl:onProperty sc:subcommodity_subCommodityId ; owl:cardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:subcommodity_subCommodityName ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:subcommodity_commodityId ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] .

sc:subcommodity_subCommodityId a owl:DatatypeProperty ;
  rdfs:label "SUB_COMMODITY_ID" ;
  rdfs:comment "DIM_SUB_COMMODITY.SUB_COMMODITY_ID" ;
  rdfs:domain sc:SubCommodity ;
  rdfs:range xsd:integer .

sc:subcommodity_subCommodityName a owl:DatatypeProperty ;
  rdfs:label "SUB_COMMODITY_NAME" ;
  rdfs:comment "DIM_SUB_COMMODITY.SUB_COMMODITY_NAME" ;
  rdfs:domain sc:SubCommodity ;
  rdfs:range xsd:string .

sc:subcommodity_commodityId a owl:ObjectProperty ;
  rdfs:label "parent commodity" ;
  rdfs:comment "Top-level commodity this sub-commodity rolls up to" ;
  rdfs:domain sc:SubCommodity ;
  rdfs:range sc:Commodity ;
  coa:fkProvenance "MANY_TO_ONE" .

#-----
# Currency <- SCS.INVENTORY.DIM_CURRENCY
#-----

sc:Currency a owl:Class ;
  rdfs:label "Currency" ;
  rdfs:comment "Currency reference with approximate USD exchange rates" ;
  skos:altLabel "fx" ;
  skos:altLabel "currency code" ;
  skos:altLabel "DIM_CURRENCY" ;
  owl:hasKey ( sc:currency_currencyCode ) ;
  rdfs:subClassOf
    [ a owl:Restriction ; owl:onProperty sc:currency_currencyCode ; owl:cardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:currency_currencyName ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:currency_toUsdRate ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] .

```

```

sc:currency_currencyCode a owl:DatatypeProperty ;
  rdfs:label "CURRENCY_CODE" ;
  rdfs:comment "DIM_CURRENCY.CURRENCY_CODE" ;
  rdfs:domain sc:Currency ;
  coa:distinctValues "BRL" ;
  coa:distinctValues "CNY" ;
  coa:distinctValues "EUR" ;
  coa:distinctValues "GBP" ;
  coa:distinctValues "INR" ;
  coa:distinctValues "JPY" ;
  coa:distinctValues "KRW" ;
  coa:distinctValues "MXN" ;
  coa:distinctValues "MYR" ;
  coa:distinctValues "THB" ;
  coa:distinctValues "TWD" ;
  coa:distinctValues "USD" ;
  rdfs:range xsd:string .

sc:currency_currencyName a owl:DatatypeProperty ;
  rdfs:label "CURRENCY_NAME" ;
  rdfs:comment "DIM_CURRENCY.CURRENCY_NAME" ;
  rdfs:domain sc:Currency ;
  rdfs:range xsd:string .

sc:currency_toUsdRate a owl:DatatypeProperty ;
  rdfs:label "TO_USD_RATE" ;
  rdfs:comment "Approximate fixed rate for demo" ;
  rdfs:domain sc:Currency ;
  rdfs:range xsd:decimal .

#-----
# UnitOfMeasure <- SCS.INVENTORY.DIM_UOM
#-----

sc:UnitOfMeasure a owl:Class ;
  rdfs:label "Unit of Measure" ;
  rdfs:comment "Units of measure" ;
  skos:altLabel "uom" ;
  skos:altLabel "unit" ;
  skos:altLabel "DIM_UOM" ;
  owl:hasKey ( sc:unitofmeasure_uomCode ) ;
  rdfs:subClassOf
    [ a owl:Restriction ; owl:onProperty sc:unitofmeasure_uomCode ; owl:cardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:unitofmeasure_uomName ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] .

sc:unitofmeasure_uomCode a owl:DatatypeProperty ;
  rdfs:label "UOM_CODE" ;
  rdfs:comment "DIM_UOM.UOM_CODE" ;
  rdfs:domain sc:UnitOfMeasure ;
  coa:distinctValues "BOX" ;
  coa:distinctValues "EA" ;
  coa:distinctValues "FT" ;
  coa:distinctValues "KG" ;
  coa:distinctValues "L" ;
  coa:distinctValues "LB" ;
  coa:distinctValues "M" ;
  coa:distinctValues "PAL" ;
  coa:distinctValues "PC" ;
  coa:distinctValues "ROL" ;
  coa:distinctValues "SET" ;
  coa:distinctValues "SHT" ;
  rdfs:range xsd:string .

sc:unitofmeasure_uomName a owl:DatatypeProperty ;
  rdfs:label "UOM_NAME" ;
  rdfs:comment "DIM_UOM.UOM_NAME" ;
  rdfs:domain sc:UnitOfMeasure ;
  rdfs:range xsd:string .

#-----
# Incoterm <- SCS.INVENTORY.DIM_INCOTERM
#-----

sc:Incoterm a owl:Class ;
  rdfs:label "Incoterm" ;
  rdfs:comment "Incoterms 2020" ;
  skos:altLabel "shipping terms" ;
  skos:altLabel "trade terms" ;

```

```

skos:altLabel "DIM_INCOTERM" ;
skos:altLabel "incoterms 2020" ;
owl:hasKey ( sc:incoterm_incotermCode ) ;
rdfs:subClassOf
  [ a owl:Restriction ; owl:onProperty sc:incoterm_incotermCode ; owl:cardinality
    "1"^^xsd:nonNegativeInteger ] ,
  [ a owl:Restriction ; owl:onProperty sc:incoterm_incotermName ; owl:minCardinality
    "1"^^xsd:nonNegativeInteger ] .

sc:incoterm_incotermCode a owl:DatatypeProperty ;
rdfs:label "INCOTERM_CODE" ;
rdfs:comment "DIM_INCOTERM.INCOTERM_CODE" ;
rdfs:domain sc:Incoterm ;
coa:distinctValues "CFR" ;
coa:distinctValues "CIF" ;
coa:distinctValues "CIP" ;
coa:distinctValues "CPT" ;
coa:distinctValues "DAP" ;
coa:distinctValues "DDP" ;
coa:distinctValues "DPU" ;
coa:distinctValues "EXW" ;
coa:distinctValues "FAS" ;
coa:distinctValues "FCA" ;
coa:distinctValues "FOB" ;
rdfs:range xsd:string .

sc:incoterm_incotermName a owl:DatatypeProperty ;
rdfs:label "INCOTERM_NAME" ;
rdfs:comment "DIM_INCOTERM.INCOTERM_NAME" ;
rdfs:domain sc:Incoterm ;
rdfs:range xsd:string .

#-----
# Port <- SCS.INVENTORY.DIM_PORT
#-----

sc:Port a owl:Class ;
rdfs:label "Port" ;
rdfs:comment "Major ports and airports for logistics" ;
skos:altLabel "seaport" ;
skos:altLabel "airport" ;
skos:altLabel "terminal" ;
skos:altLabel "DIM_PORT" ;
owl:hasKey ( sc:port_portId ) ;
rdfs:subClassOf
  [ a owl:Restriction ; owl:onProperty sc:port_portId ; owl:cardinality
    "1"^^xsd:nonNegativeInteger ] ,
  [ a owl:Restriction ; owl:onProperty sc:port_portName ; owl:minCardinality
    "1"^^xsd:nonNegativeInteger ] ,
  [ a owl:Restriction ; owl:onProperty sc:port_portCode ; owl:minCardinality
    "1"^^xsd:nonNegativeInteger ] ,
  [ a owl:Restriction ; owl:onProperty sc:port_countryId ; owl:minCardinality
    "1"^^xsd:nonNegativeInteger ] ,
  [ a owl:Restriction ; owl:onProperty sc:port_portType ; owl:minCardinality
    "1"^^xsd:nonNegativeInteger ] .

sc:port_portId a owl:DatatypeProperty ;
rdfs:label "PORT_ID" ;
rdfs:comment "DIM_PORT.PORT_ID" ;
rdfs:domain sc:Port ;
rdfs:range xsd:integer .

sc:port_portName a owl:DatatypeProperty ;
rdfs:label "PORT_NAME" ;
rdfs:comment "DIM_PORT.PORT_NAME" ;
rdfs:domain sc:Port ;
rdfs:range xsd:string .

sc:port_portCode a owl:DatatypeProperty ;
rdfs:label "PORT_CODE" ;
rdfs:comment "DIM_PORT.PORT_CODE" ;
rdfs:domain sc:Port ;
rdfs:range xsd:string .

sc:port_countryId a owl:ObjectProperty ;
rdfs:label "countryId" ;
rdfs:comment "Foreign key: DIM_PORT.COUNTRY_ID references DIM_GEO_COUNTRY.COUNTRY_ID" ;
rdfs:domain sc:Port ;
rdfs:range sc:Country ;
coa:fkProvenance "MANY_TO_ONE" .

```

```

sc:port_portType a owl:DatatypeProperty ;
  rdfs:label "PORT_TYPE" ;
  rdfs:comment "SEA, AIR, or BOTH" ;
  rdfs:domain sc:Port ;
  coa:distinctValues "AIR" ;
  coa:distinctValues "BOTH" ;
  coa:distinctValues "SEA" ;
  rdfs:range xsd:string .

#-----
# Supplier <- SCS.INVENTORY.DIM_SUPPLIER
#-----

sc:Supplier a owl:Class ;
  rdfs:label "Supplier" ;
  rdfs:comment "Supplier master – 500 suppliers across commodities and countries" ;
  skos:altLabel "vendor" ;
  skos:altLabel "supplier master" ;
  skos:altLabel "DIM_SUPPLIER" ;
  skos:altLabel "source" ;
  skos:altLabel "seller" ;
  owl:hasKey ( sc:supplier_supplierId ) ;
  rdfs:subClassOf
    [ a owl:Restriction ; owl:onProperty sc:supplier_supplierId ; owl:cardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:supplier_supplierName ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:supplier_supplierCode ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:supplier_countryId ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:supplier_commodityId ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:supplier_reliabilityRating ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] .

sc:supplier_supplierId a owl:DatatypeProperty ;
  rdfs:label "SUPPLIER_ID" ;
  rdfs:comment "DIM_SUPPLIER.SUPPLIER_ID" ;
  rdfs:domain sc:Supplier ;
  rdfs:range xsd:integer .

sc:supplier_supplierName a owl:DatatypeProperty ;
  rdfs:label "SUPPLIER_NAME" ;
  rdfs:comment "DIM_SUPPLIER.SUPPLIER_NAME" ;
  rdfs:domain sc:Supplier ;
  rdfs:range xsd:string .

sc:supplier_supplierCode a owl:DatatypeProperty ;
  rdfs:label "SUPPLIER_CODE" ;
  rdfs:comment "DIM_SUPPLIER.SUPPLIER_CODE" ;
  rdfs:domain sc:Supplier ;
  rdfs:range xsd:string .

sc:supplier_countryId a owl:ObjectProperty ;
  rdfs:label "located in country" ;
  rdfs:comment "Country the supplier is headquartered in" ;
  rdfs:domain sc:Supplier ;
  rdfs:range sc:Country ;
  coa:fkProvenance "MANY_TO_ONE" .

sc:supplier_commodityId a owl:ObjectProperty ;
  rdfs:label "supplies commodity" ;
  rdfs:comment "Primary commodity this supplier is sourced for" ;
  rdfs:domain sc:Supplier ;
  rdfs:range sc:Commodity ;
  coa:fkProvenance "MANY_TO_ONE" .

sc:supplier_reliabilityRating a owl:DatatypeProperty ;
  rdfs:label "RELIABILITY_RATING" ;
  rdfs:comment "0.00-1.00 scale" ;
  rdfs:domain sc:Supplier ;
  skos:altLabel "reliability" ;
  skos:altLabel "supplier score" ;
  skos:altLabel "otd rating" ;
  rdfs:range xsd:decimal .

sc:supplier_isProblem a owl:DatatypeProperty ;
  rdfs:label "IS_PROBLEM" ;
  rdfs:comment "Flagged problem supplier for demo scenarios" ;

```

```

    rdfs:domain sc:Supplier ;
    skos:altLabel "problem supplier" ;
    skos:altLabel "at risk supplier" ;
    skos:altLabel "flagged supplier" ;
    rdfs:range xsd:boolean .

sc:supplier_active a owl:DatatypeProperty ;
    rdfs:label "ACTIVE" ;
    rdfs:comment "DIM_SUPPLIER.ACTIVE" ;
    rdfs:domain sc:Supplier ;
    rdfs:range xsd:boolean .

sc:supplier_isSynthetic a owl:DatatypeProperty ;
    rdfs:label "IS_SYNTHETIC" ;
    rdfs:comment "DIM_SUPPLIER.IS_SYNTHETIC" ;
    rdfs:domain sc:Supplier ;
    rdfs:range xsd:boolean .

#-----
# SupplierSite  <-  SCS.INVENTORY.DIM_SUPPLIER_SITE
#-----

sc:SupplierSite a owl:Class ;
    rdfs:label "Supplier Site" ;
    rdfs:comment "Supplier manufacturing and shipping sites" ;
    skos:altLabel "supplier facility" ;
    skos:altLabel "vendor site" ;
    skos:altLabel "DIM_SUPPLIER_SITE" ;
    skos:altLabel "manufacturing site" ;
    owl:hasKey ( sc:suppliersite_siteId ) ;
    rdfs:subClassOf
        [ a owl:Restriction ; owl:onProperty sc:suppliersite_siteId ; owl:cardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:suppliersite_supplierId ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:suppliersite_siteName ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:suppliersite_city ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:suppliersite_countryId ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:suppliersite_leadTimeDays ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] .

sc:suppliersite_siteId a owl:DatatypeProperty ;
    rdfs:label "SITE_ID" ;
    rdfs:comment "DIM_SUPPLIER_SITE.SITE_ID" ;
    rdfs:domain sc:SupplierSite ;
    rdfs:range xsd:integer .

sc:suppliersite_supplierId a owl:ObjectProperty ;
    rdfs:label "belongs to supplier" ;
    rdfs:comment "Parent supplier that owns this site" ;
    rdfs:domain sc:SupplierSite ;
    rdfs:range sc:Supplier ;
    coa:fkProvenance "MANY_TO_ONE" .

sc:suppliersite_siteName a owl:DatatypeProperty ;
    rdfs:label "SITE_NAME" ;
    rdfs:comment "DIM_SUPPLIER_SITE.SITE_NAME" ;
    rdfs:domain sc:SupplierSite ;
    rdfs:range xsd:string .

sc:suppliersite_city a owl:DatatypeProperty ;
    rdfs:label "CITY" ;
    rdfs:comment "DIM_SUPPLIER_SITE.CITY" ;
    rdfs:domain sc:SupplierSite ;
    rdfs:range xsd:string .

sc:suppliersite_countryId a owl:ObjectProperty ;
    rdfs:label "countryId" ;
    rdfs:comment "Foreign key: DIM_SUPPLIER_SITE.COUNTRY_ID references DIM_GEO_COUNTRY.COUNTRY_ID" ;
    rdfs:domain sc:SupplierSite ;
    rdfs:range sc:Country ;
    coa:fkProvenance "MANY_TO_ONE" .

sc:suppliersite_leadTimeDays a owl:DatatypeProperty ;
    rdfs:label "LEAD_TIME_DAYS" ;
    rdfs:comment "Average lead time in calendar days" ;
    rdfs:domain sc:SupplierSite ;

```

```

    skos:altLabel "lead time" ;
    skos:altLabel "replenishment lead time" ;
    rdfs:range xsd:integer .

sc:suppliersite_isPrimary a owl:DatatypeProperty ;
    rdfs:label "IS_PRIMARY" ;
    rdfs:comment "DIM_SUPPLIER_SITE.IS_PRIMARY" ;
    rdfs:domain sc:SupplierSite ;
    rdfs:range xsd:boolean .

sc:suppliersite_active a owl:DatatypeProperty ;
    rdfs:label "ACTIVE" ;
    rdfs:comment "DIM_SUPPLIER_SITE.ACTIVE" ;
    rdfs:domain sc:SupplierSite ;
    rdfs:range xsd:boolean .

#-----
# Manufacturer <- SCS.INVENTORY.DIM_MANUFACTURER
#-----

sc:Manufacturer a owl:Class ;
    rdfs:label "Manufacturer" ;
    rdfs:comment "Component manufacturers" ;
    skos:altLabel "oem" ;
    skos:altLabel "component manufacturer" ;
    skos:altLabel "DIM_MANUFACTURER" ;
    skos:altLabel "mfr" ;
    owl:hasKey ( sc:manufacturer_manufacturerId ) ;
    rdfs:subClassOf
        [ a owl:Restriction ; owl:onProperty sc:manufacturer_manufacturerId ; owl:cardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:manufacturer_manufacturerName ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:manufacturer_countryId ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] .

sc:manufacturer_manufacturerId a owl:DatatypeProperty ;
    rdfs:label "MANUFACTURER_ID" ;
    rdfs:comment "DIM_MANUFACTURER.MANUFACTURER_ID" ;
    rdfs:domain sc:Manufacturer ;
    rdfs:range xsd:integer .

sc:manufacturer_manufacturerName a owl:DatatypeProperty ;
    rdfs:label "MANUFACTURER_NAME" ;
    rdfs:comment "DIM_MANUFACTURER.MANUFACTURER_NAME" ;
    rdfs:domain sc:Manufacturer ;
    rdfs:range xsd:string .

sc:manufacturer_countryId a owl:ObjectProperty ;
    rdfs:label "countryId" ;
    rdfs:comment "Foreign key: DIM_MANUFACTURER.COUNTRY_ID references DIM_GEO_COUNTRY.COUNTRY_ID" ;
    rdfs:domain sc:Manufacturer ;
    rdfs:range sc:Country ;
    coa:fkProvenance "MANY_TO_ONE" .

#-----
# Part <- SCS.INVENTORY.DIM_PART
#-----

sc:Part a owl:Class ;
    rdfs:label "Part" ;
    rdfs:comment "Part master - 30,000 parts across all commodities" ;
    skos:altLabel "component" ;
    skos:altLabel "material" ;
    skos:altLabel "sku" ;
    skos:altLabel "item" ;
    skos:altLabel "DIM_PART" ;
    skos:altLabel "part master" ;
    owl:hasKey ( sc:part_partId ) ;
    rdfs:subClassOf
        [ a owl:Restriction ; owl:onProperty sc:part_partId ; owl:cardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:part_partNumber ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:part_partName ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:part_commodityId ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:part_subCommodityId ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,

```

```

    [ a owl:Restriction ; owl:onProperty sc:part_unitCostUsd ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] .

sc:part_partId a owl:DatatypeProperty ;
  rdfs:label "PART_ID" ;
  rdfs:comment "DIM_PART.PART_ID" ;
  rdfs:domain sc:Part ;
  rdfs:range xsd:integer .

sc:part_partNumber a owl:DatatypeProperty ;
  rdfs:label "PART_NUMBER" ;
  rdfs:comment "DIM_PART.PART_NUMBER" ;
  rdfs:domain sc:Part ;
  rdfs:range xsd:string .

sc:part_partName a owl:DatatypeProperty ;
  rdfs:label "PART_NAME" ;
  rdfs:comment "DIM_PART.PART_NAME" ;
  rdfs:domain sc:Part ;
  rdfs:range xsd:string .

sc:part_commodityId a owl:ObjectProperty ;
  rdfs:label "classified as commodity" ;
  rdfs:comment "Top-level commodity classification of the part" ;
  rdfs:domain sc:Part ;
  rdfs:range sc:Commodity ;
  coa:fkProvenance "MANY_TO_ONE" .

sc:part_subCommodityId a owl:ObjectProperty ;
  rdfs:label "classified as sub-commodity" ;
  rdfs:comment "Sub-commodity classification of the part" ;
  rdfs:domain sc:Part ;
  rdfs:range sc:SubCommodity ;
  coa:fkProvenance "MANY_TO_ONE" .

sc:part_uomCode a owl:ObjectProperty ;
  rdfs:label "measured in" ;
  rdfs:comment "Unit of measure the part is transacted in" ;
  rdfs:domain sc:Part ;
  rdfs:range sc:UnitOfMeasure ;
  coa:fkProvenance "MANY_TO_ONE" .

sc:part_unitCostUsd a owl:DatatypeProperty ;
  rdfs:label "UNIT_COST_USD" ;
  rdfs:comment "Standard cost in USD" ;
  rdfs:domain sc:Part ;
  skos:altLabel "standard cost" ;
  skos:altLabel "part cost" ;
  skos:altLabel "unit cost" ;
  rdfs:range xsd:decimal .

sc:part_weightKg a owl:DatatypeProperty ;
  rdfs:label "WEIGHT_KG" ;
  rdfs:comment "DIM_PART.WEIGHT_KG" ;
  rdfs:domain sc:Part ;
  rdfs:range xsd:decimal .

sc:part_lifecycleStatus a owl:DatatypeProperty ;
  rdfs:label "LIFECYCLE_STATUS" ;
  rdfs:comment "ACTIVE, EOL, OBSOLETE, NRND" ;
  rdfs:domain sc:Part ;
  skos:altLabel "lifecycle" ;
  skos:altLabel "eol status" ;
  skos:altLabel "part status" ;
  coa:distinctValues "ACTIVE" ;
  coa:distinctValues "EOL" ;
  coa:distinctValues "NRND" ;
  coa:distinctValues "OBSOLETE" ;
  rdfs:range xsd:string .

sc:part_isSynthetic a owl:DatatypeProperty ;
  rdfs:label "IS_SYNTHETIC" ;
  rdfs:comment "DIM_PART.IS_SYNTHETIC" ;
  rdfs:domain sc:Part ;
  rdfs:range xsd:boolean .

#-----
# ProductFamily <- SCS.INVENTORY.DIM_PRODUCT_FAMILY
#-----
sc:ProductFamily a owl:Class ;

```

```

rdfs:label "Product Family" ;
rdfs:comment "Finished product families" ;
skos:altLabel "product line" ;
skos:altLabel "family" ;
skos:altLabel "DIM_PRODUCT_FAMILY" ;
owl:hasKey ( sc:productfamily_productFamilyId ) ;
rdfs:subClassOf
[ a owl:Restriction ; owl:onProperty sc:productfamily_productFamilyId ; owl:cardinality
  "1"^^xsd:nonNegativeInteger ] ,
[ a owl:Restriction ; owl:onProperty sc:productfamily_productFamilyName ; owl:minCardinality
  "1"^^xsd:nonNegativeInteger ] .

sc:productfamily_productFamilyId a owl:DatatypeProperty ;
rdfs:label "PRODUCT_FAMILY_ID" ;
rdfs:comment "DIM_PRODUCT_FAMILY.PRODUCT_FAMILY_ID" ;
rdfs:domain sc:ProductFamily ;
rdfs:range xsd:integer .

sc:productfamily_productFamilyName a owl:DatatypeProperty ;
rdfs:label "PRODUCT_FAMILY_NAME" ;
rdfs:comment "DIM_PRODUCT_FAMILY.PRODUCT_FAMILY_NAME" ;
rdfs:domain sc:ProductFamily ;
rdfs:range xsd:string .

#-----
# Product <- SCS.INVENTORY.DIM_PRODUCT
#-----

sc:Product a owl:Class ;
rdfs:label "Product" ;
rdfs:comment "Finished products – 500 SKUs" ;
skos:altLabel "finished good" ;
skos:altLabel "sku" ;
skos:altLabel "DIM_PRODUCT" ;
skos:altLabel "end item" ;
owl:hasKey ( sc:product_productId ) ;
rdfs:subClassOf
[ a owl:Restriction ; owl:onProperty sc:product_productId ; owl:cardinality
  "1"^^xsd:nonNegativeInteger ] ,
[ a owl:Restriction ; owl:onProperty sc:product_productName ; owl:minCardinality
  "1"^^xsd:nonNegativeInteger ] ,
[ a owl:Restriction ; owl:onProperty sc:product_productFamilyId ; owl:minCardinality
  "1"^^xsd:nonNegativeInteger ] ,
[ a owl:Restriction ; owl:onProperty sc:product_unitPriceUsd ; owl:minCardinality
  "1"^^xsd:nonNegativeInteger ] .

sc:product_productId a owl:DatatypeProperty ;
rdfs:label "PRODUCT_ID" ;
rdfs:comment "DIM_PRODUCT.PRODUCT_ID" ;
rdfs:domain sc:Product ;
rdfs:range xsd:integer .

sc:product_productName a owl:DatatypeProperty ;
rdfs:label "PRODUCT_NAME" ;
rdfs:comment "DIM_PRODUCT.PRODUCT_NAME" ;
rdfs:domain sc:Product ;
rdfs:range xsd:string .

sc:product_productFamilyId a owl:ObjectProperty ;
rdfs:label "in product family" ;
rdfs:comment "Product family the SKU belongs to" ;
rdfs:domain sc:Product ;
rdfs:range sc:ProductFamily ;
coa:fkProvenance "MANY_TO_ONE" .

sc:product_unitPriceUsd a owl:DatatypeProperty ;
rdfs:label "UNIT_PRICE_USD" ;
rdfs:comment "DIM_PRODUCT.UNIT_PRICE_USD" ;
rdfs:domain sc:Product ;
skos:altLabel "list price" ;
skos:altLabel "selling price" ;
rdfs:range xsd:decimal .

sc:product_isSynthetic a owl:DatatypeProperty ;
rdfs:label "IS_SYNTHETIC" ;
rdfs:comment "DIM_PRODUCT.IS_SYNTHETIC" ;
rdfs:domain sc:Product ;
rdfs:range xsd:boolean .

#-----

```

```
# BillofMaterials <- SCS.INVENTORY.BOM_HEADER
```

```
#-----
```

```
sc:BillofMaterials a owl:Class ;
  rdfs:label "Bill of Materials" ;
  rdfs:comment "Bill of Materials headers – one per product revision" ;
  skos:altLabel "bom" ;
  skos:altLabel "BOM_HEADER" ;
  skos:altLabel "product structure" ;
  skos:altLabel "recipe" ;
  owl:hasKey ( sc:billofmaterials_bomId ) ;
  rdfs:subClassOf
    [ a owl:Restriction ; owl:onProperty sc:billofmaterials_bomId ; owl:cardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:billofmaterials_productId ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:billofmaterials_bomVersion ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:billofmaterials_effectiveDate ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] .
```

```
sc:billofmaterials_bomId a owl:DatatypeProperty ;
  rdfs:label "BOM_ID" ;
  rdfs:comment "BOM_HEADER.BOM_ID" ;
  rdfs:domain sc:BillofMaterials ;
  rdfs:range xsd:integer .
```

```
sc:billofmaterials_productId a owl:ObjectProperty ;
  rdfs:label "builds product" ;
  rdfs:comment "Finished product this bill of materials builds" ;
  rdfs:domain sc:BillofMaterials ;
  rdfs:range sc:Product ;
  coa:fkProvenance "MANY_TO_ONE" .
```

```
sc:billofmaterials_bomVersion a owl:DatatypeProperty ;
  rdfs:label "BOM_VERSION" ;
  rdfs:comment "BOM_HEADER.BOM_VERSION" ;
  rdfs:domain sc:BillofMaterials ;
  coa:distinctValues "A" ;
  coa:distinctValues "B" ;
  coa:distinctValues "C" ;
  rdfs:range xsd:string .
```

```
sc:billofmaterials_effectiveDate a owl:DatatypeProperty ;
  rdfs:label "EFFECTIVE_DATE" ;
  rdfs:comment "BOM_HEADER.EFFECTIVE_DATE" ;
  rdfs:domain sc:BillofMaterials ;
  rdfs:range xsd:date .
```

```
sc:billofmaterials_status a owl:DatatypeProperty ;
  rdfs:label "STATUS" ;
  rdfs:comment "BOM_HEADER.STATUS" ;
  rdfs:domain sc:BillofMaterials ;
  coa:distinctValues "ACTIVE" ;
  rdfs:range xsd:string .
```

```
#-----
```

```
# BillofMaterialsLine <- SCS.INVENTORY.BOM_LINE
```

```
#-----
```

```
sc:BillofMaterialsLine a owl:Class ;
  rdfs:label "Bill of Materials Line" ;
  rdfs:comment "BOM line items – components per assembly" ;
  skos:altLabel "bom line" ;
  skos:altLabel "bom component" ;
  skos:altLabel "BOM_LINE" ;
  owl:hasKey ( sc:billofmaterialsline_bomLineId ) ;
  rdfs:subClassOf
    [ a owl:Restriction ; owl:onProperty sc:billofmaterialsline_bomLineId ; owl:cardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:billofmaterialsline_bomId ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:billofmaterialsline_partId ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:billofmaterialsline_quantityPer ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:billofmaterialsline_findNumber ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] .
```

```
sc:billofmaterialsline_bomLineId a owl:DatatypeProperty ;
```

```

    rdfs:label "BOM_LINE_ID" ;
    rdfs:comment "BOM_LINE.BOM_LINE_ID" ;
    rdfs:domain sc:BillofMaterialsLine ;
    rdfs:range xsd:integer .

sc:billofmaterialsline_bomId a owl:ObjectProperty ;
    rdfs:label "line of BOM" ;
    rdfs:comment "Bill of materials this line belongs to" ;
    rdfs:domain sc:BillofMaterialsLine ;
    rdfs:range sc:BillofMaterials ;
    coa:fkProvenance "MANY_TO_ONE" .

sc:billofmaterialsline_partId a owl:ObjectProperty ;
    rdfs:label "consumes part" ;
    rdfs:comment "Component part consumed by this BOM position" ;
    rdfs:domain sc:BillofMaterialsLine ;
    rdfs:range sc:Part ;
    coa:fkProvenance "MANY_TO_ONE" .

sc:billofmaterialsline_quantityPer a owl:DatatypeProperty ;
    rdfs:label "QUANTITY_PER" ;
    rdfs:comment "BOM_LINE.QUANTITY_PER" ;
    rdfs:domain sc:BillofMaterialsLine ;
    skos:altLabel "qty per" ;
    skos:altLabel "usage per assembly" ;
    skos:altLabel "quantity per assembly" ;
    rdfs:range xsd:decimal .

sc:billofmaterialsline_findNumber a owl:DatatypeProperty ;
    rdfs:label "FIND_NUMBER" ;
    rdfs:comment "Position on the BOM" ;
    rdfs:domain sc:BillofMaterialsLine ;
    rdfs:range xsd:integer .

#-----
# Plant <- SCS.INVENTORY.DIM_PLANT
#-----

sc:Plant a owl:Class ;
    rdfs:label "Plant" ;
    rdfs:comment "Manufacturing plants for Asterion Devices" ;
    skos:altLabel "factory" ;
    skos:altLabel "manufacturing site" ;
    skos:altLabel "DIM_PLANT" ;
    skos:altLabel "works" ;
    owl:hasKey ( sc:plant_plantId ) ;
    rdfs:subClassOf
        [ a owl:Restriction ; owl:onProperty sc:plant_plantId ; owl:cardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:plant_plantCode ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:plant_plantName ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:plant_city ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:plant_countryId ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] .

sc:plant_plantId a owl:DatatypeProperty ;
    rdfs:label "PLANT_ID" ;
    rdfs:comment "DIM_PLANT.PLANT_ID" ;
    rdfs:domain sc:Plant ;
    rdfs:range xsd:integer .

sc:plant_plantCode a owl:DatatypeProperty ;
    rdfs:label "PLANT_CODE" ;
    rdfs:comment "DIM_PLANT.PLANT_CODE" ;
    rdfs:domain sc:Plant ;
    rdfs:range xsd:string .

sc:plant_plantName a owl:DatatypeProperty ;
    rdfs:label "PLANT_NAME" ;
    rdfs:comment "DIM_PLANT.PLANT_NAME" ;
    rdfs:domain sc:Plant ;
    rdfs:range xsd:string .

sc:plant_city a owl:DatatypeProperty ;
    rdfs:label "CITY" ;
    rdfs:comment "DIM_PLANT.CITY" ;
    rdfs:domain sc:Plant ;

```

```

rdfs:range xsd:string .

sc:plant_countryId a owl:ObjectProperty ;
rdfs:label "countryId" ;
rdfs:comment "Foreign key: DIM_PLANT.COUNTRY_ID references DIM_GEO_COUNTRY.COUNTRY_ID" ;
rdfs:domain sc:Plant ;
rdfs:range sc:Country ;
coa:fkProvenance "MANY_TO_ONE" .

#-----
# Warehouse <- SCS.INVENTORY.DIM_WAREHOUSE
#-----

sc:Warehouse a owl:Class ;
rdfs:label "Warehouse" ;
rdfs:comment "Warehouses – plant-attached and distribution centers" ;
skos:altLabel "dc" ;
skos:altLabel "distribution center" ;
skos:altLabel "stocking location" ;
skos:altLabel "DIM_WAREHOUSE" ;
owl:hasKey ( sc:warehouse_warehouseId ) ;
rdfs:subClassOf
[ a owl:Restriction ; owl:onProperty sc:warehouse_warehouseId ; owl:cardinality
  "1"^^xsd:nonNegativeInteger ] ,
[ a owl:Restriction ; owl:onProperty sc:warehouse_warehouseCode ; owl:minCardinality
  "1"^^xsd:nonNegativeInteger ] ,
[ a owl:Restriction ; owl:onProperty sc:warehouse_warehouseName ; owl:minCardinality
  "1"^^xsd:nonNegativeInteger ] ,
[ a owl:Restriction ; owl:onProperty sc:warehouse_city ; owl:minCardinality
  "1"^^xsd:nonNegativeInteger ] ,
[ a owl:Restriction ; owl:onProperty sc:warehouse_countryId ; owl:minCardinality
  "1"^^xsd:nonNegativeInteger ] ,
[ a owl:Restriction ; owl:onProperty sc:warehouse_capacityUnits ; owl:minCardinality
  "1"^^xsd:nonNegativeInteger ] .

sc:warehouse_warehouseId a owl:DatatypeProperty ;
rdfs:label "WAREHOUSE_ID" ;
rdfs:comment "DIM_WAREHOUSE.WAREHOUSE_ID" ;
rdfs:domain sc:Warehouse ;
rdfs:range xsd:integer .

sc:warehouse_warehouseCode a owl:DatatypeProperty ;
rdfs:label "WAREHOUSE_CODE" ;
rdfs:comment "DIM_WAREHOUSE.WAREHOUSE_CODE" ;
rdfs:domain sc:Warehouse ;
rdfs:range xsd:string .

sc:warehouse_warehouseName a owl:DatatypeProperty ;
rdfs:label "WAREHOUSE_NAME" ;
rdfs:comment "DIM_WAREHOUSE.WAREHOUSE_NAME" ;
rdfs:domain sc:Warehouse ;
rdfs:range xsd:string .

sc:warehouse_plantId a owl:ObjectProperty ;
rdfs:label "attached to plant" ;
rdfs:comment "Plant this warehouse is attached to; null for standalone distribution centres" ;
rdfs:domain sc:Warehouse ;
rdfs:range sc:Plant ;
coa:fkProvenance "MANY_TO_ONE" .

sc:warehouse_city a owl:DatatypeProperty ;
rdfs:label "CITY" ;
rdfs:comment "DIM_WAREHOUSE.CITY" ;
rdfs:domain sc:Warehouse ;
rdfs:range xsd:string .

sc:warehouse_countryId a owl:ObjectProperty ;
rdfs:label "countryId" ;
rdfs:comment "Foreign key: DIM_WAREHOUSE.COUNTRY_ID references DIM_GEO_COUNTRY.COUNTRY_ID" ;
rdfs:domain sc:Warehouse ;
rdfs:range sc:Country ;
coa:fkProvenance "MANY_TO_ONE" .

sc:warehouse_capacityUnits a owl:DatatypeProperty ;
rdfs:label "CAPACITY_UNITS" ;
rdfs:comment "DIM_WAREHOUSE.CAPACITY_UNITS" ;
rdfs:domain sc:Warehouse ;
rdfs:range xsd:integer .

#-----

```

```
# Customer <- SCS.INVENTORY.DIM_CUSTOMER
```

```
#-----
```

```
sc:Customer a owl:Class ;
  rdfs:label "Customer" ;
  rdfs:comment "Customer master - 80 customers" ;
  skos:altLabel "account" ;
  skos:altLabel "buyer" ;
  skos:altLabel "DIM_CUSTOMER" ;
  skos:altLabel "client" ;
  owl:hasKey ( sc:customer_customerId ) ;
  rdfs:subClassOf
    [ a owl:Restriction ; owl:onProperty sc:customer_customerId ; owl:cardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:customer_customerName ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:customer_customerCode ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:customer_countryId ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:customer_customerTier ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] .

sc:customer_customerId a owl:DatatypeProperty ;
  rdfs:label "CUSTOMER_ID" ;
  rdfs:comment "DIM_CUSTOMER.CUSTOMER_ID" ;
  rdfs:domain sc:Customer ;
  rdfs:range xsd:integer .

sc:customer_customerName a owl:DatatypeProperty ;
  rdfs:label "CUSTOMER_NAME" ;
  rdfs:comment "DIM_CUSTOMER.CUSTOMER_NAME" ;
  rdfs:domain sc:Customer ;
  rdfs:range xsd:string .

sc:customer_customerCode a owl:DatatypeProperty ;
  rdfs:label "CUSTOMER_CODE" ;
  rdfs:comment "DIM_CUSTOMER.CUSTOMER_CODE" ;
  rdfs:domain sc:Customer ;
  rdfs:range xsd:string .

sc:customer_countryId a owl:ObjectProperty ;
  rdfs:label "countryId" ;
  rdfs:comment "Foreign key: DIM_CUSTOMER.COUNTRY_ID references DIM_GEO_COUNTRY.COUNTRY_ID" ;
  rdfs:domain sc:Customer ;
  rdfs:range sc:Country ;
  coa:fkProvenance "MANY_TO_ONE" .

sc:customer_customerTier a owl:DatatypeProperty ;
  rdfs:label "CUSTOMER_TIER" ;
  rdfs:comment "PLATINUM, GOLD, SILVER, BRONZE" ;
  rdfs:domain sc:Customer ;
  skos:altLabel "tier" ;
  skos:altLabel "segment" ;
  skos:altLabel "customer segment" ;
  coa:distinctValues "BRONZE" ;
  coa:distinctValues "GOLD" ;
  coa:distinctValues "PLATINUM" ;
  coa:distinctValues "SILVER" ;
  rdfs:range xsd:string .

sc:customer_isSynthetic a owl:DatatypeProperty ;
  rdfs:label "IS_SYNTHETIC" ;
  rdfs:comment "DIM_CUSTOMER.IS_SYNTHETIC" ;
  rdfs:domain sc:Customer ;
  rdfs:range xsd:boolean .
```

```
#-----
```

```
# Carrier <- SCS.INVENTORY.DIM_CARRIER
```

```
#-----
```

```
sc:Carrier a owl:Class ;
  rdfs:label "Carrier" ;
  rdfs:comment "Logistics carriers" ;
  skos:altLabel "freight forwarder" ;
  skos:altLabel "shipping line" ;
  skos:altLabel "DIM_CARRIER" ;
  skos:altLabel "logistics provider" ;
  owl:hasKey ( sc:carrier_carrierId ) ;
  rdfs:subClassOf
```

```

    [ a owl:Restriction ; owl:onProperty sc:carrier_carrierId ; owl:cardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:carrier_carrierName ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:carrier_carrierMode ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] .

sc:carrier_carrierId a owl:DatatypeProperty ;
  rdfs:label "CARRIER_ID" ;
  rdfs:comment "DIM_CARRIER.CARRIER_ID" ;
  rdfs:domain sc:Carrier ;
  rdfs:range xsd:integer .

sc:carrier_carrierName a owl:DatatypeProperty ;
  rdfs:label "CARRIER_NAME" ;
  rdfs:comment "DIM_CARRIER.CARRIER_NAME" ;
  rdfs:domain sc:Carrier ;
  rdfs:range xsd:string .

sc:carrier_carrierMode a owl:DatatypeProperty ;
  rdfs:label "CARRIER_MODE" ;
  rdfs:comment "OCEAN, AIR, TRUCK, RAIL, MULTIMODAL" ;
  rdfs:domain sc:Carrier ;
  coa:distinctValues "AIR" ;
  coa:distinctValues "MULTIMODAL" ;
  coa:distinctValues "OCEAN" ;
  coa:distinctValues "RAIL" ;
  coa:distinctValues "TRUCK" ;
  rdfs:range xsd:string .

#-----
# Lane <- SCS.INVENTORY.DIM_LANE
#-----

sc:Lane a owl:Class ;
  rdfs:label "Lane" ;
  rdfs:comment "Shipping lanes between ports" ;
  skos:altLabel "shipping lane" ;
  skos:altLabel "trade lane" ;
  skos:altLabel "route" ;
  skos:altLabel "DIM_LANE" ;
  skos:altLabel "corridor" ;
  owl:hasKey ( sc:lane_laneId ) ;
  rdfs:subClassOf
    [ a owl:Restriction ; owl:onProperty sc:lane_laneId ; owl:cardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:lane_originPortId ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:lane_destPortId ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:lane_transportMode ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:lane_transitDaysAvg ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] .

sc:lane_laneId a owl:DatatypeProperty ;
  rdfs:label "LANE_ID" ;
  rdfs:comment "DIM_LANE.LANE_ID" ;
  rdfs:domain sc:Lane ;
  rdfs:range xsd:integer .

sc:lane_originPortId a owl:ObjectProperty ;
  rdfs:label "origin port" ;
  rdfs:comment "Port the lane departs from" ;
  rdfs:domain sc:Lane ;
  rdfs:range sc:Port ;
  coa:fkProvenance "MANY_TO_ONE" .

sc:lane_destPortId a owl:ObjectProperty ;
  rdfs:label "destination port" ;
  rdfs:comment "Port the lane arrives at" ;
  rdfs:domain sc:Lane ;
  rdfs:range sc:Port ;
  coa:fkProvenance "MANY_TO_ONE" .

sc:lane_transportMode a owl:DatatypeProperty ;
  rdfs:label "TRANSPORT_MODE" ;
  rdfs:comment "DIM_LANE.TRANSPORT_MODE" ;
  rdfs:domain sc:Lane ;
  coa:distinctValues "AIR" ;

```

```

    coa:distinctValues "OCEAN" ;
    rdfs:range xsd:string .

sc:lane_transitDaysAvg a owl:DatatypeProperty ;
    rdfs:label "TRANSIT_DAYS_AVG" ;
    rdfs:comment "DIM_LANE.TRANSIT_DAYS_AVG" ;
    rdfs:domain sc:Lane ;
    skos:altLabel "transit time" ;
    skos:altLabel "average transit days" ;
    rdfs:range xsd:integer .

sc:lane_isProblemLane a owl:DatatypeProperty ;
    rdfs:label "IS_PROBLEM_LANE" ;
    rdfs:comment "DIM_LANE.IS_PROBLEM_LANE" ;
    rdfs:domain sc:Lane ;
    rdfs:range xsd:boolean .

#-----
# Contract <- SCS.INVENTORY.DIM_CONTRACT
#-----

sc:Contract a owl:Class ;
    rdfs:label "Contract" ;
    rdfs:comment "Supplier contracts" ;
    skos:altLabel "supplier agreement" ;
    skos:altLabel "DIM_CONTRACT" ;
    skos:altLabel "purchase agreement" ;
    owl:hasKey ( sc:contract_contractId ) ;
    rdfs:subClassOf
        [ a owl:Restriction ; owl:onProperty sc:contract_contractId ; owl:cardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:contract_supplierId ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:contract_contractCode ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:contract_startDate ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:contract_endDate ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] .

sc:contract_contractId a owl:DatatypeProperty ;
    rdfs:label "CONTRACT_ID" ;
    rdfs:comment "DIM_CONTRACT.CONTRACT_ID" ;
    rdfs:domain sc:Contract ;
    rdfs:range xsd:integer .

sc:contract_supplierId a owl:ObjectProperty ;
    rdfs:label "contracted supplier" ;
    rdfs:comment "Supplier the contract is with" ;
    rdfs:domain sc:Contract ;
    rdfs:range sc:Supplier ;
    coa:fkProvenance "MANY_TO_ONE" .

sc:contract_contractCode a owl:DatatypeProperty ;
    rdfs:label "CONTRACT_CODE" ;
    rdfs:comment "DIM_CONTRACT.CONTRACT_CODE" ;
    rdfs:domain sc:Contract ;
    rdfs:range xsd:string .

sc:contract_startDate a owl:DatatypeProperty ;
    rdfs:label "START_DATE" ;
    rdfs:comment "DIM_CONTRACT.START_DATE" ;
    rdfs:domain sc:Contract ;
    rdfs:range xsd:date .

sc:contract_endDate a owl:DatatypeProperty ;
    rdfs:label "END_DATE" ;
    rdfs:comment "DIM_CONTRACT.END_DATE" ;
    rdfs:domain sc:Contract ;
    rdfs:range xsd:date .

sc:contract_currencyCode a owl:ObjectProperty ;
    rdfs:label "currencyCode" ;
    rdfs:comment "Foreign key: DIM_CONTRACT.CURRENCY_CODE references DIM_CURRENCY.CURRENCY_CODE" ;
    rdfs:domain sc:Contract ;
    rdfs:range sc:Currency ;
    coa:fkProvenance "MANY_TO_ONE" .

sc:contract_incotermCode a owl:ObjectProperty ;
    rdfs:label "incotermCode" ;

```

```

    rdfs:comment "Foreign key: DIM_CONTRACT.INCOTERM_CODE references DIM_INCOTERM.INCOTERM_CODE" ;
    rdfs:domain sc:Contract ;
    rdfs:range sc:Incoterm ;
    coa:fkProvenance "MANY_TO_ONE" .

sc:contract_status a owl:DatatypeProperty ;
    rdfs:label "STATUS" ;
    rdfs:comment "DIM_CONTRACT.STATUS" ;
    rdfs:domain sc:Contract ;
    coa:distinctValues "ACTIVE" ;
    rdfs:range xsd:string .

#-----
# SupplierPartApproval <- SCS.INVENTORY.SUPPLIER_PART_APPROVAL
#-----

sc:SupplierPartApproval a owl:Class ;
    rdfs:label "Supplier Part Approval" ;
    rdfs:comment "Approved supplier-part combinations - 40,000 rows" ;
    skos:altLabel "approved vendor list" ;
    skos:altLabel "avl" ;
    skos:altLabel "asl" ;
    skos:altLabel "approved source" ;
    skos:altLabel "SUPPLIER_PART_APPROVAL" ;
    owl:hasKey ( sc:supplierpartapproval_approvalId ) ;
    rdfs:subClassOf
        [ a owl:Restriction ; owl:onProperty sc:supplierpartapproval_approvalId ; owl:cardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:supplierpartapproval_supplierId ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:supplierpartapproval_siteId ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:supplierpartapproval_partId ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:supplierpartapproval_approvalDate ;
          owl:minCardinality "1"^^xsd:nonNegativeInteger ] .

sc:supplierpartapproval_approvalId a owl:DatatypeProperty ;
    rdfs:label "APPROVAL_ID" ;
    rdfs:comment "SUPPLIER_PART_APPROVAL.APPROVAL_ID" ;
    rdfs:domain sc:SupplierPartApproval ;
    rdfs:range xsd:integer .

sc:supplierpartapproval_supplierId a owl:ObjectProperty ;
    rdfs:label "approved supplier" ;
    rdfs:comment "Supplier approved to provide the part" ;
    rdfs:domain sc:SupplierPartApproval ;
    rdfs:range sc:Supplier ;
    coa:fkProvenance "MANY_TO_ONE" .

sc:supplierpartapproval_siteId a owl:ObjectProperty ;
    rdfs:label "approved at site" ;
    rdfs:comment "Supplier site the approval is granted for" ;
    rdfs:domain sc:SupplierPartApproval ;
    rdfs:range sc:SupplierSite ;
    coa:fkProvenance "MANY_TO_ONE" .

sc:supplierpartapproval_partId a owl:ObjectProperty ;
    rdfs:label "approved part" ;
    rdfs:comment "Part the supplier is approved to provide" ;
    rdfs:domain sc:SupplierPartApproval ;
    rdfs:range sc:Part ;
    coa:fkProvenance "MANY_TO_ONE" .

sc:supplierpartapproval_approvalDate a owl:DatatypeProperty ;
    rdfs:label "APPROVAL_DATE" ;
    rdfs:comment "SUPPLIER_PART_APPROVAL.APPROVAL_DATE" ;
    rdfs:domain sc:SupplierPartApproval ;
    rdfs:range xsd:date .

sc:supplierpartapproval_status a owl:DatatypeProperty ;
    rdfs:label "STATUS" ;
    rdfs:comment "SUPPLIER_PART_APPROVAL.STATUS" ;
    rdfs:domain sc:SupplierPartApproval ;
    coa:distinctValues "APPROVED" ;
    rdfs:range xsd:string .

#-----
# PurchaseOrder <- SCS.INVENTORY.PO_HEADER
#-----

```

```

sc:PurchaseOrder a owl:Class ;
  rdfs:label "Purchase Order" ;
  rdfs:comment "Purchase order headers – 20,000 POs" ;
  skos:altLabel "po" ;
  skos:altLabel "PO_HEADER" ;
  skos:altLabel "procurement order" ;
  skos:altLabel "purchase requisition" ;
  owl:hasKey ( sc:purchaseorder_poId ) ;
  rdfs:subClassOf
    [ a owl:Restriction ; owl:onProperty sc:purchaseorder_poId ; owl:cardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:purchaseorder_poNumber ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:purchaseorder_supplierId ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:purchaseorder_plantId ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:purchaseorder_orderDate ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:purchaseorder_status ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] .

sc:purchaseorder_poId a owl:DatatypeProperty ;
  rdfs:label "PO_ID" ;
  rdfs:comment "PO_HEADER.PO_ID" ;
  rdfs:domain sc:PurchaseOrder ;
  rdfs:range xsd:integer .

sc:purchaseorder_poNumber a owl:DatatypeProperty ;
  rdfs:label "PO_NUMBER" ;
  rdfs:comment "PO_HEADER.PO_NUMBER" ;
  rdfs:domain sc:PurchaseOrder ;
  rdfs:range xsd:string .

sc:purchaseorder_supplierId a owl:ObjectProperty ;
  rdfs:label "ordered from supplier" ;
  rdfs:comment "Supplier the purchase order was placed with" ;
  rdfs:domain sc:PurchaseOrder ;
  rdfs:range sc:Supplier ;
  coa:fkProvenance "MANY_TO_ONE" .

sc:purchaseorder_plantId a owl:ObjectProperty ;
  rdfs:label "ordered for plant" ;
  rdfs:comment "Plant the purchase order is destined for" ;
  rdfs:domain sc:PurchaseOrder ;
  rdfs:range sc:Plant ;
  coa:fkProvenance "MANY_TO_ONE" .

sc:purchaseorder_orderDate a owl:DatatypeProperty ;
  rdfs:label "ORDER_DATE" ;
  rdfs:comment "PO_HEADER.ORDER_DATE" ;
  rdfs:domain sc:PurchaseOrder ;
  rdfs:range xsd:date .

sc:purchaseorder_currencyCode a owl:ObjectProperty ;
  rdfs:label "currencyCode" ;
  rdfs:comment "Foreign key: PO_HEADER.CURRENCY_CODE references DIM_CURRENCY.CURRENCY_CODE" ;
  rdfs:domain sc:PurchaseOrder ;
  rdfs:range sc:Currency ;
  coa:fkProvenance "MANY_TO_ONE" .

sc:purchaseorder_status a owl:DatatypeProperty ;
  rdfs:label "STATUS" ;
  rdfs:comment "OPEN, RECEIVED, PARTIAL, CANCELLED" ;
  rdfs:domain sc:PurchaseOrder ;
  coa:distinctValues "CANCELLED" ;
  coa:distinctValues "OPEN" ;
  coa:distinctValues "PARTIAL" ;
  coa:distinctValues "RECEIVED" ;
  rdfs:range xsd:string .

#-----
# PurchaseOrderLine <- SCS.INVENTORY.PO_LINE
#-----

sc:PurchaseOrderLine a owl:Class ;
  rdfs:label "Purchase Order Line" ;
  rdfs:comment "Purchase order line items – 60,000 lines" ;
  skos:altLabel "po line" ;
  skos:altLabel "po item" ;

```

```

skos:altLabel "PO_LINE" ;
owl:hasKey ( sc:purchaseorderline_poLineId ) ;
rdfs:subClassOf
[ a owl:Restriction ; owl:onProperty sc:purchaseorderline_poLineId ; owl:cardinality
  "1"^^xsd:nonNegativeInteger ] ,
[ a owl:Restriction ; owl:onProperty sc:purchaseorderline_poId ; owl:minCardinality
  "1"^^xsd:nonNegativeInteger ] ,
[ a owl:Restriction ; owl:onProperty sc:purchaseorderline_partId ; owl:minCardinality
  "1"^^xsd:nonNegativeInteger ] ,
[ a owl:Restriction ; owl:onProperty sc:purchaseorderline_quantity ; owl:minCardinality
  "1"^^xsd:nonNegativeInteger ] ,
[ a owl:Restriction ; owl:onProperty sc:purchaseorderline_unitPrice ; owl:minCardinality
  "1"^^xsd:nonNegativeInteger ] ,
[ a owl:Restriction ; owl:onProperty sc:purchaseorderline_promiseDate ; owl:minCardinality
  "1"^^xsd:nonNegativeInteger ] .

sc:purchaseorderline_poLineId a owl:DatatypeProperty ;
rdfs:label "PO_LINE_ID" ;
rdfs:comment "PO_LINE.PO_LINE_ID" ;
rdfs:domain sc:PurchaseOrderLine ;
rdfs:range xsd:integer .

sc:purchaseorderline_poId a owl:ObjectProperty ;
rdfs:label "line of purchase order" ;
rdfs:comment "Purchase order this line belongs to" ;
rdfs:domain sc:PurchaseOrderLine ;
rdfs:range sc:PurchaseOrder ;
coa:fkProvenance "MANY_TO_ONE" .

sc:purchaseorderline_partId a owl:ObjectProperty ;
rdfs:label "orders part" ;
rdfs:comment "Part being purchased on this line" ;
rdfs:domain sc:PurchaseOrderLine ;
rdfs:range sc:Part ;
coa:fkProvenance "MANY_TO_ONE" .

sc:purchaseorderline_quantity a owl:DatatypeProperty ;
rdfs:label "QUANTITY" ;
rdfs:comment "PO_LINE.QUANTITY" ;
rdfs:domain sc:PurchaseOrderLine ;
rdfs:range xsd:decimal .

sc:purchaseorderline_unitPrice a owl:DatatypeProperty ;
rdfs:label "UNIT_PRICE" ;
rdfs:comment "PO_LINE.UNIT_PRICE" ;
rdfs:domain sc:PurchaseOrderLine ;
skos:altLabel "po price" ;
skos:altLabel "purchase price" ;
rdfs:range xsd:decimal .

sc:purchaseorderline_promiseDate a owl:DatatypeProperty ;
rdfs:label "PROMISE_DATE" ;
rdfs:comment "Original supplier promise date" ;
rdfs:domain sc:PurchaseOrderLine ;
rdfs:range xsd:date .

#-----
# GoodsReceipt <- SCS.INVENTORY.GOODS_RECEIPT
#-----

sc:GoodsReceipt a owl:Class ;
rdfs:label "Goods Receipt" ;
rdfs:comment "Goods receipts – 90,000 receipts against PO lines" ;
skos:altLabel "grn" ;
skos:altLabel "receipt" ;
skos:altLabel "goods received note" ;
skos:altLabel "GOODS_RECEIPT" ;
skos:altLabel "delivery" ;
owl:hasKey ( sc:goodsreceipt_receiptId ) ;
rdfs:subClassOf
[ a owl:Restriction ; owl:onProperty sc:goodsreceipt_receiptId ; owl:cardinality
  "1"^^xsd:nonNegativeInteger ] ,
[ a owl:Restriction ; owl:onProperty sc:goodsreceipt_poLineId ; owl:minCardinality
  "1"^^xsd:nonNegativeInteger ] ,
[ a owl:Restriction ; owl:onProperty sc:goodsreceipt_receiptDate ; owl:minCardinality
  "1"^^xsd:nonNegativeInteger ] ,
[ a owl:Restriction ; owl:onProperty sc:goodsreceipt_quantityReceived ; owl:minCardinality
  "1"^^xsd:nonNegativeInteger ] ,
[ a owl:Restriction ; owl:onProperty sc:goodsreceipt_warehouseId ; owl:minCardinality
  "1"^^xsd:nonNegativeInteger ] ,

```

```

[ a owl:Restriction ; owl:onProperty sc:goodsreceipt_daysVariance ; owl:minCardinality
  "1"^^xsd:nonNegativeInteger ] .

sc:goodsreceipt_receiptId a owl:DatatypeProperty ;
  rdfs:label "RECEIPT_ID" ;
  rdfs:comment "GOODS_RECEIPT.RECEIPT_ID" ;
  rdfs:domain sc:GoodsReceipt ;
  rdfs:range xsd:integer .

sc:goodsreceipt_poLineId a owl:ObjectProperty ;
  rdfs:label "receipt against PO line" ;
  rdfs:comment "Purchase order line this receipt fulfils" ;
  rdfs:domain sc:GoodsReceipt ;
  rdfs:range sc:PurchaseOrderLine ;
  coa:fkProvenance "MANY_TO_ONE" .

sc:goodsreceipt_receiptDate a owl:DatatypeProperty ;
  rdfs:label "RECEIPT_DATE" ;
  rdfs:comment "Actual physical receipt date" ;
  rdfs:domain sc:GoodsReceipt ;
  rdfs:range xsd:date .

sc:goodsreceipt_quantityReceived a owl:DatatypeProperty ;
  rdfs:label "QUANTITY_RECEIVED" ;
  rdfs:comment "GOODS_RECEIPT.QUANTITY_RECEIVED" ;
  rdfs:domain sc:GoodsReceipt ;
  rdfs:range xsd:decimal .

sc:goodsreceipt_warehouseId a owl:ObjectProperty ;
  rdfs:label "received into warehouse" ;
  rdfs:comment "Warehouse the goods were received into" ;
  rdfs:domain sc:GoodsReceipt ;
  rdfs:range sc:Warehouse ;
  coa:fkProvenance "MANY_TO_ONE" .

sc:goodsreceipt_daysVariance a owl:DatatypeProperty ;
  rdfs:label "DAYS_VARIANCE" ;
  rdfs:comment "Days early (negative) or late (positive) vs promise date" ;
  rdfs:domain sc:GoodsReceipt ;
  skos:altLabel "delivery variance" ;
  skos:altLabel "days late" ;
  skos:altLabel "schedule variance" ;
  skos:altLabel "otd variance" ;
  rdfs:range xsd:integer .

#-----
# SalesOrder <- SCS.INVENTORY.SALES_ORDER_HEADER
#-----

sc:SalesOrder a owl:Class ;
  rdfs:label "Sales Order" ;
  rdfs:comment "Customer sales order headers – 10,000 orders" ;
  skos:altLabel "so" ;
  skos:altLabel "customer order" ;
  skos:altLabel "SALES_ORDER_HEADER" ;
  skos:altLabel "demand" ;
  owl:hasKey ( sc:salesorder_orderId ) ;
  rdfs:subClassOf
    [ a owl:Restriction ; owl:onProperty sc:salesorder_orderId ; owl:cardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:salesorder_orderNumber ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:salesorder_customerId ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:salesorder_orderDate ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:salesorder_plantId ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:salesorder_status ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] .

sc:salesorder_orderId a owl:DatatypeProperty ;
  rdfs:label "ORDER_ID" ;
  rdfs:comment "SALES_ORDER_HEADER.ORDER_ID" ;
  rdfs:domain sc:SalesOrder ;
  rdfs:range xsd:integer .

sc:salesorder_orderNumber a owl:DatatypeProperty ;
  rdfs:label "ORDER_NUMBER" ;
  rdfs:comment "SALES_ORDER_HEADER.ORDER_NUMBER" ;

```

```

    rdfs:domain sc:SalesOrder ;
    rdfs:range xsd:string .

sc:salesorder_customerId a owl:ObjectProperty ;
    rdfs:label "ordered by customer" ;
    rdfs:comment "Customer that placed the sales order" ;
    rdfs:domain sc:SalesOrder ;
    rdfs:range sc:Customer ;
    coa:fkProvenance "MANY_TO_ONE" .

sc:salesorder_orderDate a owl:DatatypeProperty ;
    rdfs:label "ORDER_DATE" ;
    rdfs:comment "SALES_ORDER_HEADER.ORDER_DATE" ;
    rdfs:domain sc:SalesOrder ;
    rdfs:range xsd:date .

sc:salesorder_plantId a owl:ObjectProperty ;
    rdfs:label "plantId" ;
    rdfs:comment "Foreign key: SALES_ORDER_HEADER.PLANT_ID references DIM_PLANT.PLANT_ID" ;
    rdfs:domain sc:SalesOrder ;
    rdfs:range sc:Plant ;
    coa:fkProvenance "MANY_TO_ONE" .

sc:salesorder_status a owl:DatatypeProperty ;
    rdfs:label "STATUS" ;
    rdfs:comment "SALES_ORDER_HEADER.STATUS" ;
    rdfs:domain sc:SalesOrder ;
    coa:distinctValues "OPEN" ;
    coa:distinctValues "PARTIAL" ;
    coa:distinctValues "SHIPPED" ;
    rdfs:range xsd:string .

#-----
# SalesOrderLine <- SCS.INVENTORY.SALES_ORDER_LINE
#-----

sc:SalesOrderLine a owl:Class ;
    rdfs:label "Sales Order Line" ;
    rdfs:comment "Sales order line items - 40,000 lines" ;
    skos:altLabel "so line" ;
    skos:altLabel "order line" ;
    skos:altLabel "SALES_ORDER_LINE" ;
    owl:hasKey ( sc:salesorderline_orderLineId ) ;
    rdfs:subClassOf
        [ a owl:Restriction ; owl:onProperty sc:salesorderline_orderLineId ; owl:cardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:salesorderline_orderId ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:salesorderline_productId ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:salesorderline_quantity ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:salesorderline_unitPrice ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:salesorderline_requestedDate ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] .

sc:salesorderline_orderLineId a owl:DatatypeProperty ;
    rdfs:label "ORDER_LINE_ID" ;
    rdfs:comment "SALES_ORDER_LINE.ORDER_LINE_ID" ;
    rdfs:domain sc:SalesOrderLine ;
    rdfs:range xsd:integer .

sc:salesorderline_orderId a owl:ObjectProperty ;
    rdfs:label "orderId" ;
    rdfs:comment "Foreign key: SALES_ORDER_LINE.ORDER_ID references SALES_ORDER_HEADER.ORDER_ID" ;
    rdfs:domain sc:SalesOrderLine ;
    rdfs:range sc:SalesOrder ;
    coa:fkProvenance "MANY_TO_ONE" .

sc:salesorderline_productId a owl:ObjectProperty ;
    rdfs:label "sells product" ;
    rdfs:comment "Finished product being sold on this line" ;
    rdfs:domain sc:SalesOrderLine ;
    rdfs:range sc:Product ;
    coa:fkProvenance "MANY_TO_ONE" .

sc:salesorderline_quantity a owl:DatatypeProperty ;
    rdfs:label "QUANTITY" ;
    rdfs:comment "SALES_ORDER_LINE.QUANTITY" ;

```

```

    rdfs:domain sc:SalesOrderLine ;
    rdfs:range xsd:decimal .

sc:salesorderline_unitPrice a owl:DatatypeProperty ;
    rdfs:label "UNIT_PRICE" ;
    rdfs:comment "SALES_ORDER_LINE.UNIT_PRICE" ;
    rdfs:domain sc:SalesOrderLine ;
    rdfs:range xsd:decimal .

sc:salesorderline_requestedDate a owl:DatatypeProperty ;
    rdfs:label "REQUESTED_DATE" ;
    rdfs:comment "SALES_ORDER_LINE.REQUESTED_DATE" ;
    rdfs:domain sc:SalesOrderLine ;
    rdfs:range xsd:date .

#-----
# Shipment <- SCS.INVENTORY.SHIPMENT_HEADER
#-----

sc:Shipment a owl:Class ;
    rdfs:label "Shipment" ;
    rdfs:comment "Shipment headers – 18,000 shipments" ;
    skos:altLabel "freight" ;
    skos:altLabel "consignment" ;
    skos:altLabel "SHIPMENT_HEADER" ;
    skos:altLabel "load" ;
    owl:hasKey ( sc:shipment_shipmentId ) ;
    rdfs:subClassOf
        [ a owl:Restriction ; owl:onProperty sc:shipment_shipmentId ; owl:cardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:shipment_carrierId ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:shipment_laneId ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:shipment_shipDate ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:shipment_status ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] .

sc:shipment_shipmentId a owl:DatatypeProperty ;
    rdfs:label "SHIPMENT_ID" ;
    rdfs:comment "SHIPMENT_HEADER.SHIPMENT_ID" ;
    rdfs:domain sc:Shipment ;
    rdfs:range xsd:integer .

sc:shipment_carrierId a owl:ObjectProperty ;
    rdfs:label "shipped by carrier" ;
    rdfs:comment "Carrier moving the shipment" ;
    rdfs:domain sc:Shipment ;
    rdfs:range sc:Carrier ;
    coa:fkProvenance "MANY_TO_ONE" .

sc:shipment_laneId a owl:ObjectProperty ;
    rdfs:label "travels lane" ;
    rdfs:comment "Shipping lane the shipment moves on" ;
    rdfs:domain sc:Shipment ;
    rdfs:range sc:Lane ;
    coa:fkProvenance "MANY_TO_ONE" .

sc:shipment_shipDate a owl:DatatypeProperty ;
    rdfs:label "SHIP_DATE" ;
    rdfs:comment "SHIPMENT_HEADER.SHIP_DATE" ;
    rdfs:domain sc:Shipment ;
    rdfs:range xsd:date .

sc:shipment_deliveryDate a owl:DatatypeProperty ;
    rdfs:label "DELIVERY_DATE" ;
    rdfs:comment "SHIPMENT_HEADER.DELIVERY_DATE" ;
    rdfs:domain sc:Shipment ;
    rdfs:range xsd:date .

sc:shipment_status a owl:DatatypeProperty ;
    rdfs:label "STATUS" ;
    rdfs:comment "SHIPMENT_HEADER.STATUS" ;
    rdfs:domain sc:Shipment ;
    coa:distinctValues "DELAYED" ;
    coa:distinctValues "DELIVERED" ;
    coa:distinctValues "IN_TRANSIT" ;
    rdfs:range xsd:string .

#-----

```

```

# ShipmentLine <- SCS.INVENTORY.SHIPMENT_LINE
#-----

sc:ShipmentLine a owl:Class ;
  rdfs:label "Shipment Line" ;
  rdfs:comment "Shipment line items linking to orders or POs" ;
  skos:altLabel "shipment item" ;
  skos:altLabel "SHIPMENT_LINE" ;
  owl:hasKey ( sc:shipmentline_shipmentLineId ) ;
  rdfs:subClassOf
    [ a owl:Restriction ; owl:onProperty sc:shipmentline_shipmentLineId ; owl:cardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:shipmentline_shipmentId ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:shipmentline_quantity ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] .

sc:shipmentline_shipmentLineId a owl:DatatypeProperty ;
  rdfs:label "SHIPMENT_LINE_ID" ;
  rdfs:comment "SHIPMENT_LINE.SHIPMENT_LINE_ID" ;
  rdfs:domain sc:ShipmentLine ;
  rdfs:range xsd:integer .

sc:shipmentline_shipmentId a owl:ObjectProperty ;
  rdfs:label "shipmentId" ;
  rdfs:comment "Foreign key: SHIPMENT_LINE.SHIPMENT_ID references SHIPMENT_HEADER.SHIPMENT_ID" ;
  rdfs:domain sc:ShipmentLine ;
  rdfs:range sc:Shipment ;
  coa:fkProvenance "MANY_TO_ONE" .

sc:shipmentline_orderLineId a owl:ObjectProperty ;
  rdfs:label "fulfils sales order line" ;
  rdfs:comment "Sales order line this shipment line fulfils; null for inbound freight" ;
  rdfs:domain sc:ShipmentLine ;
  rdfs:range sc:SalesOrderLine ;
  coa:fkProvenance "MANY_TO_ONE" .

sc:shipmentline_poLineId a owl:ObjectProperty ;
  rdfs:label "fulfils purchase order line" ;
  rdfs:comment "Purchase order line this shipment line fulfils; null for outbound freight" ;
  rdfs:domain sc:ShipmentLine ;
  rdfs:range sc:PurchaseOrderLine ;
  coa:fkProvenance "MANY_TO_ONE" .

sc:shipmentline_quantity a owl:DatatypeProperty ;
  rdfs:label "QUANTITY" ;
  rdfs:comment "SHIPMENT_LINE.QUANTITY" ;
  rdfs:domain sc:ShipmentLine ;
  rdfs:range xsd:decimal .

#-----
# ShipmentMilestoneEvent <- SCS.INVENTORY.SHIPMENT_MILESTONE_EVENT
#-----

sc:ShipmentMilestoneEvent a owl:Class ;
  rdfs:label "Shipment Milestone Event" ;
  rdfs:comment "Shipment tracking milestones – 250,000 events" ;
  skos:altLabel "tracking event" ;
  skos:altLabel "milestone" ;
  skos:altLabel "shipment status event" ;
  skos:altLabel "SHIPMENT_MILESTONE_EVENT" ;
  owl:hasKey ( sc:shipmentmilestoneevent_eventId ) ;
  rdfs:subClassOf
    [ a owl:Restriction ; owl:onProperty sc:shipmentmilestoneevent_eventId ; owl:cardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:shipmentmilestoneevent_shipmentId ;
      owl:minCardinality "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:shipmentmilestoneevent_milestone ;
      owl:minCardinality "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:shipmentmilestoneevent_eventTs ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] .

sc:shipmentmilestoneevent_eventId a owl:DatatypeProperty ;
  rdfs:label "EVENT_ID" ;
  rdfs:comment "SHIPMENT_MILESTONE_EVENT.EVENT_ID" ;
  rdfs:domain sc:ShipmentMilestoneEvent ;
  rdfs:range xsd:integer .

sc:shipmentmilestoneevent_shipmentId a owl:ObjectProperty ;
  rdfs:label "milestone of shipment" ;

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```

    rdfs:comment "Shipment this tracking milestone belongs to" ;
    rdfs:domain sc:ShipmentMilestoneEvent ;
    rdfs:range sc:Shipment ;
    coa:fkProvenance "MANY_TO_ONE" .

sc:shipmentmilestoneevent_milestone a owl:DatatypeProperty ;
    rdfs:label "MILESTONE" ;
    rdfs:comment "SHIPMENT_MILESTONE_EVENT.MILESTONE" ;
    rdfs:domain sc:ShipmentMilestoneEvent ;
    coa:distinctValues "ARRIVED_PORT" ;
    coa:distinctValues "BOOKING_CONFIRMED" ;
    coa:distinctValues "CUSTOMS_CLEARANCE" ;
    coa:distinctValues "DELIVERED" ;
    coa:distinctValues "DEPARTED" ;
    coa:distinctValues "GATE_IN" ;
    coa:distinctValues "GATE_OUT" ;
    coa:distinctValues "INVOICE_SENT" ;
    coa:distinctValues "IN_TRANSIT" ;
    coa:distinctValues "LOADED" ;
    coa:distinctValues "OUT_FOR_DELIVERY" ;
    coa:distinctValues "PICKED_UP" ;
    coa:distinctValues "POD_SIGNED" ;
    coa:distinctValues "TRANSSHIPMENT" ;
    rdfs:range xsd:string .

sc:shipmentmilestoneevent_eventTs a owl:DatatypeProperty ;
    rdfs:label "EVENT_TS" ;
    rdfs:comment "SHIPMENT_MILESTONE_EVENT.EVENT_TS" ;
    rdfs:domain sc:ShipmentMilestoneEvent ;
    rdfs:range xsd:dateTime .

sc:shipmentmilestoneevent_location a owl:DatatypeProperty ;
    rdfs:label "LOCATION" ;
    rdfs:comment "SHIPMENT_MILESTONE_EVENT.LOCATION" ;
    rdfs:domain sc:ShipmentMilestoneEvent ;
    rdfs:range xsd:string .

#-----
# SensorReading <- SCS.INVENTORY.IOT_SENSOR_READING
#-----

sc:SensorReading a owl:Class ;
    rdfs:label "Sensor Reading" ;
    rdfs:comment "IoT container sensor readings - 500,000 rows, ~1.5% breach rate" ;
    skos:altLabel "iot reading" ;
    skos:altLabel "telemetry" ;
    skos:altLabel "container sensor" ;
    skos:altLabel "IOT_SENSOR_READING" ;
    skos:altLabel "cold chain reading" ;
    owl:hasKey ( sc:sensorreading_readingId ) ;
    rdfs:subClassOf
        [ a owl:Restriction ; owl:onProperty sc:sensorreading_readingId ; owl:cardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:sensorreading_shipmentId ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:sensorreading_readingTs ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:sensorreading_breachFlag ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] .

sc:sensorreading_readingId a owl:DatatypeProperty ;
    rdfs:label "READING_ID" ;
    rdfs:comment "IOT_SENSOR_READING.READING_ID" ;
    rdfs:domain sc:SensorReading ;
    rdfs:range xsd:integer .

sc:sensorreading_shipmentId a owl:ObjectProperty ;
    rdfs:label "reading for shipment" ;
    rdfs:comment "Shipment the sensor reading was captured on" ;
    rdfs:domain sc:SensorReading ;
    rdfs:range sc:Shipment ;
    coa:fkProvenance "MANY_TO_ONE" .

sc:sensorreading_readingTs a owl:DatatypeProperty ;
    rdfs:label "READING_TS" ;
    rdfs:comment "IOT_SENSOR_READING.READING_TS" ;
    rdfs:domain sc:SensorReading ;
    rdfs:range xsd:dateTime .

sc:sensorreading_temperatureC a owl:DatatypeProperty ;

```

```

    rdfs:label "TEMPERATURE_C" ;
    rdfs:comment "IOT_SENSOR_READING.TEMPERATURE_C" ;
    rdfs:domain sc:SensorReading ;
    skos:altLabel "temperature" ;
    skos:altLabel "temp" ;
    rdfs:range xsd:decimal .

sc:sensorreading_humidityPct a owl:DatatypeProperty ;
    rdfs:label "HUMIDITY_PCT" ;
    rdfs:comment "IOT_SENSOR_READING.HUMIDITY_PCT" ;
    rdfs:domain sc:SensorReading ;
    rdfs:range xsd:decimal .

sc:sensorreading_shockG a owl:DatatypeProperty ;
    rdfs:label "SHOCK_G" ;
    rdfs:comment "IOT_SENSOR_READING.SHOCK_G" ;
    rdfs:domain sc:SensorReading ;
    rdfs:range xsd:decimal .

sc:sensorreading_tiltDeg a owl:DatatypeProperty ;
    rdfs:label "TILT_DEG" ;
    rdfs:comment "IOT_SENSOR_READING.TILT_DEG" ;
    rdfs:domain sc:SensorReading ;
    rdfs:range xsd:decimal .

sc:sensorreading_gpsLat a owl:DatatypeProperty ;
    rdfs:label "GPS_LAT" ;
    rdfs:comment "IOT_SENSOR_READING.GPS_LAT" ;
    rdfs:domain sc:SensorReading ;
    rdfs:range xsd:decimal .

sc:sensorreading_gpsLon a owl:DatatypeProperty ;
    rdfs:label "GPS_LON" ;
    rdfs:comment "IOT_SENSOR_READING.GPS_LON" ;
    rdfs:domain sc:SensorReading ;
    rdfs:range xsd:decimal .

sc:sensorreading_breachFlag a owl:DatatypeProperty ;
    rdfs:label "BREACH_FLAG" ;
    rdfs:comment "TRUE if any reading exceeds threshold" ;
    rdfs:domain sc:SensorReading ;
    skos:altLabel "breach" ;
    skos:altLabel "excursion" ;
    skos:altLabel "threshold breach" ;
    skos:altLabel "cold chain breach" ;
    rdfs:range xsd:boolean .

#-----
# WorkOrder <- SCS.INVENTORY.WORK_ORDER
#-----

sc:WorkOrder a owl:Class ;
    rdfs:label "Work Order" ;
    rdfs:comment "Manufacturing work orders - 10,000" ;
    skos:altLabel "wo" ;
    skos:altLabel "production order" ;
    skos:altLabel "manufacturing order" ;
    skos:altLabel "WORK_ORDER" ;
    owl:hasKey ( sc:workorder_workOrderId ) ;
    rdfs:subClassOf
        [ a owl:Restriction ; owl:onProperty sc:workorder_workOrderId ; owl:cardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:workorder_productId ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:workorder_plantId ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:workorder_startDate ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:workorder_quantity ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:workorder_status ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] .

sc:workorder_workOrderId a owl:DatatypeProperty ;
    rdfs:label "WORK_ORDER_ID" ;
    rdfs:comment "WORK_ORDER.WORK_ORDER_ID" ;
    rdfs:domain sc:WorkOrder ;
    rdfs:range xsd:integer .

sc:workorder_productId a owl:ObjectProperty ;

```

```

    rdfs:label "produces product" ;
    rdfs:comment "Finished product the work order builds" ;
    rdfs:domain sc:WorkOrder ;
    rdfs:range sc:Product ;
    coa:fkProvenance "MANY_TO_ONE" .

sc:workorder_plantId a owl:ObjectProperty ;
    rdfs:label "runs at plant" ;
    rdfs:comment "Plant executing the work order" ;
    rdfs:domain sc:WorkOrder ;
    rdfs:range sc:Plant ;
    coa:fkProvenance "MANY_TO_ONE" .

sc:workorder_startDate a owl:DatatypeProperty ;
    rdfs:label "START_DATE" ;
    rdfs:comment "WORK_ORDER.START_DATE" ;
    rdfs:domain sc:WorkOrder ;
    rdfs:range xsd:date .

sc:workorder_endDate a owl:DatatypeProperty ;
    rdfs:label "END_DATE" ;
    rdfs:comment "WORK_ORDER.END_DATE" ;
    rdfs:domain sc:WorkOrder ;
    rdfs:range xsd:date .

sc:workorder_quantity a owl:DatatypeProperty ;
    rdfs:label "QUANTITY" ;
    rdfs:comment "WORK_ORDER.QUANTITY" ;
    rdfs:domain sc:WorkOrder ;
    rdfs:range xsd:decimal .

sc:workorder_status a owl:DatatypeProperty ;
    rdfs:label "STATUS" ;
    rdfs:comment "WORK_ORDER.STATUS" ;
    rdfs:domain sc:WorkOrder ;
    coa:distinctValues "COMPLETED" ;
    coa:distinctValues "IN_PROGRESS" ;
    coa:distinctValues "ON_HOLD" ;
    rdfs:range xsd:string .

#-----
# QualityInspection <- SCS.INVENTORY.QUALITY_INSPECTION
#-----

sc:QualityInspection a owl:Class ;
    rdfs:label "Quality Inspection" ;
    rdfs:comment "Incoming quality inspections - 10,000" ;
    skos:altLabel "incoming inspection" ;
    skos:altLabel "iqc" ;
    skos:altLabel "QUALITY_INSPECTION" ;
    skos:altLabel "quality check" ;
    owl:hasKey ( sc:qualityinspection_inspectionId ) ;
    rdfs:subClassOf
        [ a owl:Restriction ; owl:onProperty sc:qualityinspection_inspectionId ; owl:cardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:qualityinspection_receiptId ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:qualityinspection_partId ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:qualityinspection_supplierId ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:qualityinspection_inspectionDate ;
          owl:minCardinality "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:qualityinspection_sampleSize ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:qualityinspection_defectCount ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:qualityinspection_result ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:qualityinspection_ppm ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] .

sc:qualityinspection_inspectionId a owl:DatatypeProperty ;
    rdfs:label "INSPECTION_ID" ;
    rdfs:comment "QUALITY_INSPECTION.INSPECTION_ID" ;
    rdfs:domain sc:QualityInspection ;
    rdfs:range xsd:integer .

sc:qualityinspection_receiptId a owl:ObjectProperty ;
    rdfs:label "inspects receipt" ;

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    rdfs:comment "Goods receipt that was inspected" ;
    rdfs:domain sc:QualityInspection ;
    rdfs:range sc:GoodsReceipt ;
    coa:fkProvenance "MANY_TO_ONE" .

sc:qualityinspection_partId a owl:ObjectProperty ;
    rdfs:label "inspected part" ;
    rdfs:comment "Part that was inspected" ;
    rdfs:domain sc:QualityInspection ;
    rdfs:range sc:Part ;
    coa:fkProvenance "MANY_TO_ONE" .

sc:qualityinspection_supplierId a owl:ObjectProperty ;
    rdfs:label "inspected supplier" ;
    rdfs:comment "Supplier whose material was inspected" ;
    rdfs:domain sc:QualityInspection ;
    rdfs:range sc:Supplier ;
    coa:fkProvenance "MANY_TO_ONE" .

sc:qualityinspection_inspectionDate a owl:DatatypeProperty ;
    rdfs:label "INSPECTION_DATE" ;
    rdfs:comment "QUALITY_INSPECTION.INSPECTION_DATE" ;
    rdfs:domain sc:QualityInspection ;
    rdfs:range xsd:date .

sc:qualityinspection_sampleSize a owl:DatatypeProperty ;
    rdfs:label "SAMPLE_SIZE" ;
    rdfs:comment "QUALITY_INSPECTION.SAMPLE_SIZE" ;
    rdfs:domain sc:QualityInspection ;
    rdfs:range xsd:integer .

sc:qualityinspection_defectCount a owl:DatatypeProperty ;
    rdfs:label "DEFECT_COUNT" ;
    rdfs:comment "QUALITY_INSPECTION.DEFECT_COUNT" ;
    rdfs:domain sc:QualityInspection ;
    skos:altLabel "defects" ;
    skos:altLabel "rejects" ;
    rdfs:range xsd:integer .

sc:qualityinspection_result a owl:DatatypeProperty ;
    rdfs:label "RESULT" ;
    rdfs:comment "PASS, FAIL, CONDITIONAL" ;
    rdfs:domain sc:QualityInspection ;
    coa:distinctValues "CONDITIONAL" ;
    coa:distinctValues "FAIL" ;
    coa:distinctValues "PASS" ;
    rdfs:range xsd:string .

sc:qualityinspection_ppm a owl:DatatypeProperty ;
    rdfs:label "PPM" ;
    rdfs:comment "Parts per million defect rate" ;
    rdfs:domain sc:QualityInspection ;
    skos:altLabel "defect rate" ;
    skos:altLabel "parts per million" ;
    skos:altLabel "ppm defect rate" ;
    rdfs:range xsd:decimal .

#-----
# NonconformanceReport <- SCS.INVENTORY.NONCONFORMANCE_REPORT
#-----

sc:NonconformanceReport a owl:Class ;
    rdfs:label "Nonconformance Report" ;
    rdfs:comment "Non-conformance reports – 1,000 NCRs" ;
    skos:altLabel "ncr" ;
    skos:altLabel "quality issue" ;
    skos:altLabel "defect report" ;
    skos:altLabel "NONCONFORMANCE_REPORT" ;
    skos:altLabel "car" ;
    owl:hasKey ( sc:nonconformancereport_ncrId ) ;
    rdfs:subClassOf
        [ a owl:Restriction ; owl:onProperty sc:nonconformancereport_ncrId ; owl:cardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:nonconformancereport_supplierId ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:nonconformancereport_partId ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:nonconformancereport_plantId ; owl:minCardinality
          "1"^^xsd:nonNegativeInteger ] ,
        [ a owl:Restriction ; owl:onProperty sc:nonconformancereport_ncrDate ; owl:minCardinality

```

```

        "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:nonconformancereport_severity ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:nonconformancereport_disposition ;
      owl:minCardinality "1"^^xsd:nonNegativeInteger ] .

sc:nonconformancereport_ncrId a owl:DatatypeProperty ;
  rdfs:label "NCR_ID" ;
  rdfs:comment "NONCONFORMANCE_REPORT.NCR_ID" ;
  rdfs:domain sc:NonconformanceReport ;
  rdfs:range xsd:integer .

sc:nonconformancereport_inspectionId a owl:ObjectProperty ;
  rdfs:label "raised from inspection" ;
  rdfs:comment "Inspection that triggered the NCR; null when raised in production" ;
  rdfs:domain sc:NonconformanceReport ;
  rdfs:range sc:QualityInspection ;
  coa:fkProvenance "MANY_TO_ONE" .

sc:nonconformancereport_supplierId a owl:ObjectProperty ;
  rdfs:label "supplier at fault" ;
  rdfs:comment "Supplier held responsible for the nonconformance" ;
  rdfs:domain sc:NonconformanceReport ;
  rdfs:range sc:Supplier ;
  coa:fkProvenance "MANY_TO_ONE" .

sc:nonconformancereport_partId a owl:ObjectProperty ;
  rdfs:label "nonconforming part" ;
  rdfs:comment "Part found nonconforming" ;
  rdfs:domain sc:NonconformanceReport ;
  rdfs:range sc:Part ;
  coa:fkProvenance "MANY_TO_ONE" .

sc:nonconformancereport_plantId a owl:ObjectProperty ;
  rdfs:label "raised at plant" ;
  rdfs:comment "Plant that raised the NCR" ;
  rdfs:domain sc:NonconformanceReport ;
  rdfs:range sc:Plant ;
  coa:fkProvenance "MANY_TO_ONE" .

sc:nonconformancereport_ncrDate a owl:DatatypeProperty ;
  rdfs:label "NCR_DATE" ;
  rdfs:comment "NONCONFORMANCE_REPORT.NCR_DATE" ;
  rdfs:domain sc:NonconformanceReport ;
  rdfs:range xsd:date .

sc:nonconformancereport_severity a owl:DatatypeProperty ;
  rdfs:label "SEVERITY" ;
  rdfs:comment "CRITICAL, MAJOR, MINOR" ;
  rdfs:domain sc:NonconformanceReport ;
  coa:distinctValues "CRITICAL" ;
  coa:distinctValues "MAJOR" ;
  coa:distinctValues "MINOR" ;
  rdfs:range xsd:string .

sc:nonconformancereport_rootCause a owl:DatatypeProperty ;
  rdfs:label "ROOT_CAUSE" ;
  rdfs:comment "NONCONFORMANCE_REPORT.ROOT_CAUSE" ;
  rdfs:domain sc:NonconformanceReport ;
  coa:distinctValues "Cracked substrate from shock" ;
  coa:distinctValues "Dimensional out-of-spec" ;
  coa:distinctValues "Documentation mismatch" ;
  coa:distinctValues "ESD damage in shipping" ;
  coa:distinctValues "Electrical test failure" ;
  coa:distinctValues "Material contamination" ;
  coa:distinctValues "Solder joint defect" ;
  coa:distinctValues "Surface finish degradation" ;
  coa:distinctValues "Unapproved substitution" ;
  coa:distinctValues "Wrong revision shipped" ;
  rdfs:range xsd:string .

sc:nonconformancereport_disposition a owl:DatatypeProperty ;
  rdfs:label "DISPOSITION" ;
  rdfs:comment "SCRAP, REWORK, USE_AS_IS, RETURN" ;
  rdfs:domain sc:NonconformanceReport ;
  coa:distinctValues "RETURN" ;
  coa:distinctValues "REWORK" ;
  coa:distinctValues "SCRAP" ;
  coa:distinctValues "USE_AS_IS" ;
  rdfs:range xsd:string .

```

```

sc:nonconformancereport_status a owl:DatatypeProperty ;
  rdfs:label "STATUS" ;
  rdfs:comment "NONCONFORMANCE_REPORT.STATUS" ;
  rdfs:domain sc:NonconformanceReport ;
  coa:distinctValues "CLOSED" ;
  coa:distinctValues "OPEN" ;
  rdfs:range xsd:string .

#-----
# InventorySnapshot <- SCS.INVENTORY.FACT_INVENTORY_SNAPSHOT
#-----

sc:InventorySnapshot a owl:Class ;
  rdfs:label "Inventory Snapshot" ;
  rdfs:comment "Inventory snapshots - ~1.2M rows (part x warehouse x month-end)" ;
  skos:altLabel "stock on hand" ;
  skos:altLabel "inventory position" ;
  skos:altLabel "soh" ;
  skos:altLabel "FACT_INVENTORY_SNAPSHOT" ;
  skos:altLabel "stock level" ;
  owl:hasKey ( sc:inventorysnapshot_snapshotId ) ;
  rdfs:subClassOf
    [ a owl:Restriction ; owl:onProperty sc:inventorysnapshot_snapshotId ; owl:cardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:inventorysnapshot_snapshotDate ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:inventorysnapshot_partId ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:inventorysnapshot_warehouseId ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:inventorysnapshot_onHandQty ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:inventorysnapshot_unitCostUsd ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] ,
    [ a owl:Restriction ; owl:onProperty sc:inventorysnapshot_onHandValue ; owl:minCardinality
      "1"^^xsd:nonNegativeInteger ] .

sc:inventorysnapshot_snapshotId a owl:DatatypeProperty ;
  rdfs:label "SNAPSHOT_ID" ;
  rdfs:comment "FACT_INVENTORY_SNAPSHOT.SNAPSHOT_ID" ;
  rdfs:domain sc:InventorySnapshot ;
  rdfs:range xsd:integer .

sc:inventorysnapshot_snapshotDate a owl:ObjectProperty ;
  rdfs:label "as of date" ;
  rdfs:comment "Calendar date the snapshot was taken; join here for year, quarter and fiscal
    period" ;
  rdfs:domain sc:InventorySnapshot ;
  rdfs:range sc:CalendarDate ;
  coa:fkProvenance "MANY_TO_ONE" .

sc:inventorysnapshot_partId a owl:ObjectProperty ;
  rdfs:label "stock of part" ;
  rdfs:comment "Part the stock position is measured for" ;
  rdfs:domain sc:InventorySnapshot ;
  rdfs:range sc:Part ;
  coa:fkProvenance "MANY_TO_ONE" .

sc:inventorysnapshot_warehouseId a owl:ObjectProperty ;
  rdfs:label "stock at warehouse" ;
  rdfs:comment "Warehouse holding the stock" ;
  rdfs:domain sc:InventorySnapshot ;
  rdfs:range sc:Warehouse ;
  coa:fkProvenance "MANY_TO_ONE" .

sc:inventorysnapshot_onHandQty a owl:DatatypeProperty ;
  rdfs:label "ON_HAND_QTY" ;
  rdfs:comment "FACT_INVENTORY_SNAPSHOT.ON_HAND_QTY" ;
  rdfs:domain sc:InventorySnapshot ;
  skos:altLabel "stock on hand" ;
  skos:altLabel "soh" ;
  skos:altLabel "on hand quantity" ;
  skos:altLabel "inventory quantity" ;
  rdfs:range xsd:decimal .

sc:inventorysnapshot_unitCostUsd a owl:DatatypeProperty ;
  rdfs:label "UNIT_COST_USD" ;
  rdfs:comment "FACT_INVENTORY_SNAPSHOT.UNIT_COST_USD" ;
  rdfs:domain sc:InventorySnapshot ;
  rdfs:range xsd:decimal .

```

```
sc:inventorysnapshot_onHandValue a owl:DatatypeProperty ;
  rdfs:label "ON_HAND_VALUE" ;
  rdfs:comment "on_hand_qty * unit_cost_usd" ;
  rdfs:domain sc:InventorySnapshot ;
  skos:altLabel "inventory value" ;
  skos:altLabel "stock value" ;
  skos:altLabel "tied up capital" ;
  rdfs:range xsd:decimal .

sc:inventorysnapshot_dailyUsageQty a owl:DatatypeProperty ;
  rdfs:label "DAILY_USAGE_QTY" ;
  rdfs:comment "Average daily consumption" ;
  rdfs:domain sc:InventorySnapshot ;
  skos:altLabel "daily usage" ;
  skos:altLabel "consumption rate" ;
  skos:altLabel "burn rate" ;
  rdfs:range xsd:decimal .

sc:inventorysnapshot_doi a owl:DatatypeProperty ;
  rdfs:label "DOI" ;
  rdfs:comment "Days of inventory = on_hand / daily_usage" ;
  rdfs:domain sc:InventorySnapshot ;
  skos:altLabel "days of inventory" ;
  skos:altLabel "days of supply" ;
  skos:altLabel "dos" ;
  skos:altLabel "coverage days" ;
  rdfs:range xsd:decimal .
```

supply_chain_r2rml.ttl

R2RML mapping to Snowflake (SCS.INVENTORY) — target: UpdateProposal.r2rmlTurtle

```
# R2RML mapping — Asterion Supply Chain -> SCS.INVENTORY (Snowflake)
# Generated by generate_ontology.py. Do not hand-edit.
#
# Target: POST /namespaces/{ns}/proposals/update with r2rmlTurtle
# This is what makes classes coa:isMapped and therefore answerable by
# Tier-2 structured (NL->SPARQL->SQL) queries. Without it, an uploaded
# T-Box serves Tier-3 retrieval only.

@prefix rr:    <http://www.w3.org/ns/r2rml#> .
@prefix xsd:   <http://www.w3.org/2001/XMLSchema#> .
@prefix coa:   <http://coa.amazon.com/vocab/coa#> .
@prefix sc:    <https://asterion.example.com/ontology/supplychain#> .

sc:TriplesMap_CalendarDate
  rr:logicalTable [ rr:tableName "SCS.INVENTORY.DIM_DATE" ] ;
  rr:subjectMap [
    rr:template "https://asterion.example.com/ontology/supplychain/calendardate/{DATE_KEY}" ;
    rr:class sc:CalendarDate ;
  ] ;
  coa:datasourceId "ds-scs-snowflake" ;
  coa:sourceSchema "SCS.INVENTORY" ;
  rr:predicateObjectMap [
    rr:predicate sc:calendardate_dateKey ;
    rr:objectMap [ rr:column "DATE_KEY" ; rr:datatype xsd:date ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:calendardate_year ;
    rr:objectMap [ rr:column "YEAR" ; rr:datatype xsd:integer ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:calendardate_quarter ;
    rr:objectMap [ rr:column "QUARTER" ; rr:datatype xsd:integer ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:calendardate_month ;
    rr:objectMap [ rr:column "MONTH" ; rr:datatype xsd:integer ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:calendardate_monthName ;
    rr:objectMap [ rr:column "MONTH_NAME" ; rr:datatype xsd:string ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:calendardate_weekOfYear ;
    rr:objectMap [ rr:column "WEEK_OF_YEAR" ; rr:datatype xsd:integer ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:calendardate_dayOfWeek ;
    rr:objectMap [ rr:column "DAY_OF_WEEK" ; rr:datatype xsd:integer ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:calendardate_dayName ;
    rr:objectMap [ rr:column "DAY_NAME" ; rr:datatype xsd:string ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:calendardate_isWeekend ;
    rr:objectMap [ rr:column "IS_WEEKEND" ; rr:datatype xsd:boolean ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:calendardate_fiscalYear ;
    rr:objectMap [ rr:column "FISCAL_YEAR" ; rr:datatype xsd:integer ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:calendardate_fiscalQuarter ;
    rr:objectMap [ rr:column "FISCAL_QUARTER" ; rr:datatype xsd:integer ] ;
  ] .

sc:TriplesMap_Region
  rr:logicalTable [ rr:tableName "SCS.INVENTORY.DIM_GEO_REGION" ] ;
  rr:subjectMap [
    rr:template "https://asterion.example.com/ontology/supplychain/region/{REGION_ID}" ;
    rr:class sc:Region ;
  ] ;
  coa:datasourceId "ds-scs-snowflake" ;
  coa:sourceSchema "SCS.INVENTORY" ;
```

```

rr:predicateObjectMap [
  rr:predicate sc:region_regionId ;
  rr:objectMap [ rr:column "REGION_ID" ; rr:datatype xsd:integer ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:region_regionName ;
  rr:objectMap [ rr:column "REGION_NAME" ; rr:datatype xsd:string ] ;
] .

sc:TriplesMap_Country
rr:logicalTable [ rr:tableName "SCS.INVENTORY.DIM_GEO_COUNTRY" ] ;
rr:subjectMap [
  rr:template "https://asterion.example.com/ontology/supplychain/country/{COUNTRY_ID}" ;
  rr:class sc:Country ;
] ;
coa:datasourceId "ds-scs-snowflake" ;
coa:sourceSchema "SCS.INVENTORY" ;
rr:predicateObjectMap [
  rr:predicate sc:country_countryId ;
  rr:objectMap [ rr:column "COUNTRY_ID" ; rr:datatype xsd:integer ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:country_countryName ;
  rr:objectMap [ rr:column "COUNTRY_NAME" ; rr:datatype xsd:string ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:country_countryCode ;
  rr:objectMap [ rr:column "COUNTRY_CODE" ; rr:datatype xsd:string ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:country_regionId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Region ;
    rr:joinCondition [ rr:child "REGION_ID" ; rr:parent "REGION_ID" ] ;
  ] ;
] .

sc:TriplesMap_Commodity
rr:logicalTable [ rr:tableName "SCS.INVENTORY.DIM_COMMODITY" ] ;
rr:subjectMap [
  rr:template "https://asterion.example.com/ontology/supplychain/commodity/{COMMODITY_ID}" ;
  rr:class sc:Commodity ;
] ;
coa:datasourceId "ds-scs-snowflake" ;
coa:sourceSchema "SCS.INVENTORY" ;
rr:predicateObjectMap [
  rr:predicate sc:commodity_commodityId ;
  rr:objectMap [ rr:column "COMMODITY_ID" ; rr:datatype xsd:integer ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:commodity_commodityName ;
  rr:objectMap [ rr:column "COMMODITY_NAME" ; rr:datatype xsd:string ] ;
] .

sc:TriplesMap_SubCommodity
rr:logicalTable [ rr:tableName "SCS.INVENTORY.DIM_SUB_COMMODITY" ] ;
rr:subjectMap [
  rr:template
    "https://asterion.example.com/ontology/supplychain/subcommodity/{SUB_COMMODITY_ID}" ;
  rr:class sc:SubCommodity ;
] ;
coa:datasourceId "ds-scs-snowflake" ;
coa:sourceSchema "SCS.INVENTORY" ;
rr:predicateObjectMap [
  rr:predicate sc:subcommodity_subCommodityId ;
  rr:objectMap [ rr:column "SUB_COMMODITY_ID" ; rr:datatype xsd:integer ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:subcommodity_subCommodityName ;
  rr:objectMap [ rr:column "SUB_COMMODITY_NAME" ; rr:datatype xsd:string ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:subcommodity_commodityId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Commodity ;
    rr:joinCondition [ rr:child "COMMODITY_ID" ; rr:parent "COMMODITY_ID" ] ;
  ] ;
] .

sc:TriplesMap_Currency

```

```

rr:logicalTable [ rr:tableName "SCS.INVENTORY.DIM_CURRENCY" ] ;
rr:subjectMap [
  rr:template "https://asterion.example.com/ontology/supplychain/currency/{CURRENCY_CODE}" ;
  rr:class sc:Currency ;
] ;
coa:datasourceId "ds-scs-snowflake" ;
coa:sourceSchema "SCS.INVENTORY" ;
rr:predicateObjectMap [
  rr:predicate sc:currency_currencyCode ;
  rr:objectMap [ rr:column "CURRENCY_CODE" ; rr:datatype xsd:string ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:currency_currencyName ;
  rr:objectMap [ rr:column "CURRENCY_NAME" ; rr:datatype xsd:string ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:currency_toUsdRate ;
  rr:objectMap [ rr:column "TO_USD_RATE" ; rr:datatype xsd:decimal ] ;
] .

sc:TriplesMap_UnitOfMeasure
rr:logicalTable [ rr:tableName "SCS.INVENTORY.DIM_UOM" ] ;
rr:subjectMap [
  rr:template "https://asterion.example.com/ontology/supplychain/unitofmeasure/{UOM_CODE}" ;
  rr:class sc:UnitOfMeasure ;
] ;
coa:datasourceId "ds-scs-snowflake" ;
coa:sourceSchema "SCS.INVENTORY" ;
rr:predicateObjectMap [
  rr:predicate sc:unitofmeasure_uomCode ;
  rr:objectMap [ rr:column "UOM_CODE" ; rr:datatype xsd:string ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:unitofmeasure_uomName ;
  rr:objectMap [ rr:column "UOM_NAME" ; rr:datatype xsd:string ] ;
] .

sc:TriplesMap_Incoterms
rr:logicalTable [ rr:tableName "SCS.INVENTORY.DIM_INCOTERM" ] ;
rr:subjectMap [
  rr:template "https://asterion.example.com/ontology/supplychain/incoterm/{INCOTERM_CODE}" ;
  rr:class sc:Incoterms ;
] ;
coa:datasourceId "ds-scs-snowflake" ;
coa:sourceSchema "SCS.INVENTORY" ;
rr:predicateObjectMap [
  rr:predicate sc:incoterms_incotermCode ;
  rr:objectMap [ rr:column "INCOTERM_CODE" ; rr:datatype xsd:string ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:incoterms_incotermName ;
  rr:objectMap [ rr:column "INCOTERM_NAME" ; rr:datatype xsd:string ] ;
] .

sc:TriplesMap_Port
rr:logicalTable [ rr:tableName "SCS.INVENTORY.DIM_PORT" ] ;
rr:subjectMap [
  rr:template "https://asterion.example.com/ontology/supplychain/port/{PORT_ID}" ;
  rr:class sc:Port ;
] ;
coa:datasourceId "ds-scs-snowflake" ;
coa:sourceSchema "SCS.INVENTORY" ;
rr:predicateObjectMap [
  rr:predicate sc:port_portId ;
  rr:objectMap [ rr:column "PORT_ID" ; rr:datatype xsd:integer ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:port_portName ;
  rr:objectMap [ rr:column "PORT_NAME" ; rr:datatype xsd:string ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:port_portCode ;
  rr:objectMap [ rr:column "PORT_CODE" ; rr:datatype xsd:string ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:port_countryId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Country ;
    rr:joinCondition [ rr:child "COUNTRY_ID" ; rr:parent "COUNTRY_ID" ] ;
] ;
] ;

```

```

];
rr:predicateObjectMap [
  rr:predicate sc:port_portType ;
  rr:objectMap [ rr:column "PORT_TYPE" ; rr:datatype xsd:string ] ;
] .

sc:TriplesMap_Supplier
rr:logicalTable [ rr:tableName "SCS.INVENTORY.DIM_SUPPLIER" ] ;
rr:subjectMap [
  rr:template "https://asterion.example.com/ontology/supplychain/supplier/{SUPPLIER_ID}" ;
  rr:class sc:Supplier ;
] ;
coa:datasourceId "ds-scs-snowflake" ;
coa:sourceSchema "SCS.INVENTORY" ;
rr:predicateObjectMap [
  rr:predicate sc:supplier_supplierId ;
  rr:objectMap [ rr:column "SUPPLIER_ID" ; rr:datatype xsd:integer ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:supplier_supplierName ;
  rr:objectMap [ rr:column "SUPPLIER_NAME" ; rr:datatype xsd:string ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:supplier_supplierCode ;
  rr:objectMap [ rr:column "SUPPLIER_CODE" ; rr:datatype xsd:string ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:supplier_countryId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Country ;
    rr:joinCondition [ rr:child "COUNTRY_ID" ; rr:parent "COUNTRY_ID" ] ;
  ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:supplier_commodityId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Commodity ;
    rr:joinCondition [ rr:child "COMMODITY_ID" ; rr:parent "COMMODITY_ID" ] ;
  ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:supplier_reliabilityRating ;
  rr:objectMap [ rr:column "RELIABILITY_RATING" ; rr:datatype xsd:decimal ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:supplier_isProblem ;
  rr:objectMap [ rr:column "IS_PROBLEM" ; rr:datatype xsd:boolean ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:supplier_active ;
  rr:objectMap [ rr:column "ACTIVE" ; rr:datatype xsd:boolean ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:supplier_isSynthetic ;
  rr:objectMap [ rr:column "IS_SYNTHETIC" ; rr:datatype xsd:boolean ] ;
] .

sc:TriplesMap_SupplierSite
rr:logicalTable [ rr:tableName "SCS.INVENTORY.DIM_SUPPLIER_SITE" ] ;
rr:subjectMap [
  rr:template "https://asterion.example.com/ontology/supplychain/suppliersite/{SITE_ID}" ;
  rr:class sc:SupplierSite ;
] ;
coa:datasourceId "ds-scs-snowflake" ;
coa:sourceSchema "SCS.INVENTORY" ;
rr:predicateObjectMap [
  rr:predicate sc:suppliersite_siteId ;
  rr:objectMap [ rr:column "SITE_ID" ; rr:datatype xsd:integer ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:suppliersite_supplierId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Supplier ;
    rr:joinCondition [ rr:child "SUPPLIER_ID" ; rr:parent "SUPPLIER_ID" ] ;
  ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:suppliersite_siteName ;
  rr:objectMap [ rr:column "SITE_NAME" ; rr:datatype xsd:string ] ;
] ;

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rr:predicateObjectMap [
  rr:predicate sc:suppliersite_city ;
  rr:objectMap [ rr:column "CITY" ; rr:datatype xsd:string ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:suppliersite_countryId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Country ;
    rr:joinCondition [ rr:child "COUNTRY_ID" ; rr:parent "COUNTRY_ID" ] ;
  ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:suppliersite_leadTimeDays ;
  rr:objectMap [ rr:column "LEAD_TIME_DAYS" ; rr:datatype xsd:integer ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:suppliersite_isPrimary ;
  rr:objectMap [ rr:column "IS_PRIMARY" ; rr:datatype xsd:boolean ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:suppliersite_active ;
  rr:objectMap [ rr:column "ACTIVE" ; rr:datatype xsd:boolean ] ;
] .

sc:TriplesMap_Manufacturer
rr:logicalTable [ rr:tableName "SCS.INVENTORY.DIM_MANUFACTURER" ] ;
rr:subjectMap [
  rr:template
    "https://asterion.example.com/ontology/supplychain/manufacturer/{MANUFACTURER_ID}" ;
  rr:class sc:Manufacturer ;
] ;
coa:datasourceId "ds-scs-snowflake" ;
coa:sourceSchema "SCS.INVENTORY" ;
rr:predicateObjectMap [
  rr:predicate sc:manufacturer_manufacturerId ;
  rr:objectMap [ rr:column "MANUFACTURER_ID" ; rr:datatype xsd:integer ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:manufacturer_manufacturerName ;
  rr:objectMap [ rr:column "MANUFACTURER_NAME" ; rr:datatype xsd:string ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:manufacturer_countryId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Country ;
    rr:joinCondition [ rr:child "COUNTRY_ID" ; rr:parent "COUNTRY_ID" ] ;
  ] ;
] .

sc:TriplesMap_Part
rr:logicalTable [ rr:tableName "SCS.INVENTORY.DIM_PART" ] ;
rr:subjectMap [
  rr:template "https://asterion.example.com/ontology/supplychain/part/{PART_ID}" ;
  rr:class sc:Part ;
] ;
coa:datasourceId "ds-scs-snowflake" ;
coa:sourceSchema "SCS.INVENTORY" ;
rr:predicateObjectMap [
  rr:predicate sc:part_partId ;
  rr:objectMap [ rr:column "PART_ID" ; rr:datatype xsd:integer ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:part_partNumber ;
  rr:objectMap [ rr:column "PART_NUMBER" ; rr:datatype xsd:string ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:part_partName ;
  rr:objectMap [ rr:column "PART_NAME" ; rr:datatype xsd:string ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:part_commodityId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Commodity ;
    rr:joinCondition [ rr:child "COMMODITY_ID" ; rr:parent "COMMODITY_ID" ] ;
  ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:part_subCommodityId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_SubCommodity ;

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        rr:joinCondition [ rr:child "SUB_COMMODITY_ID" ; rr:parent "SUB_COMMODITY_ID" ] ;
    ] ;
rr:predicateObjectMap [
    rr:predicate sc:part_uomCode ;
    rr:objectMap [
        rr:parentTriplesMap sc:TriplesMap_UnitOfMeasure ;
        rr:joinCondition [ rr:child "UOM_CODE" ; rr:parent "UOM_CODE" ] ;
    ] ;
] ;
rr:predicateObjectMap [
    rr:predicate sc:part_unitCostUsd ;
    rr:objectMap [ rr:column "UNIT_COST_USD" ; rr:datatype xsd:decimal ] ;
] ;
rr:predicateObjectMap [
    rr:predicate sc:part_weightKg ;
    rr:objectMap [ rr:column "WEIGHT_KG" ; rr:datatype xsd:decimal ] ;
] ;
rr:predicateObjectMap [
    rr:predicate sc:part_lifecycleStatus ;
    rr:objectMap [ rr:column "LIFECYCLE_STATUS" ; rr:datatype xsd:string ] ;
] ;
rr:predicateObjectMap [
    rr:predicate sc:part_isSynthetic ;
    rr:objectMap [ rr:column "IS_SYNTHETIC" ; rr:datatype xsd:boolean ] ;
] .

sc:TriplesMap_ProductFamily
rr:logicalTable [ rr:tableName "SCS.INVENTORY.DIM_PRODUCT_FAMILY" ] ;
rr:subjectMap [
    rr:template
        "https://asterion.example.com/ontology/supplychain/productfamily/{PRODUCT_FAMILY_ID}" ;
    rr:class sc:ProductFamily ;
] ;
coa:datasourceId "ds-scs-snowflake" ;
coa:sourceSchema "SCS.INVENTORY" ;
rr:predicateObjectMap [
    rr:predicate sc:productfamily_productFamilyId ;
    rr:objectMap [ rr:column "PRODUCT_FAMILY_ID" ; rr:datatype xsd:integer ] ;
] ;
rr:predicateObjectMap [
    rr:predicate sc:productfamily_productFamilyName ;
    rr:objectMap [ rr:column "PRODUCT_FAMILY_NAME" ; rr:datatype xsd:string ] ;
] .

sc:TriplesMap_Product
rr:logicalTable [ rr:tableName "SCS.INVENTORY.DIM_PRODUCT" ] ;
rr:subjectMap [
    rr:template "https://asterion.example.com/ontology/supplychain/product/{PRODUCT_ID}" ;
    rr:class sc:Product ;
] ;
coa:datasourceId "ds-scs-snowflake" ;
coa:sourceSchema "SCS.INVENTORY" ;
rr:predicateObjectMap [
    rr:predicate sc:product_productId ;
    rr:objectMap [ rr:column "PRODUCT_ID" ; rr:datatype xsd:integer ] ;
] ;
rr:predicateObjectMap [
    rr:predicate sc:product_productName ;
    rr:objectMap [ rr:column "PRODUCT_NAME" ; rr:datatype xsd:string ] ;
] ;
rr:predicateObjectMap [
    rr:predicate sc:product_productFamilyId ;
    rr:objectMap [
        rr:parentTriplesMap sc:TriplesMap_ProductFamily ;
        rr:joinCondition [ rr:child "PRODUCT_FAMILY_ID" ; rr:parent "PRODUCT_FAMILY_ID" ] ;
    ] ;
] ;
rr:predicateObjectMap [
    rr:predicate sc:product_unitPriceUsd ;
    rr:objectMap [ rr:column "UNIT_PRICE_USD" ; rr:datatype xsd:decimal ] ;
] ;
rr:predicateObjectMap [
    rr:predicate sc:product_isSynthetic ;
    rr:objectMap [ rr:column "IS_SYNTHETIC" ; rr:datatype xsd:boolean ] ;
] .

sc:TriplesMap_BillOfMaterials
rr:logicalTable [ rr:tableName "SCS.INVENTORY.BOM_HEADER" ] ;
rr:subjectMap [

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        rr:template "https://asterion.example.com/ontology/supplychain/billofmaterials/{BOM_ID}" ;
        rr:class sc:BillofMaterials ;
    ] ;
    coa:datasourceId "ds-scs-snowflake" ;
    coa:sourceSchema "SCS.INVENTORY" ;
    rr:predicateObjectMap [
        rr:predicate sc:billofmaterials_bomId ;
        rr:objectMap [ rr:column "BOM_ID" ; rr:datatype xsd:integer ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:billofmaterials_productId ;
        rr:objectMap [
            rr:parentTriplesMap sc:TriplesMap_Product ;
            rr:joinCondition [ rr:child "PRODUCT_ID" ; rr:parent "PRODUCT_ID" ] ;
        ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:billofmaterials_bomVersion ;
        rr:objectMap [ rr:column "BOM_VERSION" ; rr:datatype xsd:string ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:billofmaterials_effectiveDate ;
        rr:objectMap [ rr:column "EFFECTIVE_DATE" ; rr:datatype xsd:date ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:billofmaterials_status ;
        rr:objectMap [ rr:column "STATUS" ; rr:datatype xsd:string ] ;
    ] .

sc:TriplesMap_BillofMaterialsLine
    rr:logicalTable [ rr:tableName "SCS.INVENTORY.BOM_LINE" ] ;
    rr:subjectMap [
        rr:template
            "https://asterion.example.com/ontology/supplychain/billofmaterialsline/{BOM_LINE_ID}" ;
        rr:class sc:BillofMaterialsLine ;
    ] ;
    coa:datasourceId "ds-scs-snowflake" ;
    coa:sourceSchema "SCS.INVENTORY" ;
    rr:predicateObjectMap [
        rr:predicate sc:billofmaterialsline_bomLineId ;
        rr:objectMap [ rr:column "BOM_LINE_ID" ; rr:datatype xsd:integer ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:billofmaterialsline_bomId ;
        rr:objectMap [
            rr:parentTriplesMap sc:TriplesMap_BillofMaterials ;
            rr:joinCondition [ rr:child "BOM_ID" ; rr:parent "BOM_ID" ] ;
        ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:billofmaterialsline_partId ;
        rr:objectMap [
            rr:parentTriplesMap sc:TriplesMap_Part ;
            rr:joinCondition [ rr:child "PART_ID" ; rr:parent "PART_ID" ] ;
        ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:billofmaterialsline_quantityPer ;
        rr:objectMap [ rr:column "QUANTITY_PER" ; rr:datatype xsd:decimal ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:billofmaterialsline_findNumber ;
        rr:objectMap [ rr:column "FIND_NUMBER" ; rr:datatype xsd:integer ] ;
    ] .

sc:TriplesMap_Plant
    rr:logicalTable [ rr:tableName "SCS.INVENTORY.DIM_PLANT" ] ;
    rr:subjectMap [
        rr:template "https://asterion.example.com/ontology/supplychain/plant/{PLANT_ID}" ;
        rr:class sc:Plant ;
    ] ;
    coa:datasourceId "ds-scs-snowflake" ;
    coa:sourceSchema "SCS.INVENTORY" ;
    rr:predicateObjectMap [
        rr:predicate sc:plant_plantId ;
        rr:objectMap [ rr:column "PLANT_ID" ; rr:datatype xsd:integer ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:plant_plantCode ;
        rr:objectMap [ rr:column "PLANT_CODE" ; rr:datatype xsd:string ] ;
    ] ;

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];
rr:predicateObjectMap [
  rr:predicate sc:plant_plantName ;
  rr:objectMap [ rr:column "PLANT_NAME" ; rr:datatype xsd:string ] ;
];
rr:predicateObjectMap [
  rr:predicate sc:plant_city ;
  rr:objectMap [ rr:column "CITY" ; rr:datatype xsd:string ] ;
];
rr:predicateObjectMap [
  rr:predicate sc:plant_countryId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Country ;
    rr:joinCondition [ rr:child "COUNTRY_ID" ; rr:parent "COUNTRY_ID" ] ;
  ] ;
] .

sc:TriplesMap_Warehouse
rr:logicalTable [ rr:tableName "SCS.INVENTORY.DIM_WAREHOUSE" ] ;
rr:subjectMap [
  rr:template "https://asterion.example.com/ontology/supplychain/warehouse/{WAREHOUSE_ID}" ;
  rr:class sc:Warehouse ;
];
coa:datasourceId "ds-scs-snowflake" ;
coa:sourceSchema "SCS.INVENTORY" ;
rr:predicateObjectMap [
  rr:predicate sc:warehouse_warehouseId ;
  rr:objectMap [ rr:column "WAREHOUSE_ID" ; rr:datatype xsd:integer ] ;
];
rr:predicateObjectMap [
  rr:predicate sc:warehouse_warehouseCode ;
  rr:objectMap [ rr:column "WAREHOUSE_CODE" ; rr:datatype xsd:string ] ;
];
rr:predicateObjectMap [
  rr:predicate sc:warehouse_warehouseName ;
  rr:objectMap [ rr:column "WAREHOUSE_NAME" ; rr:datatype xsd:string ] ;
];
rr:predicateObjectMap [
  rr:predicate sc:warehouse_plantId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Plant ;
    rr:joinCondition [ rr:child "PLANT_ID" ; rr:parent "PLANT_ID" ] ;
  ] ;
];
rr:predicateObjectMap [
  rr:predicate sc:warehouse_city ;
  rr:objectMap [ rr:column "CITY" ; rr:datatype xsd:string ] ;
];
rr:predicateObjectMap [
  rr:predicate sc:warehouse_countryId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Country ;
    rr:joinCondition [ rr:child "COUNTRY_ID" ; rr:parent "COUNTRY_ID" ] ;
  ] ;
];
rr:predicateObjectMap [
  rr:predicate sc:warehouse_capacityUnits ;
  rr:objectMap [ rr:column "CAPACITY_UNITS" ; rr:datatype xsd:integer ] ;
] .

sc:TriplesMap_Customer
rr:logicalTable [ rr:tableName "SCS.INVENTORY.DIM_CUSTOMER" ] ;
rr:subjectMap [
  rr:template "https://asterion.example.com/ontology/supplychain/customer/{CUSTOMER_ID}" ;
  rr:class sc:Customer ;
];
coa:datasourceId "ds-scs-snowflake" ;
coa:sourceSchema "SCS.INVENTORY" ;
rr:predicateObjectMap [
  rr:predicate sc:customer_customerId ;
  rr:objectMap [ rr:column "CUSTOMER_ID" ; rr:datatype xsd:integer ] ;
];
rr:predicateObjectMap [
  rr:predicate sc:customer_customerName ;
  rr:objectMap [ rr:column "CUSTOMER_NAME" ; rr:datatype xsd:string ] ;
];
rr:predicateObjectMap [
  rr:predicate sc:customer_customerCode ;
  rr:objectMap [ rr:column "CUSTOMER_CODE" ; rr:datatype xsd:string ] ;
];

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rr:predicateObjectMap [
  rr:predicate sc:customer_countryId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Country ;
    rr:joinCondition [ rr:child "COUNTRY_ID" ; rr:parent "COUNTRY_ID" ] ;
  ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:customer_customerTier ;
  rr:objectMap [ rr:column "CUSTOMER_TIER" ; rr:datatype xsd:string ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:customer_isSynthetic ;
  rr:objectMap [ rr:column "IS_SYNTHETIC" ; rr:datatype xsd:boolean ] ;
] .

sc:TriplesMap_Carrier
rr:logicalTable [ rr:tableName "SCS.INVENTORY.DIM_CARRIER" ] ;
rr:subjectMap [
  rr:template "https://asterion.example.com/ontology/supplychain/carrier/{CARRIER_ID}" ;
  rr:class sc:Carrier ;
] ;
coa:datasourceId "ds-scs-snowflake" ;
coa:sourceSchema "SCS.INVENTORY" ;
rr:predicateObjectMap [
  rr:predicate sc:carrier_carrierId ;
  rr:objectMap [ rr:column "CARRIER_ID" ; rr:datatype xsd:integer ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:carrier_carrierName ;
  rr:objectMap [ rr:column "CARRIER_NAME" ; rr:datatype xsd:string ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:carrier_carrierMode ;
  rr:objectMap [ rr:column "CARRIER_MODE" ; rr:datatype xsd:string ] ;
] .

sc:TriplesMap_Lane
rr:logicalTable [ rr:tableName "SCS.INVENTORY.DIM_LANE" ] ;
rr:subjectMap [
  rr:template "https://asterion.example.com/ontology/supplychain/lane/{LANE_ID}" ;
  rr:class sc:Lane ;
] ;
coa:datasourceId "ds-scs-snowflake" ;
coa:sourceSchema "SCS.INVENTORY" ;
rr:predicateObjectMap [
  rr:predicate sc:lane_laneId ;
  rr:objectMap [ rr:column "LANE_ID" ; rr:datatype xsd:integer ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:lane_originPortId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Port ;
    rr:joinCondition [ rr:child "ORIGIN_PORT_ID" ; rr:parent "PORT_ID" ] ;
  ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:lane_destPortId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Port ;
    rr:joinCondition [ rr:child "DEST_PORT_ID" ; rr:parent "PORT_ID" ] ;
  ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:lane_transportMode ;
  rr:objectMap [ rr:column "TRANSPORT_MODE" ; rr:datatype xsd:string ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:lane_transitDaysAvg ;
  rr:objectMap [ rr:column "TRANSIT_DAYS_AVG" ; rr:datatype xsd:integer ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:lane_isProblemLane ;
  rr:objectMap [ rr:column "IS_PROBLEM_LANE" ; rr:datatype xsd:boolean ] ;
] .

sc:TriplesMap_Contract
rr:logicalTable [ rr:tableName "SCS.INVENTORY.DIM_CONTRACT" ] ;
rr:subjectMap [
  rr:template "https://asterion.example.com/ontology/supplychain/contract/{CONTRACT_ID}" ;

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        rr:class sc:Contract ;
    ] ;
    coa:datasourceId "ds-scs-snowflake" ;
    coa:sourceSchema "SCS.INVENTORY" ;
    rr:predicateObjectMap [
        rr:predicate sc:contract_contractId ;
        rr:objectMap [ rr:column "CONTRACT_ID" ; rr:datatype xsd:integer ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:contract_supplierId ;
        rr:objectMap [
            rr:parentTriplesMap sc:TriplesMap_Supplier ;
            rr:joinCondition [ rr:child "SUPPLIER_ID" ; rr:parent "SUPPLIER_ID" ] ;
        ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:contract_contractCode ;
        rr:objectMap [ rr:column "CONTRACT_CODE" ; rr:datatype xsd:string ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:contract_startDate ;
        rr:objectMap [ rr:column "START_DATE" ; rr:datatype xsd:date ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:contract_endDate ;
        rr:objectMap [ rr:column "END_DATE" ; rr:datatype xsd:date ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:contract_currencyCode ;
        rr:objectMap [
            rr:parentTriplesMap sc:TriplesMap_Currency ;
            rr:joinCondition [ rr:child "CURRENCY_CODE" ; rr:parent "CURRENCY_CODE" ] ;
        ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:contract_incotermCode ;
        rr:objectMap [
            rr:parentTriplesMap sc:TriplesMap_Incoterm ;
            rr:joinCondition [ rr:child "INCOTERM_CODE" ; rr:parent "INCOTERM_CODE" ] ;
        ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:contract_status ;
        rr:objectMap [ rr:column "STATUS" ; rr:datatype xsd:string ] ;
    ] .

sc:TriplesMap_SupplierPartApproval
    rr:logicalTable [ rr:tableName "SCS.INVENTORY.SUPPLIER_PART_APPROVAL" ] ;
    rr:subjectMap [
        rr:template
            "https://asterion.example.com/ontology/supplychain/supplierpartapproval/{APPROVAL_ID}" ;
        rr:class sc:SupplierPartApproval ;
    ] ;
    coa:datasourceId "ds-scs-snowflake" ;
    coa:sourceSchema "SCS.INVENTORY" ;
    rr:predicateObjectMap [
        rr:predicate sc:supplierpartapproval_approvalId ;
        rr:objectMap [ rr:column "APPROVAL_ID" ; rr:datatype xsd:integer ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:supplierpartapproval_supplierId ;
        rr:objectMap [
            rr:parentTriplesMap sc:TriplesMap_Supplier ;
            rr:joinCondition [ rr:child "SUPPLIER_ID" ; rr:parent "SUPPLIER_ID" ] ;
        ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:supplierpartapproval_siteId ;
        rr:objectMap [
            rr:parentTriplesMap sc:TriplesMap_SupplierSite ;
            rr:joinCondition [ rr:child "SITE_ID" ; rr:parent "SITE_ID" ] ;
        ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:supplierpartapproval_partId ;
        rr:objectMap [
            rr:parentTriplesMap sc:TriplesMap_Part ;
            rr:joinCondition [ rr:child "PART_ID" ; rr:parent "PART_ID" ] ;
        ] ;
    ] ;

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rr:predicateObjectMap [
  rr:predicate sc:supplierpartapproval_approvalDate ;
  rr:objectMap [ rr:column "APPROVAL_DATE" ; rr:datatype xsd:date ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:supplierpartapproval_status ;
  rr:objectMap [ rr:column "STATUS" ; rr:datatype xsd:string ] ;
] .

sc:TriplesMap_PurchaseOrder
rr:logicalTable [ rr:tableName "SCS.INVENTORY.PO_HEADER" ] ;
rr:subjectMap [
  rr:template "https://asterion.example.com/ontology/supplychain/purchaseorder/{PO_ID}" ;
  rr:class sc:PurchaseOrder ;
] ;
coa:datasourceId "ds-scs-snowflake" ;
coa:sourceSchema "SCS.INVENTORY" ;
rr:predicateObjectMap [
  rr:predicate sc:purchaseorder_poId ;
  rr:objectMap [ rr:column "PO_ID" ; rr:datatype xsd:integer ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:purchaseorder_poNumber ;
  rr:objectMap [ rr:column "PO_NUMBER" ; rr:datatype xsd:string ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:purchaseorder_supplierId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Supplier ;
    rr:joinCondition [ rr:child "SUPPLIER_ID" ; rr:parent "SUPPLIER_ID" ] ;
  ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:purchaseorder_plantId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Plant ;
    rr:joinCondition [ rr:child "PLANT_ID" ; rr:parent "PLANT_ID" ] ;
  ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:purchaseorder_orderDate ;
  rr:objectMap [ rr:column "ORDER_DATE" ; rr:datatype xsd:date ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:purchaseorder_currencyCode ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Currency ;
    rr:joinCondition [ rr:child "CURRENCY_CODE" ; rr:parent "CURRENCY_CODE" ] ;
  ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:purchaseorder_status ;
  rr:objectMap [ rr:column "STATUS" ; rr:datatype xsd:string ] ;
] .

sc:TriplesMap_PurchaseOrderLine
rr:logicalTable [ rr:tableName "SCS.INVENTORY.PO_LINE" ] ;
rr:subjectMap [
  rr:template
    "https://asterion.example.com/ontology/supplychain/purchaseorderline/{PO_LINE_ID}" ;
  rr:class sc:PurchaseOrderLine ;
] ;
coa:datasourceId "ds-scs-snowflake" ;
coa:sourceSchema "SCS.INVENTORY" ;
rr:predicateObjectMap [
  rr:predicate sc:purchaseorderline_poLineId ;
  rr:objectMap [ rr:column "PO_LINE_ID" ; rr:datatype xsd:integer ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:purchaseorderline_poId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_PurchaseOrder ;
    rr:joinCondition [ rr:child "PO_ID" ; rr:parent "PO_ID" ] ;
  ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:purchaseorderline_partId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Part ;
    rr:joinCondition [ rr:child "PART_ID" ; rr:parent "PART_ID" ] ;
  ] ;
] ;

```

```

    ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:purchaseorderline_quantity ;
    rr:objectMap [ rr:column "QUANTITY" ; rr:datatype xsd:decimal ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:purchaseorderline_unitPrice ;
    rr:objectMap [ rr:column "UNIT_PRICE" ; rr:datatype xsd:decimal ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:purchaseorderline_promiseDate ;
    rr:objectMap [ rr:column "PROMISE_DATE" ; rr:datatype xsd:date ] ;
  ] .

sc:TriplesMap_GoodsReceipt
  rr:logicalTable [ rr:tableName "SCS.INVENTORY.GOODS_RECEIPT" ] ;
  rr:subjectMap [
    rr:template "https://asterion.example.com/ontology/supplychain/goodsreceipt/{RECEIPT_ID}" ;
    rr:class sc:GoodsReceipt ;
  ] ;
  coa:datasourceId "ds-scs-snowflake" ;
  coa:sourceSchema "SCS.INVENTORY" ;
  rr:predicateObjectMap [
    rr:predicate sc:goodsreceipt_receiptId ;
    rr:objectMap [ rr:column "RECEIPT_ID" ; rr:datatype xsd:integer ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:goodsreceipt_poLineId ;
    rr:objectMap [
      rr:parentTriplesMap sc:TriplesMap_PurchaseOrderLine ;
      rr:joinCondition [ rr:child "PO_LINE_ID" ; rr:parent "PO_LINE_ID" ] ;
    ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:goodsreceipt_receiptDate ;
    rr:objectMap [ rr:column "RECEIPT_DATE" ; rr:datatype xsd:date ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:goodsreceipt_quantityReceived ;
    rr:objectMap [ rr:column "QUANTITY_RECEIVED" ; rr:datatype xsd:decimal ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:goodsreceipt_warehouseId ;
    rr:objectMap [
      rr:parentTriplesMap sc:TriplesMap_Warehouse ;
      rr:joinCondition [ rr:child "WAREHOUSE_ID" ; rr:parent "WAREHOUSE_ID" ] ;
    ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:goodsreceipt_daysVariance ;
    rr:objectMap [ rr:column "DAYS_VARIANCE" ; rr:datatype xsd:integer ] ;
  ] .

sc:TriplesMap_SalesOrder
  rr:logicalTable [ rr:tableName "SCS.INVENTORY.SALES_ORDER_HEADER" ] ;
  rr:subjectMap [
    rr:template "https://asterion.example.com/ontology/supplychain/salesorder/{ORDER_ID}" ;
    rr:class sc:SalesOrder ;
  ] ;
  coa:datasourceId "ds-scs-snowflake" ;
  coa:sourceSchema "SCS.INVENTORY" ;
  rr:predicateObjectMap [
    rr:predicate sc:salesorder_orderId ;
    rr:objectMap [ rr:column "ORDER_ID" ; rr:datatype xsd:integer ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:salesorder_orderNumber ;
    rr:objectMap [ rr:column "ORDER_NUMBER" ; rr:datatype xsd:string ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:salesorder_customerId ;
    rr:objectMap [
      rr:parentTriplesMap sc:TriplesMap_Customer ;
      rr:joinCondition [ rr:child "CUSTOMER_ID" ; rr:parent "CUSTOMER_ID" ] ;
    ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:salesorder_orderDate ;
    rr:objectMap [ rr:column "ORDER_DATE" ; rr:datatype xsd:date ] ;
  ] ;

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] ;
rr:predicateObjectMap [
  rr:predicate sc:salesorder_plantId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Plant ;
    rr:joinCondition [ rr:child "PLANT_ID" ; rr:parent "PLANT_ID" ] ;
  ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:salesorder_status ;
  rr:objectMap [ rr:column "STATUS" ; rr:datatype xsd:string ] ;
] .

sc:TriplesMap_SalesOrderLine
rr:logicalTable [ rr:tableName "SCS.INVENTORY.SALES_ORDER_LINE" ] ;
rr:subjectMap [
  rr:template
    "https://asterion.example.com/ontology/supplychain/salesorderline/{ORDER_LINE_ID}" ;
  rr:class sc:SalesOrderLine ;
] ;
coa:datasourceId "ds-scs-snowflake" ;
coa:sourceSchema "SCS.INVENTORY" ;
rr:predicateObjectMap [
  rr:predicate sc:salesorderline_orderLineId ;
  rr:objectMap [ rr:column "ORDER_LINE_ID" ; rr:datatype xsd:integer ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:salesorderline_orderId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_SalesOrder ;
    rr:joinCondition [ rr:child "ORDER_ID" ; rr:parent "ORDER_ID" ] ;
  ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:salesorderline_productId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Product ;
    rr:joinCondition [ rr:child "PRODUCT_ID" ; rr:parent "PRODUCT_ID" ] ;
  ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:salesorderline_quantity ;
  rr:objectMap [ rr:column "QUANTITY" ; rr:datatype xsd:decimal ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:salesorderline_unitPrice ;
  rr:objectMap [ rr:column "UNIT_PRICE" ; rr:datatype xsd:decimal ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:salesorderline_requestedDate ;
  rr:objectMap [ rr:column "REQUESTED_DATE" ; rr:datatype xsd:date ] ;
] .

sc:TriplesMap_Shipment
rr:logicalTable [ rr:tableName "SCS.INVENTORY.SHIPMENT_HEADER" ] ;
rr:subjectMap [
  rr:template "https://asterion.example.com/ontology/supplychain/shipment/{SHIPMENT_ID}" ;
  rr:class sc:Shipment ;
] ;
coa:datasourceId "ds-scs-snowflake" ;
coa:sourceSchema "SCS.INVENTORY" ;
rr:predicateObjectMap [
  rr:predicate sc:shipment_shipmentId ;
  rr:objectMap [ rr:column "SHIPMENT_ID" ; rr:datatype xsd:integer ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:shipment_carrierId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Carrier ;
    rr:joinCondition [ rr:child "CARRIER_ID" ; rr:parent "CARRIER_ID" ] ;
  ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:shipment_laneId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Lane ;
    rr:joinCondition [ rr:child "LANE_ID" ; rr:parent "LANE_ID" ] ;
  ] ;
] ;
rr:predicateObjectMap [

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        rr:predicate sc:shipment_shipDate ;
        rr:objectMap [ rr:column "SHIP_DATE" ; rr:datatype xsd:date ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:shipment_deliveryDate ;
        rr:objectMap [ rr:column "DELIVERY_DATE" ; rr:datatype xsd:date ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:shipment_status ;
        rr:objectMap [ rr:column "STATUS" ; rr:datatype xsd:string ] ;
    ] .

sc:TriplesMap_ShipmentLine
    rr:logicalTable [ rr:tableName "SCS.INVENTORY.SHIPMENT_LINE" ] ;
    rr:subjectMap [
        rr:template
            "https://asterion.example.com/ontology/supplychain/shipmentline/{SHIPMENT_LINE_ID}" ;
        rr:class sc:ShipmentLine ;
    ] ;
    coa:datasourceId "ds-scs-snowflake" ;
    coa:sourceSchema "SCS.INVENTORY" ;
    rr:predicateObjectMap [
        rr:predicate sc:shipmentline_shipmentLineId ;
        rr:objectMap [ rr:column "SHIPMENT_LINE_ID" ; rr:datatype xsd:integer ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:shipmentline_shipmentId ;
        rr:objectMap [
            rr:parentTriplesMap sc:TriplesMap_Shipment ;
            rr:joinCondition [ rr:child "SHIPMENT_ID" ; rr:parent "SHIPMENT_ID" ] ;
        ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:shipmentline_orderLineId ;
        rr:objectMap [
            rr:parentTriplesMap sc:TriplesMap_SalesOrderLine ;
            rr:joinCondition [ rr:child "ORDER_LINE_ID" ; rr:parent "ORDER_LINE_ID" ] ;
        ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:shipmentline_poLineId ;
        rr:objectMap [
            rr:parentTriplesMap sc:TriplesMap_PurchaseOrderLine ;
            rr:joinCondition [ rr:child "PO_LINE_ID" ; rr:parent "PO_LINE_ID" ] ;
        ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:shipmentline_quantity ;
        rr:objectMap [ rr:column "QUANTITY" ; rr:datatype xsd:decimal ] ;
    ] .

sc:TriplesMap_ShipmentMilestoneEvent
    rr:logicalTable [ rr:tableName "SCS.INVENTORY.SHIPMENT_MILESTONE_EVENT" ] ;
    rr:subjectMap [
        rr:template
            "https://asterion.example.com/ontology/supplychain/shipmentmilestoneevent/{EVENT_ID}" ;
        rr:class sc:ShipmentMilestoneEvent ;
    ] ;
    coa:datasourceId "ds-scs-snowflake" ;
    coa:sourceSchema "SCS.INVENTORY" ;
    rr:predicateObjectMap [
        rr:predicate sc:shipmentmilestoneevent_eventId ;
        rr:objectMap [ rr:column "EVENT_ID" ; rr:datatype xsd:integer ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:shipmentmilestoneevent_shipmentId ;
        rr:objectMap [
            rr:parentTriplesMap sc:TriplesMap_Shipment ;
            rr:joinCondition [ rr:child "SHIPMENT_ID" ; rr:parent "SHIPMENT_ID" ] ;
        ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:shipmentmilestoneevent_milestone ;
        rr:objectMap [ rr:column "MILESTONE" ; rr:datatype xsd:string ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:shipmentmilestoneevent_eventTs ;
        rr:objectMap [ rr:column "EVENT_TS" ; rr:datatype xsd:dateTime ] ;
    ] ;
    rr:predicateObjectMap [

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        rr:predicate sc:shipmentmilestoneevent_location ;
        rr:objectMap [ rr:column "LOCATION" ; rr:datatype xsd:string ] ;
    ] .

sc:TriplesMap_SensorReading
    rr:logicalTable [ rr:tableName "SCS.INVENTORY.IOT_SENSOR_READING" ] ;
    rr:subjectMap [
        rr:template "https://asterion.example.com/ontology/supplychain/sensorreading/{READING_ID}" ;
        rr:class sc:SensorReading ;
    ] ;
    coa:datasourceId "ds-scs-snowflake" ;
    coa:sourceSchema "SCS.INVENTORY" ;
    rr:predicateObjectMap [
        rr:predicate sc:sensorreading_readingId ;
        rr:objectMap [ rr:column "READING_ID" ; rr:datatype xsd:integer ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:sensorreading_shipmentId ;
        rr:objectMap [
            rr:parentTriplesMap sc:TriplesMap_Shipment ;
            rr:joinCondition [ rr:child "SHIPMENT_ID" ; rr:parent "SHIPMENT_ID" ] ;
        ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:sensorreading_readingTs ;
        rr:objectMap [ rr:column "READING_TS" ; rr:datatype xsd:dateTime ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:sensorreading_temperatureC ;
        rr:objectMap [ rr:column "TEMPERATURE_C" ; rr:datatype xsd:decimal ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:sensorreading_humidityPct ;
        rr:objectMap [ rr:column "HUMIDITY_PCT" ; rr:datatype xsd:decimal ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:sensorreading_shockG ;
        rr:objectMap [ rr:column "SHOCK_G" ; rr:datatype xsd:decimal ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:sensorreading_tiltDeg ;
        rr:objectMap [ rr:column "TILT_DEG" ; rr:datatype xsd:decimal ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:sensorreading_gpsLat ;
        rr:objectMap [ rr:column "GPS_LAT" ; rr:datatype xsd:decimal ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:sensorreading_gpsLon ;
        rr:objectMap [ rr:column "GPS_LON" ; rr:datatype xsd:decimal ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:sensorreading_breachFlag ;
        rr:objectMap [ rr:column "BREACH_FLAG" ; rr:datatype xsd:boolean ] ;
    ] .

sc:TriplesMap_WorkOrder
    rr:logicalTable [ rr:tableName "SCS.INVENTORY.WORK_ORDER" ] ;
    rr:subjectMap [
        rr:template "https://asterion.example.com/ontology/supplychain/workorder/{WORK_ORDER_ID}" ;
        rr:class sc:WorkOrder ;
    ] ;
    coa:datasourceId "ds-scs-snowflake" ;
    coa:sourceSchema "SCS.INVENTORY" ;
    rr:predicateObjectMap [
        rr:predicate sc:workorder_workOrderId ;
        rr:objectMap [ rr:column "WORK_ORDER_ID" ; rr:datatype xsd:integer ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:workorder_productId ;
        rr:objectMap [
            rr:parentTriplesMap sc:TriplesMap_Product ;
            rr:joinCondition [ rr:child "PRODUCT_ID" ; rr:parent "PRODUCT_ID" ] ;
        ] ;
    ] ;
    rr:predicateObjectMap [
        rr:predicate sc:workorder_plantId ;
        rr:objectMap [
            rr:parentTriplesMap sc:TriplesMap_Plant ;
            rr:joinCondition [ rr:child "PLANT_ID" ; rr:parent "PLANT_ID" ] ;
        ] ;
    ] ;

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    ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:workorder_startDate ;
    rr:objectMap [ rr:column "START_DATE" ; rr:datatype xsd:date ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:workorder_endDate ;
    rr:objectMap [ rr:column "END_DATE" ; rr:datatype xsd:date ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:workorder_quantity ;
    rr:objectMap [ rr:column "QUANTITY" ; rr:datatype xsd:decimal ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:workorder_status ;
    rr:objectMap [ rr:column "STATUS" ; rr:datatype xsd:string ] ;
  ] .

sc:TriplesMap_QualityInspection
  rr:logicalTable [ rr:tableName "SCS.INVENTORY.QUALITY_INSPECTION" ] ;
  rr:subjectMap [
    rr:template
      "https://asterion.example.com/ontology/supplychain/qualityinspection/{INSPECTION_ID}" ;
    rr:class sc:QualityInspection ;
  ] ;
  coa:datasourceId "ds-scs-snowflake" ;
  coa:sourceSchema "SCS.INVENTORY" ;
  rr:predicateObjectMap [
    rr:predicate sc:qualityinspection_inspectionId ;
    rr:objectMap [ rr:column "INSPECTION_ID" ; rr:datatype xsd:integer ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:qualityinspection_receiptId ;
    rr:objectMap [
      rr:parentTriplesMap sc:TriplesMap_GoodsReceipt ;
      rr:joinCondition [ rr:child "RECEIPT_ID" ; rr:parent "RECEIPT_ID" ] ;
    ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:qualityinspection_partId ;
    rr:objectMap [
      rr:parentTriplesMap sc:TriplesMap_Part ;
      rr:joinCondition [ rr:child "PART_ID" ; rr:parent "PART_ID" ] ;
    ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:qualityinspection_supplierId ;
    rr:objectMap [
      rr:parentTriplesMap sc:TriplesMap_Supplier ;
      rr:joinCondition [ rr:child "SUPPLIER_ID" ; rr:parent "SUPPLIER_ID" ] ;
    ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:qualityinspection_inspectionDate ;
    rr:objectMap [ rr:column "INSPECTION_DATE" ; rr:datatype xsd:date ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:qualityinspection_sampleSize ;
    rr:objectMap [ rr:column "SAMPLE_SIZE" ; rr:datatype xsd:integer ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:qualityinspection_defectCount ;
    rr:objectMap [ rr:column "DEFECT_COUNT" ; rr:datatype xsd:integer ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:qualityinspection_result ;
    rr:objectMap [ rr:column "RESULT" ; rr:datatype xsd:string ] ;
  ] ;
  rr:predicateObjectMap [
    rr:predicate sc:qualityinspection_ppm ;
    rr:objectMap [ rr:column "PPM" ; rr:datatype xsd:decimal ] ;
  ] .

sc:TriplesMap_NonconformanceReport
  rr:logicalTable [ rr:tableName "SCS.INVENTORY.NONCONFORMANCE_REPORT" ] ;
  rr:subjectMap [
    rr:template
      "https://asterion.example.com/ontology/supplychain/nonconformancereport/{NCR_ID}" ;
    rr:class sc:NonconformanceReport ;
  ] ;

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] ;
coa:datasourceId "ds-scs-snowflake" ;
coa:sourceSchema "SCS.INVENTORY" ;
rr:predicateObjectMap [
  rr:predicate sc:nonconformancereport_ncrId ;
  rr:objectMap [ rr:column "NCR_ID" ; rr:datatype xsd:integer ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:nonconformancereport_inspectionId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_QualityInspection ;
    rr:joinCondition [ rr:child "INSPECTION_ID" ; rr:parent "INSPECTION_ID" ] ;
  ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:nonconformancereport_supplierId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Supplier ;
    rr:joinCondition [ rr:child "SUPPLIER_ID" ; rr:parent "SUPPLIER_ID" ] ;
  ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:nonconformancereport_partId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Part ;
    rr:joinCondition [ rr:child "PART_ID" ; rr:parent "PART_ID" ] ;
  ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:nonconformancereport_plantId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Plant ;
    rr:joinCondition [ rr:child "PLANT_ID" ; rr:parent "PLANT_ID" ] ;
  ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:nonconformancereport_ncrDate ;
  rr:objectMap [ rr:column "NCR_DATE" ; rr:datatype xsd:date ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:nonconformancereport_severity ;
  rr:objectMap [ rr:column "SEVERITY" ; rr:datatype xsd:string ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:nonconformancereport_rootCause ;
  rr:objectMap [ rr:column "ROOT_CAUSE" ; rr:datatype xsd:string ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:nonconformancereport_disposition ;
  rr:objectMap [ rr:column "DISPOSITION" ; rr:datatype xsd:string ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:nonconformancereport_status ;
  rr:objectMap [ rr:column "STATUS" ; rr:datatype xsd:string ] ;
] .

sc:TriplesMap_InventorySnapshot
rr:logicalTable [ rr:tableName "SCS.INVENTORY.FACT_INVENTORY_SNAPSHOT" ] ;
rr:subjectMap [
  rr:template
    "https://asterion.example.com/ontology/supplychain/inventorysnapshot/{SNAPSHOT_ID}" ;
  rr:class sc:InventorySnapshot ;
] ;
coa:datasourceId "ds-scs-snowflake" ;
coa:sourceSchema "SCS.INVENTORY" ;
rr:predicateObjectMap [
  rr:predicate sc:inventorysnapshot_snapshotId ;
  rr:objectMap [ rr:column "SNAPSHOT_ID" ; rr:datatype xsd:integer ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:inventorysnapshot_snapshotDate ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_CalendarDate ;
    rr:joinCondition [ rr:child "SNAPSHOT_DATE" ; rr:parent "DATE_KEY" ] ;
  ] ;
] ;
rr:predicateObjectMap [
  rr:predicate sc:inventorysnapshot_partId ;
  rr:objectMap [
    rr:parentTriplesMap sc:TriplesMap_Part ;

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        rr:joinCondition [ rr:child "PART_ID" ; rr:parent "PART_ID" ] ;
    ] ;
rr:predicateObjectMap [
    rr:predicate sc:inventorysnapshot_warehouseId ;
    rr:objectMap [
        rr:parentTriplesMap sc:TriplesMap_Warehouse ;
        rr:joinCondition [ rr:child "WAREHOUSE_ID" ; rr:parent "WAREHOUSE_ID" ] ;
    ] ;
] ;
rr:predicateObjectMap [
    rr:predicate sc:inventorysnapshot_onHandQty ;
    rr:objectMap [ rr:column "ON_HAND_QTY" ; rr:datatype xsd:decimal ] ;
] ;
rr:predicateObjectMap [
    rr:predicate sc:inventorysnapshot_unitCostUsd ;
    rr:objectMap [ rr:column "UNIT_COST_USD" ; rr:datatype xsd:decimal ] ;
] ;
rr:predicateObjectMap [
    rr:predicate sc:inventorysnapshot_onHandValue ;
    rr:objectMap [ rr:column "ON_HAND_VALUE" ; rr:datatype xsd:decimal ] ;
] ;
rr:predicateObjectMap [
    rr:predicate sc:inventorysnapshot_dailyUsageQty ;
    rr:objectMap [ rr:column "DAILY_USAGE_QTY" ; rr:datatype xsd:decimal ] ;
] ;
rr:predicateObjectMap [
    rr:predicate sc:inventorysnapshot_doi ;
    rr:objectMap [ rr:column "DOI" ; rr:datatype xsd:decimal ] ;
] .

```

supply_chain_catalog.json

Data-catalog / schema-induction input

```
{
  "namespaceId": "ns-asterion-supplychain",
  "sources": [
    {
      "datasourceId": "ds-scs-snowflake",
      "sourceType": "JDBC_DATABASE",
      "databases": [
        {
          "name": "SCS",
          "description": "Asterion Devices synthetic supply-chain warehouse. Snowflake; serve path  
is Athena-federated (no direct-SQL driver for Snowflake).",
          "tables": [
            {
              "id": "SCS.INVENTORY.DIM_DATE",
              "name": "DIM_DATE",
              "fullyQualifiedName": "SCS.INVENTORY.DIM_DATE",
              "description": "Calendar dimension from 2022-01-01 to 2026-12-31",
              "synonyms": [
                "Calendar Date",
                "date",
                "day",
                "calendar",
                "DIM_DATE",
                "time dimension"
              ],
              "datasourceId": "ds-scs-snowflake",
              "sourceSchema": "SCS.INVENTORY",
              "columns": [
                {
                  "name": "DATE_KEY",
                  "dataType": "DATE",
                  "description": "",
                  "constraint": "PRIMARY_KEY",
                  "ordinalPosition": 1
                },
                {
                  "name": "YEAR",
                  "dataType": "INT",
                  "description": "",
                  "constraint": "NOT_NULL",
                  "ordinalPosition": 2
                },
                {
                  "name": "QUARTER",
                  "dataType": "INT",
                  "description": "",
                  "constraint": "NOT_NULL",
                  "ordinalPosition": 3
                },
                {
                  "name": "MONTH",
                  "dataType": "INT",
                  "description": "",
                  "constraint": "NOT_NULL",
                  "ordinalPosition": 4
                },
                {
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                  "ordinalPosition": 5,
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                    "Aug",
                    "Dec",
                    "Feb",
                    "Jan",
                    "Jul",
                    "Jun",
                    "Mar",
                    "May",
                    "Nov",
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        "Oct",
        "Sep"
    ]
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{
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    "ordinalPosition": 6
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    "dataType": "INT",
    "description": "",
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    "ordinalPosition": 7
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{
    "name": "DAY_NAME",
    "dataType": "VARCHAR",
    "description": "",
    "constraint": "NOT_NULL",
    "ordinalPosition": 8,
    "dataLength": 10,
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        "Fri",
        "Mon",
        "Sat",
        "Sun",
        "Thu",
        "Tue",
        "Wed"
    ]
},
{
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    "dataType": "BOOLEAN",
    "description": "",
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    "ordinalPosition": 9
},
{
    "name": "FISCAL_YEAR",
    "dataType": "INT",
    "description": "",
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    "ordinalPosition": 10
},
{
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    "dataType": "INT",
    "description": "",
    "constraint": "NOT_NULL",
    "ordinalPosition": 11
}
],
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        "columns": [
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        ]
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    "name": "DIM_GEO_REGION",
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    "description": "Geographic regions",
    "synonyms": [
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        "geography",
        "geo region",
        "DIM_GEO_REGION",
        "world region"
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    "name": "REGION_NAME",
    "dataType": "VARCHAR",
    "description": "",
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    "ordinalPosition": 2,
    "dataLength": 50
  }
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    "columns": [
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    ]
  }
]
},
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  "id": "SCS.INVENTORY.DIM_GEO_COUNTRY",
  "name": "DIM_GEO_COUNTRY",
  "fullyQualifiedName": "SCS.INVENTORY.DIM_GEO_COUNTRY",
  "description": "Countries mapped to regions",
  "synonyms": [
    "Country",
    "nation",
    "DIM_GEO_COUNTRY",
    "country of origin"
  ],
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      "description": "",
      "constraint": "NOT_NULL",
      "ordinalPosition": 2,
      "dataLength": 50
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    {
      "name": "COUNTRY_CODE",
      "dataType": "VARCHAR",
      "description": "",
      "constraint": "NOT_NULL",
      "ordinalPosition": 3,
      "dataLength": 3
    },
    {
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      "dataType": "INT",
      "description": "",
      "constraint": "NOT_NULL",
      "ordinalPosition": 4
    }
  ],
  "tableConstraints": [
    {
      "constraintType": "PRIMARY_KEY",
      "columns": [
        "COUNTRY_ID"
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    },
    {
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        "columns": [
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        ],
        "referredColumns": [
            "DIM_GEO_REGION.REGION_ID"
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    "name": "DIM_COMMODITY",
    "fullyQualifiedName": "SCS.INVENTORY.DIM_COMMODITY",
    "description": "Top-level commodity classification",
    "synonyms": [
        "Commodity",
        "category",
        "commodity group",
        "DIM_COMMODITY",
        "spend category"
    ],
    "datasourceId": "ds-scs-snowflake",
    "sourceSchema": "SCS.INVENTORY",
    "columns": [
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            "dataType": "INT",
            "description": "",
            "constraint": "PRIMARY_KEY",
            "ordinalPosition": 1
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        {
            "name": "COMMODITY_NAME",
            "dataType": "VARCHAR",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 2,
            "dataLength": 50
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},
{
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    "name": "DIM_SUB_COMMODITY",
    "fullyQualifiedName": "SCS.INVENTORY.DIM_SUB_COMMODITY",
    "description": "Sub-commodity classification",
    "synonyms": [
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        "subcategory",
        "sub commodity",
        "DIM_SUB_COMMODITY"
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    "sourceSchema": "SCS.INVENTORY",
    "columns": [
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            "name": "SUB_COMMODITY_ID",
            "dataType": "INT",
            "description": "",
            "constraint": "PRIMARY_KEY",
            "ordinalPosition": 1
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        {
            "name": "SUB_COMMODITY_NAME",
            "dataType": "VARCHAR",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 2,
            "dataLength": 80
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    ],
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        "columns": [
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    {
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        "columns": [
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        ],
        "referredColumns": [
            "DIM_COMMODITY.COMMODITY_ID"
        ],
        "relationshipType": "MANY_TO_ONE"
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]
},
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    "name": "DIM_CURRENCY",
    "fullyQualifiedName": "SCS.INVENTORY.DIM_CURRENCY",
    "description": "Currency reference with approximate USD exchange rates",
    "synonyms": [
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        "fx",
        "currency code",
        "DIM_CURRENCY"
    ],
    "datasourceId": "ds-scs-snowflake",
    "sourceSchema": "SCS.INVENTORY",
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            "dataLength": 3,
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                "CNY",
                "EUR",
                "GBP",
                "INR",
                "JPY",
                "KRW",
                "MXN",
                "MYR",
                "THB",
                "TWD",
                "USD"
            ]
        },
        {
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            "dataType": "VARCHAR",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 2,
            "dataLength": 50
        },
        {
            "name": "TO_USD_RATE",
            "dataType": "DECIMAL",
            "description": "Approximate fixed rate for demo",
            "constraint": "NOT_NULL",
            "ordinalPosition": 3,
            "precision": 10,
            "scale": 4
        }
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},
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  "description": "Units of measure",
  "synonyms": [
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    "uom",
    "unit",
    "DIM_UOM"
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  "sourceSchema": "SCS.INVENTORY",
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      "dataType": "VARCHAR",
      "description": "",
      "constraint": "PRIMARY_KEY",
      "ordinalPosition": 1,
      "dataLength": 5,
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        "BOX",
        "EA",
        "FT",
        "KG",
        "L",
        "LB",
        "M",
        "PAL",
        "PC",
        "ROL",
        "SET",
        "SHT"
      ]
    },
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      "dataType": "VARCHAR",
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  "name": "DIM_INCOTERM",
  "fullyQualifiedName": "SCS.INVENTORY.DIM_INCOTERM",
  "description": "Incoterms 2020",
  "synonyms": [
    "Incoterm",
    "shipping terms",
    "trade terms",
    "DIM_INCOTERM",
    "incoterms 2020"
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  "datasourceId": "ds-scs-snowflake",
  "sourceSchema": "SCS.INVENTORY",
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            "CIP",
            "CPT",
            "DAP",
            "DDP",
            "DPU",
            "EXW",
            "FAS",
            "FCA",
            "FOB"
        ]
    },
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        "columns": [
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},
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    "description": "Major ports and airports for logistics",
    "synonyms": [
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        "seaport",
        "airport",
        "terminal",
        "DIM_PORT"
    ],
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            "description": "",
            "constraint": "PRIMARY_KEY",
            "ordinalPosition": 1
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        {
            "name": "PORT_NAME",
            "dataType": "VARCHAR",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 2,
            "dataLength": 80
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        {
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            "dataType": "VARCHAR",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 3,
            "dataLength": 10
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        {
            "name": "COUNTRY_ID",
            "dataType": "INT",
            "description": "",
            "constraint": "NOT_NULL",

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        "name": "PORT_TYPE",
        "dataType": "VARCHAR",
        "description": "SEA, AIR, or BOTH",
        "constraint": "NOT_NULL",
        "ordinalPosition": 5,
        "dataLength": 10,
        "distinctValues": [
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            "BOTH",
            "SEA"
        ]
    }
],
"tableConstraints": [
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        "constraintType": "PRIMARY_KEY",
        "columns": [
            "PORT_ID"
        ]
    },
    {
        "constraintType": "FOREIGN_KEY",
        "columns": [
            "COUNTRY_ID"
        ],
        "referredColumns": [
            "DIM_GEO_COUNTRY.COUNTRY_ID"
        ],
        "relationshipType": "MANY_TO_ONE"
    }
]
},
{
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    "name": "DIM_SUPPLIER",
    "fullyQualifiedName": "SCS.INVENTORY.DIM_SUPPLIER",
    "description": "Supplier master \u2014 500 suppliers across commodities and countries",
    "synonyms": [
        "Supplier",
        "vendor",
        "supplier master",
        "DIM_SUPPLIER",
        "source",
        "seller"
    ],
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            "dataType": "INT",
            "description": "",
            "constraint": "PRIMARY_KEY",
            "ordinalPosition": 1
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        {
            "name": "SUPPLIER_NAME",
            "dataType": "VARCHAR",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 2,
            "dataLength": 200
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        {
            "name": "SUPPLIER_CODE",
            "dataType": "VARCHAR",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 3,
            "dataLength": 20
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        {
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            "dataType": "INT",
            "description": "",
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        "dataType": "INT",
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        "ordinalPosition": 5
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        "dataType": "DECIMAL",
        "description": "0.00-1.00 scale",
        "constraint": "NOT_NULL",
        "ordinalPosition": 6,
        "precision": 3,
        "scale": 2,
        "synonyms": [
            "reliability",
            "supplier score",
            "otd rating"
        ]
    },
    {
        "name": "IS_PROBLEM",
        "dataType": "BOOLEAN",
        "description": "Flagged problem supplier for demo scenarios",
        "constraint": "NULL",
        "ordinalPosition": 7,
        "synonyms": [
            "problem supplier",
            "at risk supplier",
            "flagged supplier"
        ]
    },
    {
        "name": "ACTIVE",
        "dataType": "BOOLEAN",
        "description": "",
        "constraint": "NULL",
        "ordinalPosition": 8
    },
    {
        "name": "IS_SYNTHETIC",
        "dataType": "BOOLEAN",
        "description": "",
        "constraint": "NULL",
        "ordinalPosition": 9
    }
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"tableConstraints": [
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        "constraintType": "PRIMARY_KEY",
        "columns": [
            "SUPPLIER_ID"
        ]
    },
    {
        "constraintType": "FOREIGN_KEY",
        "columns": [
            "COUNTRY_ID"
        ],
        "referredColumns": [
            "DIM_GEO_COUNTRY.COUNTRY_ID"
        ],
        "relationshipType": "MANY_TO_ONE"
    },
    {
        "constraintType": "FOREIGN_KEY",
        "columns": [
            "COMMODITY_ID"
        ],
        "referredColumns": [
            "DIM_COMMODITY.COMMODITY_ID"
        ],
        "relationshipType": "MANY_TO_ONE"
    }
]
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{

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"name": "DIM_SUPPLIER_SITE",
"fullyQualified_name": "SCS.INVENTORY.DIM_SUPPLIER_SITE",
"description": "Supplier manufacturing and shipping sites",
"synonyms": [
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  "supplier facility",
  "vendor site",
  "DIM_SUPPLIER_SITE",
  "manufacturing site"
],
"dataSourceId": "ds-scs-snowflake",
"sourceSchema": "SCS.INVENTORY",
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    "description": "",
    "constraint": "PRIMARY_KEY",
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  {
    "name": "SUPPLIER_ID",
    "dataType": "INT",
    "description": "",
    "constraint": "NOT_NULL",
    "ordinalPosition": 2
  },
  {
    "name": "SITE_NAME",
    "dataType": "VARCHAR",
    "description": "",
    "constraint": "NOT_NULL",
    "ordinalPosition": 3,
    "dataLength": 200
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    "name": "CITY",
    "dataType": "VARCHAR",
    "description": "",
    "constraint": "NOT_NULL",
    "ordinalPosition": 4,
    "dataLength": 100
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  {
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    "dataType": "INT",
    "description": "",
    "constraint": "NOT_NULL",
    "ordinalPosition": 5
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  {
    "name": "LEAD_TIME_DAYS",
    "dataType": "INT",
    "description": "Average lead time in calendar days",
    "constraint": "NOT_NULL",
    "ordinalPosition": 6,
    "synonyms": [
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      "replenishment lead time"
    ]
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  {
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    "dataType": "BOOLEAN",
    "description": "",
    "constraint": "NULL",
    "ordinalPosition": 7
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    "name": "ACTIVE",
    "dataType": "BOOLEAN",
    "description": "",
    "constraint": "NULL",
    "ordinalPosition": 8
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],
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    {
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        "columns": [
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        "referredColumns": [
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        ],
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        "constraintType": "FOREIGN_KEY",
        "columns": [
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        "referredColumns": [
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        ],
        "relationshipType": "MANY_TO_ONE"
    }
],
},
{
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    "name": "DIM_MANUFACTURER",
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    "description": "Component manufacturers",
    "synonyms": [
        "Manufacturer",
        "oem",
        "component manufacturer",
        "DIM_MANUFACTURER",
        "mfr"
    ],
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            "description": "",
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            "ordinalPosition": 1
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        {
            "name": "MANUFACTURER_NAME",
            "dataType": "VARCHAR",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 2,
            "dataLength": 200
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        {
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            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 3
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            "columns": [
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            "columns": [
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            ],
            "relationshipType": "MANY_TO_ONE"
        }
    ]
}

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    }
  ]
},
{
  "id": "SCS.INVENTORY.DIM_PART",
  "name": "DIM_PART",
  "fullyQualifiedName": "SCS.INVENTORY.DIM_PART",
  "description": "Part master \u2014 30,000 parts across all commodities",
  "synonyms": [
    "Part",
    "component",
    "material",
    "sku",
    "item",
    "DIM_PART",
    "part master"
  ],
  "datasourceId": "ds-scs-snowflake",
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      "constraint": "PRIMARY_KEY",
      "ordinalPosition": 1
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    {
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      "dataType": "VARCHAR",
      "description": "",
      "constraint": "NOT_NULL",
      "ordinalPosition": 2,
      "dataLength": 30
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    {
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      "dataType": "VARCHAR",
      "description": "",
      "constraint": "NOT_NULL",
      "ordinalPosition": 3,
      "dataLength": 200
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      "dataType": "INT",
      "description": "",
      "constraint": "NOT_NULL",
      "ordinalPosition": 4
    },
    {
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      "dataType": "INT",
      "description": "",
      "constraint": "NOT_NULL",
      "ordinalPosition": 5
    },
    {
      "name": "UOM_CODE",
      "dataType": "VARCHAR",
      "description": "",
      "constraint": "NULL",
      "ordinalPosition": 6,
      "dataLength": 5
    },
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      "dataType": "DECIMAL",
      "description": "Standard cost in USD",
      "constraint": "NOT_NULL",
      "ordinalPosition": 7,
      "precision": 10,
      "scale": 2,
      "synonyms": [
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        "part cost",
        "unit cost"
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    }
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{

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        "constraint": "NULL",
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        "precision": 8,
        "scale": 3
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        "description": "ACTIVE, EOL, OBSOLETE, NRND",
        "constraint": "NULL",
        "ordinalPosition": 9,
        "dataLength": 20,
        "synonyms": [
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            "eol status",
            "part status"
        ],
        "distinctValues": [
            "ACTIVE",
            "EOL",
            "NRND",
            "OBSOLETE"
        ]
    },
    {
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        "dataType": "BOOLEAN",
        "description": "",
        "constraint": "NULL",
        "ordinalPosition": 10
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        "columns": [
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        "relationshipType": "MANY_TO_ONE"
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    "fullyQualified_name": "SCS.INVENTORY.DIM_PRODUCT_FAMILY",
    "description": "Finished product families",
    "synonyms": [

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        "family",
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            "ordinalPosition": 2,
            "dataLength": 100
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},
{
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    "name": "DIM_PRODUCT",
    "fullyQualifiedName": "SCS.INVENTORY.DIM_PRODUCT",
    "description": "Finished products \u2014 500 SKUs",
    "synonyms": [
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        "finished good",
        "sku",
        "DIM_PRODUCT",
        "end item"
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            "synonyms": [
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        "selling price"
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        "referredColumns": [
            "DIM_PRODUCT_FAMILY.PRODUCT_FAMILY_ID"
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        "relationshipType": "MANY_TO_ONE"
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]
},
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    "id": "SCS.INVENTORY.BOM_HEADER",
    "name": "BOM_HEADER",
    "fullyQualifiedName": "SCS.INVENTORY.BOM_HEADER",
    "description": "Bill of Materials headers \u2014 one per product revision",
    "synonyms": [
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        "bom",
        "BOM_HEADER",
        "product structure",
        "recipe"
    ],
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            "description": "",
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            "ordinalPosition": 2
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            "name": "BOM_VERSION",
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            "description": "",
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                "B",
                "C"
            ]
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            "columns": [
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            ],
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            "relationshipType": "MANY_TO_ONE"
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        "name": "BOM_LINE",
        "fullyQualifiedName": "SCS.INVENTORY.BOM_LINE",
        "description": "BOM line items \u2014 components per assembly",
        "synonyms": [
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            "bom line",
            "bom component",
            "BOM_LINE"
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                "name": "BOM_ID",
                "dataType": "INT",
                "description": "",
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                "ordinalPosition": 2
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            {
                "name": "PART_ID",
                "dataType": "INT",
                "description": "",
                "constraint": "NOT_NULL",
                "ordinalPosition": 3
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            {
                "name": "QUANTITY_PER",
                "dataType": "DECIMAL",
                "description": "",
                "constraint": "NOT_NULL",
                "ordinalPosition": 4,
                "precision": 10,
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                    "usage per assembly",
                    "quantity per assembly"
                ]
            }
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    },
    {

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        "columns": [
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        "referredColumns": [
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        "relationshipType": "MANY_TO_ONE"
    },
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        "columns": [
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        ],
        "referredColumns": [
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        ],
        "relationshipType": "MANY_TO_ONE"
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]
},
{
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    "name": "DIM_PLANT",
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    "description": "Manufacturing plants for Asterion Devices",
    "synonyms": [
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        "factory",
        "manufacturing site",
        "DIM_PLANT",
        "works"
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            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 2,
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        "columns": [
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        ],
        "relationshipType": "MANY_TO_ONE"
    }
],
},
{
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    "fullyQualifiedName": "SCS.INVENTORY.DIM_WAREHOUSE",
    "description": "Warehouses \u2014 plant-attached and distribution centers",
    "synonyms": [
        "Warehouse",
        "dc",
        "distribution center",
        "stocking location",
        "DIM_WAREHOUSE"
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            "dataType": "VARCHAR",
            "description": "",
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            "ordinalPosition": 2,
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            "dataType": "VARCHAR",
            "description": "",
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            "ordinalPosition": 3,
            "dataLength": 100
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            "dataType": "INT",
            "description": "",
            "constraint": "NULL",
            "ordinalPosition": 4
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            "dataType": "VARCHAR",
            "description": "",
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        "ordinalPosition": 6
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        "dataType": "INT",
        "description": "",
        "constraint": "NOT_NULL",
        "ordinalPosition": 7
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        "columns": [
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        ],
        "relationshipType": "MANY_TO_ONE"
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        "relationshipType": "MANY_TO_ONE"
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],
},
{
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    "fullyQualifiedName": "SCS.INVENTORY.DIM_CUSTOMER",
    "description": "Customer master \u2014 80 customers",
    "synonyms": [
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        "account",
        "buyer",
        "DIM_CUSTOMER",
        "client"
    ],
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            "dataType": "INT",
            "description": "",
            "constraint": "PRIMARY_KEY",
            "ordinalPosition": 1
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        {
            "name": "CUSTOMER_NAME",
            "dataType": "VARCHAR",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 2,
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        "description": "PLATINUM, GOLD, SILVER, BRONZE",
        "constraint": "NOT_NULL",
        "ordinalPosition": 5,
        "dataLength": 10,
        "synonyms": [
            "tier",
            "segment",
            "customer segment"
        ],
        "distinctValues": [
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            "GOLD",
            "PLATINUM",
            "SILVER"
        ]
    },
    {
        "name": "IS_SYNTHETIC",
        "dataType": "BOOLEAN",
        "description": "",
        "constraint": "NULL",
        "ordinalPosition": 6
    }
],
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        "columns": [
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    },
    {
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        "columns": [
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        ],
        "referredColumns": [
            "DIM_GEO_COUNTRY.COUNTRY_ID"
        ],
        "relationshipType": "MANY_TO_ONE"
    }
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{
    "id": "SCS.INVENTORY.DIM_CARRIER",
    "name": "DIM_CARRIER",
    "fullyQualifiedName": "SCS.INVENTORY.DIM_CARRIER",
    "description": "Logistics carriers",
    "synonyms": [
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        "freight forwarder",
        "shipping line",
        "DIM_CARRIER",
        "logistics provider"
    ],
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    "sourceSchema": "SCS.INVENTORY",
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            "dataType": "INT",
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            "constraint": "PRIMARY_KEY",
            "ordinalPosition": 1
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    ]
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        "dataType": "VARCHAR",
        "description": "OCEAN, AIR, TRUCK, RAIL, MULTIMODAL",
        "constraint": "NOT_NULL",
        "ordinalPosition": 3,
        "dataLength": 20,
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            "MULTIMODAL",
            "OCEAN",
            "RAIL",
            "TRUCK"
        ]
    }
],
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        "columns": [
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        ]
    }
]
},
{
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    "name": "DIM_LANE",
    "fullyQualifiedName": "SCS.INVENTORY.DIM_LANE",
    "description": "Shipping lanes between ports",
    "synonyms": [
        "Lane",
        "shipping lane",
        "trade lane",
        "route",
        "DIM_LANE",
        "corridor"
    ],
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            "constraint": "PRIMARY_KEY",
            "ordinalPosition": 1
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        {
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            "dataType": "INT",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 2
        },
        {
            "name": "DEST_PORT_ID",
            "dataType": "INT",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 3
        },
        {
            "name": "TRANSPORT_MODE",
            "dataType": "VARCHAR",
            "description": "",
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            "ordinalPosition": 4,
            "dataLength": 20,
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                "OCEAN"
            ]
        }
    ]
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      "description": "",
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      "columns": [
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      ],
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        "DIM_PORT.PORT_ID"
      ],
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},
{
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  "name": "DIM_CONTRACT",
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  "description": "Supplier contracts",
  "synonyms": [
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    "supplier agreement",
    "DIM_CONTRACT",
    "purchase agreement"
  ],
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  "sourceSchema": "SCS.INVENTORY",
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      "description": "",
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    },
    {
      "name": "SUPPLIER_ID",
      "dataType": "INT",
      "description": "",
      "constraint": "NOT_NULL",
      "ordinalPosition": 2
    },
    {
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      "dataType": "VARCHAR",

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        "description": "",
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    {
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        "ordinalPosition": 4
    },
    {
        "name": "END_DATE",
        "dataType": "DATE",
        "description": "",
        "constraint": "NOT_NULL",
        "ordinalPosition": 5
    },
    {
        "name": "CURRENCY_CODE",
        "dataType": "VARCHAR",
        "description": "",
        "constraint": "NULL",
        "ordinalPosition": 6,
        "dataLength": 3
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    {
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        "dataType": "VARCHAR",
        "description": "",
        "constraint": "NULL",
        "ordinalPosition": 7,
        "dataLength": 3
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    {
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        "dataType": "VARCHAR",
        "description": "",
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        "ordinalPosition": 8,
        "dataLength": 20,
        "distinctValues": [
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        ]
    }
],
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        "columns": [
            "CONTRACT_ID"
        ]
    },
    {
        "constraintType": "FOREIGN_KEY",
        "columns": [
            "SUPPLIER_ID"
        ],
        "referredColumns": [
            "DIM_SUPPLIER.SUPPLIER_ID"
        ],
        "relationshipType": "MANY_TO_ONE"
    },
    {
        "constraintType": "FOREIGN_KEY",
        "columns": [
            "CURRENCY_CODE"
        ],
        "referredColumns": [
            "DIM_CURRENCY.CURRENCY_CODE"
        ],
        "relationshipType": "MANY_TO_ONE"
    },
    {
        "constraintType": "FOREIGN_KEY",
        "columns": [
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        ],
        "referredColumns": [

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    "relationshipType": "MANY_TO_ONE"
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],
},
{
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    "name": "SUPPLIER_PART_APPROVAL",
    "fullyQualifiedName": "SCS.INVENTORY.SUPPLIER_PART_APPROVAL",
    "description": "Approved supplier-part combinations \u2014 40,000 rows",
    "synonyms": [
        "Supplier Part Approval",
        "approved vendor list",
        "avl",
        "asl",
        "approved source",
        "SUPPLIER_PART_APPROVAL"
    ],
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            "dataType": "INT",
            "description": "",
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            "ordinalPosition": 2
        },
        {
            "name": "SITE_ID",
            "dataType": "INT",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 3
        },
        {
            "name": "PART_ID",
            "dataType": "INT",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 4
        },
        {
            "name": "APPROVAL_DATE",
            "dataType": "DATE",
            "description": "",
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            "ordinalPosition": 5
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            "dataType": "VARCHAR",
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        {
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            "columns": [
                "SUPPLIER_ID"
            ]
        }
    ]
}

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    ],
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{
    "constraintType": "FOREIGN_KEY",
    "columns": [
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}
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},
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    "description": "Purchase order headers \u2014 20,000 POs",
    "synonyms": [
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        "po",
        "PO_HEADER",
        "procurement order",
        "purchase requisition"
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            "dataType": "INT",
            "description": "",
            "constraint": "PRIMARY_KEY",
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        {
            "name": "PO_NUMBER",
            "dataType": "VARCHAR",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 2,
            "dataLength": 20
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            "ordinalPosition": 3
        },
        {
            "name": "PLANT_ID",
            "dataType": "INT",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 4
        },
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            "name": "ORDER_DATE",
            "dataType": "DATE",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 5
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    ],
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        "constraint": "NOT_NULL",
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            "PARTIAL",
            "RECEIVED"
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    "name": "PO_LINE",
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    "description": "Purchase order line items \u2014 60,000 lines",
    "synonyms": [
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        "po line",
        "po item",
        "PO_LINE"
    ],
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    "sourceSchema": "SCS.INVENTORY",
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            "description": "",
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        "name": "PART_ID",
        "dataType": "INT",
        "description": "",
        "constraint": "NOT_NULL",
        "ordinalPosition": 3
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        "dataType": "DECIMAL",
        "description": "",
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        "ordinalPosition": 4,
        "precision": 12,
        "scale": 2
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        "description": "",
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        "ordinalPosition": 5,
        "precision": 10,
        "scale": 2,
        "synonyms": [
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            "purchase price"
        ]
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        "columns": [
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        "columns": [
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        ],
        "referredColumns": [
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        "relationshipType": "MANY_TO_ONE"
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        "columns": [
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        ],
        "referredColumns": [
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        ],
        "relationshipType": "MANY_TO_ONE"
    }
],
{
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    "name": "GOODS_RECEIPT",
    "fullyQualifiedName": "SCS.INVENTORY.GOODS_RECEIPT",
    "description": "Goods receipts \u2014 90,000 receipts against PO lines",

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"synonyms": [
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  "grn",
  "receipt",
  "goods received note",
  "GOODS_RECEIPT",
  "delivery"
],
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  },
  {
    "name": "PO_LINE_ID",
    "dataType": "INT",
    "description": "",
    "constraint": "NOT_NULL",
    "ordinalPosition": 2
  },
  {
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    "dataType": "DATE",
    "description": "Actual physical receipt date",
    "constraint": "NOT_NULL",
    "ordinalPosition": 3
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    "precision": 12,
    "scale": 2
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    "description": "",
    "constraint": "NOT_NULL",
    "ordinalPosition": 5
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    "dataType": "INT",
    "description": "Days early (negative) or late (positive) vs promise date",
    "constraint": "NOT_NULL",
    "ordinalPosition": 6,
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      "days late",
      "schedule variance",
      "otd variance"
    ]
  }
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    ]
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    "columns": [
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    "referredColumns": [
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    "relationshipType": "MANY_TO_ONE"
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],
{

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        "relationshipType": "MANY_TO_ONE"
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    "description": "Customer sales order headers \u2014 10,000 orders",
    "synonyms": [
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        "so",
        "customer order",
        "SALES_ORDER_HEADER",
        "demand"
    ],
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    "sourceSchema": "SCS.INVENTORY",
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            "dataType": "VARCHAR",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 2,
            "dataLength": 20
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            "dataType": "INT",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 3
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        {
            "name": "ORDER_DATE",
            "dataType": "DATE",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 4
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        {
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            "dataType": "INT",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 5
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            "description": "",
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            "ordinalPosition": 6,
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                "PARTIAL",
                "SHIPPED"
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        "ORDER_ID"
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    ],
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},
{
    "constraintType": "FOREIGN_KEY",
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    "referredColumns": [
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    "relationshipType": "MANY_TO_ONE"
}
]
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    "synonyms": [
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        "so line",
        "order line",
        "SALES_ORDER_LINE"
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    "sourceSchema": "SCS.INVENTORY",
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            "dataType": "INT",
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        {
            "name": "ORDER_ID",
            "dataType": "INT",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 2
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            "dataType": "INT",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 3
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            "ordinalPosition": 4,
            "precision": 12,
            "scale": 2
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            "dataType": "DECIMAL",
            "description": "",
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            "ordinalPosition": 5,
            "precision": 10,
            "scale": 2
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        "columns": [
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        ],
        "referredColumns": [
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        ],
        "relationshipType": "MANY_TO_ONE"
    },
    {
        "constraintType": "FOREIGN_KEY",
        "columns": [
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        ],
        "referredColumns": [
            "DIM_PRODUCT.PRODUCT_ID"
        ],
        "relationshipType": "MANY_TO_ONE"
    }
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{
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    "synonyms": [
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        "freight",
        "consignment",
        "SHIPMENT_HEADER",
        "load"
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    "sourceSchema": "SCS.INVENTORY",
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            "dataType": "INT",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 2
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        {
            "name": "LANE_ID",
            "dataType": "INT",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 3
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            "description": "",
            "constraint": "NOT_NULL",
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        {
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            "DELIVERED",
            "IN_TRANSIT"
        ]
    }
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        "referredColumns": [
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        ],
        "relationshipType": "MANY_TO_ONE"
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{
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    "fullyQualifiedName": "SCS.INVENTORY.SHIPMENT_LINE",
    "description": "Shipment line items linking to orders or POs",
    "synonyms": [
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        "shipment item",
        "SHIPMENT_LINE"
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    "sourceSchema": "SCS.INVENTORY",
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        {
            "name": "SHIPMENT_ID",
            "dataType": "INT",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 2
        },
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            "name": "ORDER_LINE_ID",
            "dataType": "INT",
            "description": "",
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        "ordinalPosition": 3
    },
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        "name": "PO_LINE_ID",
        "dataType": "INT",
        "description": "",
        "constraint": "NULL",
        "ordinalPosition": 4
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        "name": "QUANTITY",
        "dataType": "DECIMAL",
        "description": "",
        "constraint": "NOT_NULL",
        "ordinalPosition": 5,
        "precision": 12,
        "scale": 2
    }
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        "columns": [
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        ],
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        "constraintType": "FOREIGN_KEY",
        "columns": [
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        "referredColumns": [
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        ],
        "relationshipType": "MANY_TO_ONE"
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        "constraintType": "FOREIGN_KEY",
        "columns": [
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        "referredColumns": [
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        ],
        "relationshipType": "MANY_TO_ONE"
    }
],
{
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    "name": "SHIPMENT_MILESTONE_EVENT",
    "fullyQualifiedName": "SCS.INVENTORY.SHIPMENT_MILESTONE_EVENT",
    "description": "Shipment tracking milestones \u2014 250,000 events",
    "synonyms": [
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        "tracking event",
        "milestone",
        "shipment status event",
        "SHIPMENT_MILESTONE_EVENT"
    ],
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    "sourceSchema": "SCS.INVENTORY",
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            "ordinalPosition": 1
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    ]
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  "ordinalPosition": 2
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  "description": "",
  "constraint": "NOT_NULL",
  "ordinalPosition": 3,
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    "BOOKING_CONFIRMED",
    "CUSTOMS_CLEARANCE",
    "DELIVERED",
    "DEPARTED",
    "GATE_IN",
    "GATE_OUT",
    "INVOICE_SENT",
    "IN_TRANSIT",
    "LOADED",
    "OUT_FOR_DELIVERY",
    "PICKED_UP",
    "POD_SIGNED",
    "TRANSSHIPMENT"
  ]
},
{
  "name": "EVENT_TS",
  "dataType": "TIMESTAMP",
  "description": "",
  "constraint": "NOT_NULL",
  "ordinalPosition": 4
},
{
  "name": "LOCATION",
  "dataType": "VARCHAR",
  "description": "",
  "constraint": "NULL",
  "ordinalPosition": 5,
  "dataLength": 100
}
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    "columns": [
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    ]
  },
  {
    "constraintType": "FOREIGN_KEY",
    "columns": [
      "SHIPMENT_ID"
    ],
    "referredColumns": [
      "SHIPMENT_HEADER.SHIPMENT_ID"
    ],
    "relationshipType": "MANY_TO_ONE"
  }
]
},
{
  "id": "SCS.INVENTORY.IOT_SENSOR_READING",
  "name": "IOT_SENSOR_READING",
  "fullyQualifiedName": "SCS.INVENTORY.IOT_SENSOR_READING",
  "description": "IoT container sensor readings \u2014 500,000 rows, ~1.5% breach rate",
  "synonyms": [
    "Sensor Reading",
    "iot reading",
    "telemetry",
    "container sensor",
    "IOT_SENSOR_READING",
    "cold chain reading"
  ]
},
"datasourceId": "ds-scs-snowflake",

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  {
    "name": "SHIPMENT_ID",
    "dataType": "INT",
    "description": "",
    "constraint": "NOT_NULL",
    "ordinalPosition": 2
  },
  {
    "name": "READING_TS",
    "dataType": "TIMESTAMP",
    "description": "",
    "constraint": "NOT_NULL",
    "ordinalPosition": 3
  },
  {
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    "dataType": "DECIMAL",
    "description": "",
    "constraint": "NULL",
    "ordinalPosition": 4,
    "precision": 5,
    "scale": 1,
    "synonyms": [
      "temperature",
      "temp"
    ]
  },
  {
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    "dataType": "DECIMAL",
    "description": "",
    "constraint": "NULL",
    "ordinalPosition": 5,
    "precision": 5,
    "scale": 1
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    "dataType": "DECIMAL",
    "description": "",
    "constraint": "NULL",
    "ordinalPosition": 6,
    "precision": 5,
    "scale": 2
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  {
    "name": "TILT_DEG",
    "dataType": "DECIMAL",
    "description": "",
    "constraint": "NULL",
    "ordinalPosition": 7,
    "precision": 5,
    "scale": 1
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    "dataType": "DECIMAL",
    "description": "",
    "constraint": "NULL",
    "ordinalPosition": 8,
    "precision": 9,
    "scale": 6
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  {
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    "dataType": "DECIMAL",
    "description": "",
    "constraint": "NULL",
    "ordinalPosition": 9,
    "precision": 9,
    "scale": 6
  }
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      "dataType": "BOOLEAN",
      "description": "TRUE if any reading exceeds threshold",
      "constraint": "NOT_NULL",
      "ordinalPosition": 10,
      "synonyms": [
        "breach",
        "excursion",
        "threshold breach",
        "cold chain breach"
      ]
    }
  ],
  "tableConstraints": [
    {
      "constraintType": "PRIMARY_KEY",
      "columns": [
        "READING_ID"
      ]
    },
    {
      "constraintType": "FOREIGN_KEY",
      "columns": [
        "SHIPMENT_ID"
      ],
      "referredColumns": [
        "SHIPMENT_HEADER.SHIPMENT_ID"
      ],
      "relationshipType": "MANY_TO_ONE"
    }
  ]
},
{
  "id": "SCS.INVENTORY.WORK_ORDER",
  "name": "WORK_ORDER",
  "fullyQualifiedName": "SCS.INVENTORY.WORK_ORDER",
  "description": "Manufacturing work orders \u2014 10,000",
  "synonyms": [
    "Work Order",
    "wo",
    "production order",
    "manufacturing order",
    "WORK_ORDER"
  ],
  "datasourceId": "ds-scs-snowflake",
  "sourceSchema": "SCS.INVENTORY",
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      "description": "",
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      "ordinalPosition": 1
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    {
      "name": "PRODUCT_ID",
      "dataType": "INT",
      "description": "",
      "constraint": "NOT_NULL",
      "ordinalPosition": 2
    },
    {
      "name": "PLANT_ID",
      "dataType": "INT",
      "description": "",
      "constraint": "NOT_NULL",
      "ordinalPosition": 3
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      "dataType": "DATE",
      "description": "",
      "constraint": "NOT_NULL",
      "ordinalPosition": 4
    },
    {
      "name": "END_DATE",
      "dataType": "DATE",

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        "constraint": "NULL",
        "ordinalPosition": 5
    },
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        "dataType": "DECIMAL",
        "description": "",
        "constraint": "NOT_NULL",
        "ordinalPosition": 6,
        "precision": 12,
        "scale": 2
    },
    {
        "name": "STATUS",
        "dataType": "VARCHAR",
        "description": "",
        "constraint": "NOT_NULL",
        "ordinalPosition": 7,
        "dataLength": 20,
        "distinctValues": [
            "COMPLETED",
            "IN_PROGRESS",
            "ON_HOLD"
        ]
    }
],
"tableConstraints": [
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        "constraintType": "PRIMARY_KEY",
        "columns": [
            "WORK_ORDER_ID"
        ]
    },
    {
        "constraintType": "FOREIGN_KEY",
        "columns": [
            "PRODUCT_ID"
        ],
        "referredColumns": [
            "DIM_PRODUCT.PRODUCT_ID"
        ],
        "relationshipType": "MANY_TO_ONE"
    },
    {
        "constraintType": "FOREIGN_KEY",
        "columns": [
            "PLANT_ID"
        ],
        "referredColumns": [
            "DIM_PLANT.PLANT_ID"
        ],
        "relationshipType": "MANY_TO_ONE"
    }
],
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    "id": "SCS.INVENTORY.QUALITY_INSPECTION",
    "name": "QUALITY_INSPECTION",
    "fullyQualifiedName": "SCS.INVENTORY.QUALITY_INSPECTION",
    "description": "Incoming quality inspections \u2014 10,000",
    "synonyms": [
        "Quality Inspection",
        "incoming inspection",
        "iqc",
        "QUALITY_INSPECTION",
        "quality check"
    ],
    "datasourceId": "ds-scs-snowflake",
    "sourceSchema": "SCS.INVENTORY",
    "columns": [
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            "dataType": "INT",
            "description": "",
            "constraint": "PRIMARY_KEY",
            "ordinalPosition": 1
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        {
            "name": "RECEIPT_ID",

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        "constraint": "NOT_NULL",
        "ordinalPosition": 2
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        "dataType": "INT",
        "description": "",
        "constraint": "NOT_NULL",
        "ordinalPosition": 3
    },
    {
        "name": "SUPPLIER_ID",
        "dataType": "INT",
        "description": "",
        "constraint": "NOT_NULL",
        "ordinalPosition": 4
    },
    {
        "name": "INSPECTION_DATE",
        "dataType": "DATE",
        "description": "",
        "constraint": "NOT_NULL",
        "ordinalPosition": 5
    },
    {
        "name": "SAMPLE_SIZE",
        "dataType": "INT",
        "description": "",
        "constraint": "NOT_NULL",
        "ordinalPosition": 6
    },
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        "name": "DEFECT_COUNT",
        "dataType": "INT",
        "description": "",
        "constraint": "NOT_NULL",
        "ordinalPosition": 7,
        "synonyms": [
            "defects",
            "rejects"
        ]
    },
    {
        "name": "RESULT",
        "dataType": "VARCHAR",
        "description": "PASS, FAIL, CONDITIONAL",
        "constraint": "NOT_NULL",
        "ordinalPosition": 8,
        "dataLength": 20,
        "distinctValues": [
            "CONDITIONAL",
            "FAIL",
            "PASS"
        ]
    },
    {
        "name": "PPM",
        "dataType": "DECIMAL",
        "description": "Parts per million defect rate",
        "constraint": "NOT_NULL",
        "ordinalPosition": 9,
        "precision": 10,
        "scale": 1,
        "synonyms": [
            "defect rate",
            "parts per million",
            "ppm defect rate"
        ]
    }
],
"tableConstraints": [
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        "constraintType": "PRIMARY_KEY",
        "columns": [
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        ]
    }
],
{

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            "RECEIPT_ID"
        ],
        "referredColumns": [
            "GOODS_RECEIPT.RECEIPT_ID"
        ],
        "relationshipType": "MANY_TO_ONE"
    },
    {
        "constraintType": "FOREIGN_KEY",
        "columns": [
            "PART_ID"
        ],
        "referredColumns": [
            "DIM_PART.PART_ID"
        ],
        "relationshipType": "MANY_TO_ONE"
    },
    {
        "constraintType": "FOREIGN_KEY",
        "columns": [
            "SUPPLIER_ID"
        ],
        "referredColumns": [
            "DIM_SUPPLIER.SUPPLIER_ID"
        ],
        "relationshipType": "MANY_TO_ONE"
    }
    ]
},
{
    "id": "SCS.INVENTORY.NONCONFORMANCE_REPORT",
    "name": "NONCONFORMANCE_REPORT",
    "fullyQualifiedNames": "SCS.INVENTORY.NONCONFORMANCE_REPORT",
    "description": "Non-conformance reports \u2014 1,000 NCRs",
    "synonyms": [
        "Nonconformance Report",
        "ncr",
        "quality issue",
        "defect report",
        "NONCONFORMANCE_REPORT",
        "car"
    ],
    "datasourceId": "ds-scs-snowflake",
    "sourceSchema": "SCS.INVENTORY",
    "columns": [
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            "name": "NCR_ID",
            "dataType": "INT",
            "description": "",
            "constraint": "PRIMARY_KEY",
            "ordinalPosition": 1
        },
        {
            "name": "INSPECTION_ID",
            "dataType": "INT",
            "description": "",
            "constraint": "NULL",
            "ordinalPosition": 2
        },
        {
            "name": "SUPPLIER_ID",
            "dataType": "INT",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 3
        },
        {
            "name": "PART_ID",
            "dataType": "INT",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 4
        },
        {
            "name": "PLANT_ID",
            "dataType": "INT",
            "description": "",
            "constraint": "NOT_NULL",

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    "name": "NCR_DATE",
    "dataType": "DATE",
    "description": "",
    "constraint": "NOT_NULL",
    "ordinalPosition": 6
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    "dataType": "VARCHAR",
    "description": "CRITICAL, MAJOR, MINOR",
    "constraint": "NOT_NULL",
    "ordinalPosition": 7,
    "dataLength": 20,
    "distinctValues": [
      "CRITICAL",
      "MAJOR",
      "MINOR"
    ]
  },
  {
    "name": "ROOT_CAUSE",
    "dataType": "VARCHAR",
    "description": "",
    "constraint": "NULL",
    "ordinalPosition": 8,
    "dataLength": 500,
    "distinctValues": [
      "Cracked substrate from shock",
      "Dimensional out-of-spec",
      "Documentation mismatch",
      "ESD damage in shipping",
      "Electrical test failure",
      "Material contamination",
      "Solder joint defect",
      "Surface finish degradation",
      "Unapproved substitution",
      "Wrong revision shipped"
    ]
  },
  {
    "name": "DISPOSITION",
    "dataType": "VARCHAR",
    "description": "SCRAP, REWORK, USE_AS_IS, RETURN",
    "constraint": "NOT_NULL",
    "ordinalPosition": 9,
    "dataLength": 20,
    "distinctValues": [
      "RETURN",
      "REWORK",
      "SCRAP",
      "USE_AS_IS"
    ]
  },
  {
    "name": "STATUS",
    "dataType": "VARCHAR",
    "description": "",
    "constraint": "NULL",
    "ordinalPosition": 10,
    "dataLength": 20,
    "distinctValues": [
      "CLOSED",
      "OPEN"
    ]
  }
],
"tableConstraints": [
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    "constraintType": "PRIMARY_KEY",
    "columns": [
      "NCR_ID"
    ]
  },
  {
    "constraintType": "FOREIGN_KEY",
    "columns": [
      "INSPECTION_ID"
    ]
  }
]

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    ],
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    ],
    "relationshipType": "MANY_TO_ONE"
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{
    "constraintType": "FOREIGN_KEY",
    "columns": [
        "SUPPLIER_ID"
    ],
    "referredColumns": [
        "DIM_SUPPLIER.SUPPLIER_ID"
    ],
    "relationshipType": "MANY_TO_ONE"
},
{
    "constraintType": "FOREIGN_KEY",
    "columns": [
        "PART_ID"
    ],
    "referredColumns": [
        "DIM_PART.PART_ID"
    ],
    "relationshipType": "MANY_TO_ONE"
},
{
    "constraintType": "FOREIGN_KEY",
    "columns": [
        "PLANT_ID"
    ],
    "referredColumns": [
        "DIM_PLANT.PLANT_ID"
    ],
    "relationshipType": "MANY_TO_ONE"
}
]
},
{
    "id": "SCS.INVENTORY.FACT_INVENTORY_SNAPSHOT",
    "name": "FACT_INVENTORY_SNAPSHOT",
    "fullyQualifiedName": "SCS.INVENTORY.FACT_INVENTORY_SNAPSHOT",
    "description": "Inventory snapshots \u2014 ~1.2M rows (part x warehouse x month-end)",
    "synonyms": [
        "Inventory Snapshot",
        "stock on hand",
        "inventory position",
        "soh",
        "FACT_INVENTORY_SNAPSHOT",
        "stock level"
    ],
    "datasourceId": "ds-scs-snowflake",
    "sourceSchema": "SCS.INVENTORY",
    "columns": [
        {
            "name": "SNAPSHOT_ID",
            "dataType": "INT",
            "description": "",
            "constraint": "PRIMARY_KEY",
            "ordinalPosition": 1
        },
        {
            "name": "SNAPSHOT_DATE",
            "dataType": "DATE",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 2
        },
        {
            "name": "PART_ID",
            "dataType": "INT",
            "description": "",
            "constraint": "NOT_NULL",
            "ordinalPosition": 3
        },
        {
            "name": "WAREHOUSE_ID",
            "dataType": "INT",
            "description": "",
            "constraint": "NOT_NULL",

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    "ordinalPosition": 4
  },
  {
    "name": "ON_HAND_QTY",
    "dataType": "DECIMAL",
    "description": "",
    "constraint": "NOT_NULL",
    "ordinalPosition": 5,
    "precision": 12,
    "scale": 2,
    "synonyms": [
      "stock on hand",
      "soh",
      "on hand quantity",
      "inventory quantity"
    ]
  },
  {
    "name": "UNIT_COST_USD",
    "dataType": "DECIMAL",
    "description": "",
    "constraint": "NOT_NULL",
    "ordinalPosition": 6,
    "precision": 10,
    "scale": 2
  },
  {
    "name": "ON_HAND_VALUE",
    "dataType": "DECIMAL",
    "description": "on_hand_qty * unit_cost_usd",
    "constraint": "NOT_NULL",
    "ordinalPosition": 7,
    "precision": 14,
    "scale": 2,
    "synonyms": [
      "inventory value",
      "stock value",
      "tied up capital"
    ]
  },
  {
    "name": "DAILY_USAGE_QTY",
    "dataType": "DECIMAL",
    "description": "Average daily consumption",
    "constraint": "NULL",
    "ordinalPosition": 8,
    "precision": 10,
    "scale": 2,
    "synonyms": [
      "daily usage",
      "consumption rate",
      "burn rate"
    ]
  },
  {
    "name": "DOI",
    "dataType": "DECIMAL",
    "description": "Days of inventory = on_hand / daily_usage",
    "constraint": "NULL",
    "ordinalPosition": 9,
    "precision": 10,
    "scale": 1,
    "synonyms": [
      "days of inventory",
      "days of supply",
      "dos",
      "coverage days"
    ]
  }
],
"tableConstraints": [
  {
    "constraintType": "PRIMARY_KEY",
    "columns": [
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    ]
  },
  {
    "constraintType": "FOREIGN_KEY",
    "columns": [

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[{"name": "PART_ID",  
    "referredColumns": [  
        "DIM_PART.PART_ID"  
    ],  
    "relationshipType": "MANY_TO_ONE"  
},  
{  
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    "columns": [  
        "WAREHOUSE_ID"  
    ],  
    "referredColumns": [  
        "DIM_WAREHOUSE.WAREHOUSE_ID"  
    ],  
    "relationshipType": "MANY_TO_ONE"  
},  
{  
    "constraintType": "FOREIGN_KEY",  
    "columns": [  
        "SNAPSHOT_DATE"  
    ],  
    "referredColumns": [  
        "DIM_DATE.DATE_KEY"  
    ],  
    "relationshipType": "MANY_TO_ONE"  
}]
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